

Overflow metabolism originates from growth optimization and cell heterogeneity

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Abstract

A classic problem in metabolism is that fast-proliferating cells use seemingly wasteful fermentation to generate energy in the presence of sufficient oxygen. This counterintuitive phenomenon, known as overflow metabolism, or the Warburg effect in cancer, is universal across various organisms. Despite extensive research, its origin and function remain unclear. Here, we take *Escherichia coli* as a typical example and show that overflow metabolism can be understood through growth optimization combined with cell heterogeneity. A model of optimal protein allocation, coupled with heterogeneity in enzyme catalytic rates among cells, quantitatively explains why and how cells make the choice between respiration and fermentation under different nutrient conditions. Our model quantitatively illustrates the growth rate dependence of fermentation flux and enzyme allocation under various perturbations, which is fully validated by experimental results. Our work solves the long-standing puzzle of overflow metabolism and can be broadly used to address heterogeneity-related challenges in metabolism.

eLife assessment

This **useful** study tackles the well-established overflow metabolism issue by applying a coarse-grained metabolic flux model to predict how individual cells execute various energy strategies, such as respiration versus fermentation. While the model's population average is **convincing** enough to align with experimental observations on overflow metabolism, the overall assertions to enhance our comprehension of this biological phenomenon are **incomplete**.

Introduction

A prominent feature of cancer metabolism is that tumor cells excrete large quantities of fermentation products in the presence of sufficient oxygen (Hanahan and Weinberg, 2011 ; Liberti and Locasale, 2016 ; Vander Heiden et al., 2009 ). This process, discovered by Otto Warburg in the 1920s (Warburg et al., 1924 ) and known as the Warburg effect, aerobic glycolysis or overflow metabolism, (Basan et al., 2015 ; Hanahan and Weinberg, 2011 ; Liberti and

Locasale, 2016 [\[1\]](#); Vander Heiden et al., 2009 [\[2\]](#)) is ubiquitous for fast-proliferating cells across a broad spectrum of organisms (Vander Heiden et al., 2009 [\[2\]](#)), ranging from bacteria (Basan et al., 2015 [\[3\]](#); Holms, 1996 [\[4\]](#); Meyer et al., 1984 [\[5\]](#); Nanchen et al., 2006 [\[6\]](#); Neidhardt et al., 1990 [\[7\]](#)) and fungi (De Deken, 1966 [\[8\]](#)) to mammalian cells (Hanahan and Weinberg, 2011 [\[9\]](#); Liberti and Locasale, 2016 [\[10\]](#); Vander Heiden et al., 2009 [\[2\]](#)). For microbes, cells use standard respiration when nutrients are scarce, while they use the counterintuitive aerobic glycolysis when nutrients are adequate, just analogous to normal tissues and cancer cells, respectively (Vander Heiden et al., 2009 [\[2\]](#)). Over the past century, especially with extensive studies in the last two decades (Liberti and Locasale, 2016 [\[10\]](#)), various rationales for overflow metabolism have been proposed (Basan et al., 2015 [\[3\]](#); Chen and Nielsen, 2019 [\[11\]](#); Majewski and Domach, 1990 [\[12\]](#); Molenaar et al., 2009 [\[13\]](#); Niebel et al., 2019 [\[14\]](#); Pfeiffer et al., 2001 [\[15\]](#); Shlomi et al., 2011 [\[16\]](#); Vander Heiden et al., 2009 [\[2\]](#); Varma and Palsson, 1994 [\[17\]](#); Vazquez et al., 2010 [\[18\]](#); Vazquez and Oltvai, 2016 [\[19\]](#)). In particular, Basan et al. (Basan et al., 2015 [\[3\]](#)) provided a systematic characterization of this process, including various types of perturbations in experiments. However, the origin and function of overflow metabolism still remain unclear (DeBerardinis and Chandel, 2020 [\[20\]](#); Hanahan and Weinberg, 2011 [\[9\]](#); Liberti and Locasale, 2016 [\[10\]](#); Vander Heiden et al., 2009 [\[2\]](#)).

Why have microbes and cancer cells evolved to possess the seemingly wasteful aerobic glycolysis strategy? For unicellular organisms, there is evolutionary pressure (Vander Heiden et al., 2009 [\[2\]](#)) to optimize cellular resources for rapid growth (Dekel and Alon, 2005 [\[21\]](#); Edwards et al., 2001 [\[22\]](#); Hui et al., 2015 [\[23\]](#); Li et al., 2018 [\[24\]](#); Scott et al., 2010 [\[25\]](#); Towbin et al., 2017 [\[26\]](#); Wang et al., 2019 [\[27\]](#); You et al., 2013 [\[28\]](#)). In particular, it has been shown that cells allocate protein resources for optimal growth (Hui et al., 2015 [\[23\]](#); Scott et al., 2010 [\[25\]](#); Wang et al., 2019 [\[27\]](#); You et al., 2013 [\[28\]](#)), and the most efficient protein allocation corresponds to elementary flux mode (Müller et al., 2014 [\[29\]](#); Wortel et al., 2014 [\[30\]](#)). In this study, we extend these approaches in a heterogeneous framework to address the puzzle of aerobic glycolysis. We use *Escherichia coli* as a typical example and show that overflow metabolism can be understood from optimal protein allocation combined with the heterogeneity in enzyme catalytic rates. The optimal growth strategy varies between respiration and fermentation depending on the concentration and type of the nutrient, and the combination with cell heterogeneity results in the standard picture (Basan et al., 2015 [\[3\]](#); Holms, 1996 [\[4\]](#); Meyer et al., 1984 [\[5\]](#); Nanchen et al., 2006 [\[6\]](#)) of overflow metabolism. Our model quantitatively illustrates the growth rate dependence of fermentation/respiration flux and enzyme allocation under various types of perturbations, and can be used to explain the Warburg effect in tumor cells.

Results

Coarse-grained model

Based on the topology of the metabolic network (Neidhardt et al., 1990 [\[7\]](#); Nelson et al., 2008 [\[31\]](#)) (Fig. 1A [\[32\]](#)), we classify the carbon sources that enter from the upper part of glycolysis into Group A (Wang et al., 2019 [\[27\]](#)), and the precursors of biomass components (such as amino acids) into five pools (see Appendix 1.2 [\[33\]](#) for details): a1 (entry point: G6P/F6P), a2 (entry point: GA3P/3PG/PEP), b (entry point: pyruvate/acetyl-CoA), c (entry point: α -ketoglutarate), and d (entry point: oxaloacetate). Pools a1 and a2 are also combined as Pool a due to joint synthesis of precursors. Then, the metabolic network for Group A carbon source utilization (Fig. 1A [\[32\]](#)) is coarse-grained into a model shown in Fig. 1B [\[34\]](#) (see Appendix 2.1 [\[35\]](#) for details), where node A represents an arbitrary carbon source of Group A. Evidently, Fig. 1B [\[34\]](#) is topologically identical to Fig. 1A [\[32\]](#). Each coarse-grained arrow in Fig. 1B [\[34\]](#) carries a stoichiometric flux J_i , which delivers carbon flux and may consume or produce energy (e.g., J_1, J_{a1} , see Figs. 1A-B [\[32\]](#) and Appendix-fig. 1A [\[36\]](#)).

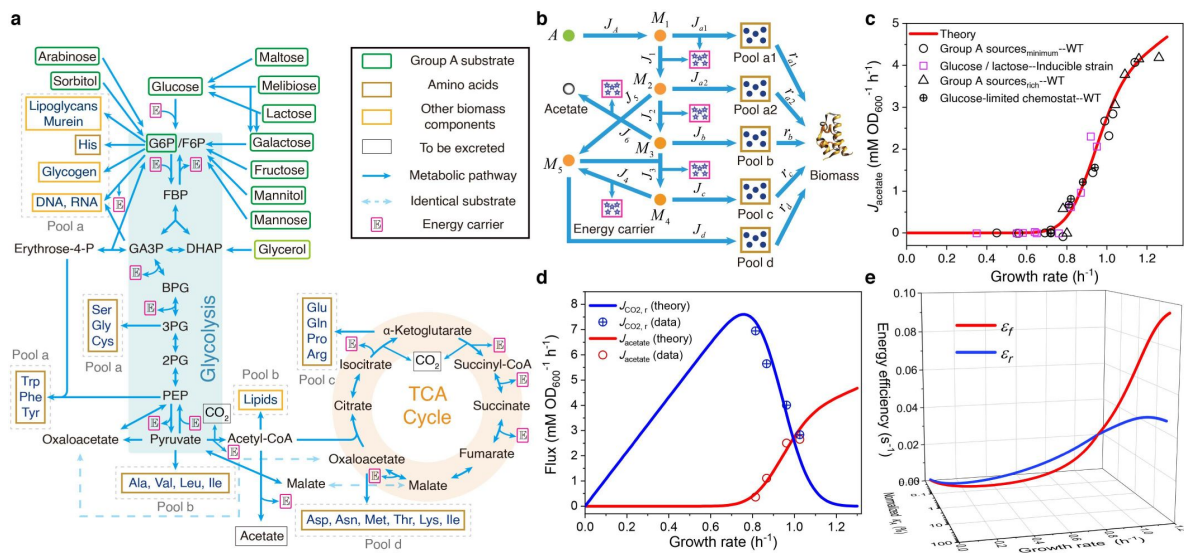


Fig 1.

Model and results of overflow metabolism.

(A) The central metabolic network of carbon source utilization. The Group A carbon sources (Wang et al., 2019) are labeled with green squares. (B) Coarse-grained model for Group A carbon source utilization. (C) Model predictions (Eqs. S47 and S160) and experimental results (Basan et al., 2015; Holms, 1996) of overflow metabolism, covering the data for all the Group A carbon sources shown in (A). (D) Growth rate dependence of respiration and fermentation fluxes (Eqs. S47 and S160). (E) The energy efficiencies of respiration and fermentation pathways vary with the growth rate as functions of the substrate quality of a Group A carbon source (Eqs. S31 and S36). See Appendix 8 for model parameter settings and experimental data sources (Basan et al., 2015; Holms, 1996; Hui et al., 2015) of Figs. 1–4.

In fact, the stoichiometric flux J_i scales with the cell population. For comparison with experiments, we define the normalized flux $J_i^{(N)} \equiv J_i \cdot m_0 / M_{\text{carbon}}$, which can be regarded as the flux per unit of biomass (the superscript “(N)” stands for normalized; see [Appendix 1.3–1.4](#) for details). Here, M_{carbon} represents the carbon mass of the cell population, and m_0 is the weighted average carbon mass of metabolite molecules at the entrance of precursor pools (see [Eq. S17](#)).

Then, the cell growth rate λ can be represented by $\lambda = \sum_i^{a1,a2,b,c,d} J_i^{(N)}$ (see [Appendix 1.4](#)), and

the normalized fluxes of respiration and fermentation are $J_r^{(N)} \equiv J_4^{(N)}$ and $J_f^{(N)} \equiv J_6^{(N)}$,

respectively. In practice, each $J_i^{(N)}$ is characterized by two quantities: the proteomic mass fraction ϕ_i of the enzyme dedicated to carrying the flux and the substrate quality κ_i , such that $J_i^{(N)} = \phi_i \cdot \kappa_i$. We take the Michaelis-Menten form for the enzyme kinetics, and then

$$\kappa_i \equiv k_i \cdot \frac{[S_i]}{[S_i] + K_i} \quad (\text{see [Eq. S12](#) and [Appendix 1.4](#) for details}), \text{ where } [S_i] \text{ is the}$$

concentration of substrate S_i . For each intermediate node and reaction along the pathway (e.g., node M_1 in J_{a1}), the substrate quality κ_i can be approximated as a constant (see [Appendix 1.5](#)):

$$\kappa_i \equiv k_i \cdot \frac{[S_i]}{[S_i] + K_i} \approx k_i, \text{ where } [S_i] \geq K_i \text{ generally holds true in bacteria ([Bennett et al., 2009](#);$$

[Park et al., 2016](#)). However, κ_A is a variable that depends on the concentration and nutrient type of a Group A carbon source (see [Eq. S27](#)).

Generally, there are three distinct destinies of a Group A carbon source in the metabolic network ([Appendix-fig. 1C-E](#)): fermentation, respiration, and biomass generation. Each draws a proteome fraction of ϕ_f , ϕ_r and ϕ_{BM} . The net effect of the first two destinies is energy production, while the last one generates precursors of biomass accompanied by energy production. By applying the proteomic constraint ([Scott et al., 2010](#)) that there is a maximum fraction ϕ_{max} for proteome allocation ($\phi_{\text{max}} \approx 0.48$ ([Scott et al., 2010](#))), we have:

$$\phi_f + \phi_r + \phi_{\text{BM}} = \phi_{\text{max}}. \quad (1)$$

In fact, [Eq. 1](#) is equivalent to $\phi_R + \phi_A + \sum_{j=1}^6 \phi_j + \sum_i^{a1,a2,b,c,d} \phi_i = \phi_{\text{max}}$ (see [Appendix 2.1](#)),

where ϕ_R stands for the mass fraction of the active ribosome-affiliated proteins. In cell proliferation, ribosomes constitute the majority of enzymes for protein synthesis ([Neidhardt et al., 1990](#); [Nelson et al., 2008](#)), while other biomass components such as RNA are optimally produced ([Kostinski and Reuveni, 2020](#)) following the growth rate set by protein synthesis. Thus, $\lambda = \phi_R \cdot \kappa_t$ where κ_t is a parameter determined by the translation rate ([Scott et al., 2010](#)) (see [Appendix 1.1](#) for details), which can be approximated as a constant in the growth rate range of concern ([Dai et al., 2016](#)).

For balanced cell growth, the energy demand is generally proportional to the biomass production rate. Thus, the normalized energy production rate $J_E^{(N)}$ is proportional to the growth rate λ (see [Appendix 2.1](#) for details):

$$J_E^{(N)} = \eta_E \cdot \lambda, \quad (2)$$

where η_E is the energy coefficient. By converting all the energy currencies into ATPs, the normalized energy fluxes of respiration and fermentation are $J_r^{(E)} = \beta_r^{(A)} \cdot J_r^{(N)} / 2$ and

$$J_f^{(E)} = \beta_f^{(A)} \cdot J_f^{(N)} / 2, \text{ where } \beta_r^{(A)} \text{ and } \beta_f^{(A)} \text{ are the stoichiometric coefficients of ATP}$$

production per glucose in each pathway (see [Appendix-fig. 1C-E](#) and [Appendix 2.1](#) for details). The denominator coefficient (“2”) is derived from the stoichiometry of the coarse-grained reaction $M_1 \rightarrow M_2$ (see [Fig. 1A-B](#)). Using flux balance analysis at each intermediate node (M_i , $i = 1, \dots, 5$) and precursor pool (Pool i , $i = a1, a2, b, c, d$), combined with the constraints of proteome allocation ([Eq. 1](#)) and energy demand ([Eq. 2](#)), we obtain the relations between energy fluxes and growth rate for a given nutrient condition with fixed κ_A (see [Appendix 2.1](#) for details):

$$\begin{cases} J_r^{(E)} + J_f^{(E)} = \phi \cdot \lambda, \\ \frac{J_r^{(E)}}{\varepsilon_r} + \frac{J_f^{(E)}}{\varepsilon_f} = \phi_{\max} - \psi \cdot \lambda, \end{cases} \quad (3)$$

where ϕ is a constant coefficient mainly determined by η_E (see [Eq. S33](#)), and $\phi \cdot \lambda$ represents the normalized energy demand other than the biomass pathway. The coefficients ψ , ε_r and ε_f are all functions of κ_A . ψ^{-1} is the proteome efficiency of the biomass pathway (see [Eq. S32](#)), with $\psi^{-1} = \lambda / \phi_{\text{BM}} \cdot \varepsilon_r$ and ε_r and ε_f are the proteome energy efficiencies of the respiration and fermentation

pathways, with $\varepsilon_r \equiv J_r^{(E)} / \phi_r$ and $\varepsilon_f \equiv J_f^{(E)} / \phi_f$:

$$\begin{cases} \varepsilon_r = \frac{\beta_r^{(A)}}{1/\kappa_A + 1/\kappa_r^{(A)}}, \\ \varepsilon_f = \frac{\beta_f^{(A)}}{1/\kappa_A + 1/\kappa_f^{(A)}}, \end{cases} \quad (4)$$

where both $\kappa_r^{(A)}$ and $\kappa_f^{(A)}$ are constants, with $1/\kappa_r^{(A)} \equiv 1/\kappa_1 + 2/\kappa_2 + 2/\kappa_3 + 2/\kappa_4$ and

$$1/\kappa_f^{(A)} \equiv 1/\kappa_1 + 2/\kappa_2 + 2/\kappa_6 \text{ (see [Appendix 2.1](#) for details).}$$

Origin of overflow metabolism

The standard picture of overflow metabolism ([Basan et al., 2015](#); [Holms, 1996](#); [Meyer et al., 1984](#); [Nanchen et al., 2006](#)) is exemplified by the experimental data ([Basan et al., 2015](#)) shown in [Fig. 1C](#): the fermentation flux exhibits a threshold-analog dependence on the growth rate λ . It is well known that respiration is far more efficient than fermentation in terms of energy production per unit carbon (i.e., $\beta_r^{(A)} > \beta_f^{(A)}$) ([Nelson et al., 2008](#); [Vander Heiden et al.,](#)

2009). Then, why do cells bother to use the seemingly wasteful fermentation pathway? We proceed to resolve these issues using the optimal protein allocation hypothesis (Scott et al., 2010; You et al., 2013).

For cell proliferation in a given nutrient with fixed κ_A , the values of ε_r , ε_f and ψ are determined (Eqs. 4 and S32). However, the growth rate λ can be influenced by protein allocation between φ_r and φ_f with the governing equation Eq. 3. If $\varepsilon_r < \varepsilon_f$, then

$$\lambda = \frac{\phi_{\max} - J_f^{(E)} (1/\varepsilon_f - 1/\varepsilon_r)}{\psi + \varphi/\varepsilon_r} \leq \frac{\phi_{\max}}{\psi + \varphi/\varepsilon_r}. \text{ The optimal solution is } \varepsilon_f = J_f^{(E)} = 0, \text{ which}$$

means that the cell only uses respiration. Conversely, if $\varepsilon_f > \varepsilon_r$, $\varphi_r = J_r^{(E)} = 0$ is optimal, and the cell only uses fermentation. In either case, the optimal choice between respiration and fermentation is determined by weighing the proteome energy efficiencies.

In practice, both ε_r and ε_f are functions of κ_A (Eq. 4), and therefore the optimal choice may vary depending on the nutrient conditions. In nutrient-poor conditions where $\kappa_A \ll \kappa_r^{(A)}$ and $\kappa_A \ll$

$\kappa_f^{(A)}$, the proteome energy efficiencies are reduced to $\varepsilon_r \approx \beta_r^{(A)} \cdot \kappa_A$ and $\varepsilon_f \approx \beta_f^{(A)} \cdot \kappa_A$ (see

Eq. 4), where $\varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A)$. In rich media, however, using the parameters of κ_i derived from in vivo/in vitro experimental data of *Escherichia coli* (Appendix-table 1, see also Appendix

6.1–6.2 and Appendix-table 2), we confirm that $\varepsilon_r \left(\kappa_{\text{glucose}}^{(\text{ST})} \right) < \varepsilon_f \left(\kappa_{\text{glucose}}^{(\text{ST})} \right)$ with Eq.

4 (see also Eqs. S39–S40), where $\kappa_{\text{glucose}}^{(\text{ST})}$ represents the substrate quality of glucose at saturated concentration (abbreviated as “ST” in the superscript). Indeed, recent experimental studies have validated that $\varepsilon_r \left(\kappa_{\text{lactose}}^{(\text{ST})} \right) < \varepsilon_f \left(\kappa_{\text{lactose}}^{(\text{ST})} \right)$ for lactose (Basan et al., 2015). In Fig.

1E, we present the growth rate dependence of proteome energy efficiencies ε_r and ε_f in a three-dimensional (3D) form using the data from Appendix-table 1, where ε_r , ε_f and the growth rate λ all vary as functions of κ_A . Therefore, the ratio $\varepsilon_f/\varepsilon_r$ (defined as $\Delta(\kappa_A)$) is a monotonically increasing function of κ_A , and there exists a critical value of κ_A (denoted as $\kappa_A^{(C)}$, see Appendix

2.2 for details) satisfying $\Delta(\kappa_A^{(C)}) = 1$. Below $\kappa_A^{(C)}$, the cell grows slowly and the optimal

choice is respiration (with $\varepsilon_f < \varepsilon_r$, $\lambda = \phi_{\max} \cdot (\psi + \phi/\varepsilon_r)^{-1}$), while above $\kappa_A^{(C)}$, the cell grows faster and the optimal growth strategy is fermentation (with $\varepsilon_f > \varepsilon_r$, $\lambda = \phi_{\max} \cdot (\psi + \phi/\varepsilon_f)^{-1}$). The above analysis qualitatively explains the phenomenon of aerobic glycolysis.

For a quantitative understanding of overflow metabolism, let us first consider the homogeneous case, where all cells share identical biochemical parameters. For optimal protein allocation, the relation between fermentation flux and growth rate is $J_f^{(E)} = \phi \cdot \theta(\phi - \phi_C)$, where “ θ ”

represents the Heaviside step function, and λ_C is the critical growth rate for $\kappa_A^{(C)}$ (i.e., $\lambda_C \equiv \lambda$

$(\kappa_A^{(C)})$). Similarly, we can determine the growth rate dependence of respiration flux, which is

$J_r^{(E)} = \phi \cdot \lambda \cdot [1 - \theta(\lambda - \lambda_C)]$. These digital response outcomes are consistent with the numerical

simulation findings of Molenaar *et al.* (Molenaar *et al.*, 2009). However, they are clearly incompatible with the threshold-analog response in the standard picture of overflow metabolism (Basan *et al.*, 2015; Holms, 1996; Meyer *et al.*, 1984; Nanchen *et al.*, 2006).

To address this issue, we take into account cell heterogeneity, which is ubiquitous in both microbes (Ackermann, 2015; Bagamery *et al.*, 2020; Balaban *et al.*, 2004; Nikolic *et al.*, 2013; Solopova *et al.*, 2014; Wallden *et al.*, 2016) and tumor cells (Duraj *et al.*, 2021; Hanahan and Weinberg, 2011; Hensley *et al.*, 2016). For the Warburg effect or overflow metabolism of our concern, experimental studies have reported significant metabolic heterogeneity in the choice between respiration and fermentation within a cell population (Bagamery *et al.*, 2020; Duraj *et al.*, 2021; Hensley *et al.*, 2016; Nikolic *et al.*, 2013). Motivated by the fact that the turnover number (k_{cat} value) of a catalytic enzyme varies considerably between in vitro and in vivo measurements (Davidi *et al.*, 2016; García - Contreras *et al.*, 2012), we note that the concentrations of potassium and phosphate, which vary from cell to cell, have a significant impact on the k_{cat} values of the metabolic enzymes (García - Contreras *et al.*, 2012). Therefore, in a cell population, there is a distribution of values for k_{cat} , which is commonly referred to as extrinsic noise (Elowitz *et al.*, 2002). For simplicity, we assume that each k_{cat} value follows a Gaussian distribution. This gives the distributions of single-cell growth rate in various types of carbon sources (see Eqs. S155–S157, S163–S165), which are fully verified by recent experiments using isogenic *Escherichia coli* with single-cell resolution (Wallden *et al.*, 2016) (**Appendix-fig.**

2B). Then, the critical growth rate λ_c should follow a Gaussian distribution $\mathcal{N}(\mu_{\lambda_c}, \sigma_{\lambda_c}^2)$

in the cell population (see **Appendix 7** for details), where μ_{λ_c} is approximated by the deterministic result of λ_c (Eq. S43). We assume that the coefficient of variation (CV) is $\sigma_{\lambda_c}/\mu_{\lambda_c} = 12\%$, or equivalently, the CV for the catalytic rate of each metabolic enzyme is 25%. Thus, we have the growth rate dependence of fermentation and respiration fluxes (see **Appendix 2.3** for details):

$$\begin{cases} J_f^{(N)}(\lambda) = \frac{\varphi \cdot \lambda}{\beta_f^{(A)}} \cdot \left[\text{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) + 1 \right], \\ J_r^{(N)}(\lambda) = \frac{\varphi \cdot \lambda}{\beta_r^{(A)}} \cdot \left[1 - \text{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) \right], \end{cases} \quad (5)$$

where “erf” represents the error function. The fermentation flux exhibits a threshold-analog relation with the growth rate (the red curves in **Figs. 1C–D**, **2B–C** and **3B, D, F**), while the respiration flux (the blue curve in **Fig. 1D**) decreases with an increase in fermentation flux. In **Fig. 1C–D**, we see that the model results (see Eq. 5 and **Appendix 8**; parameters are set by the experimental data shown in Appendix-table S1) agree quantitatively with the experimental data of *Escherichia coli* (Basan *et al.*, 2015; Holms, 1996). The fermentation flux was

determined by the acetate secretion rate $J_{\text{acetate}}^{(M)} = 2J_f^{(N)}$, and the respiration flux was deduced

from the carbon dioxide flux $J_{\text{CO}_2, r}^{(M)} = 6J_r^{(N)}$ (the superscript “(M)” represents the measurable

flux in the unit of mM/OD600/h; see **Appendix 8.1** for details). Therefore, with the integration of cell heterogeneity, our model of optimal protein allocation quantitatively explains overflow metabolism.

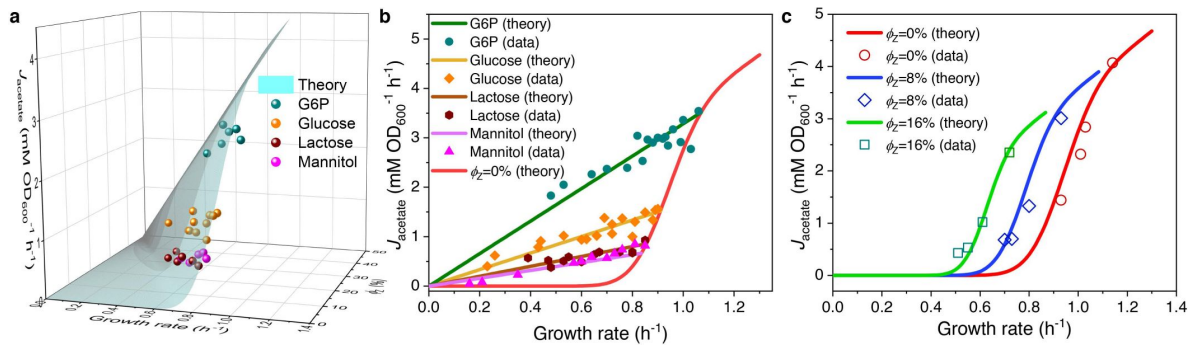


Fig 2.

Influence of protein overexpression on overflow metabolism.

(A) A 3D plot of the relations among fermentation flux, growth rate, and the expression level of useless proteins. In this plot, both the acetate excretion rate and growth rate vary as bivariate functions of the substrate quality of a Group A carbon source (denoted as κ_A) and the useless protein expression encoded by LacZ (denoted as ϕ_Z perturbation, see Eqs. S57 and S160). (B) Growth rate dependence of the acetate excretion rate upon ϕ_Z perturbation for each fixed nutrient condition (Eq. S58 and S160). (C) Growth rate dependence of the acetate excretion rate as κ_A varies (Eqs. S57 and S160), with each fixed expression level of LacZ.

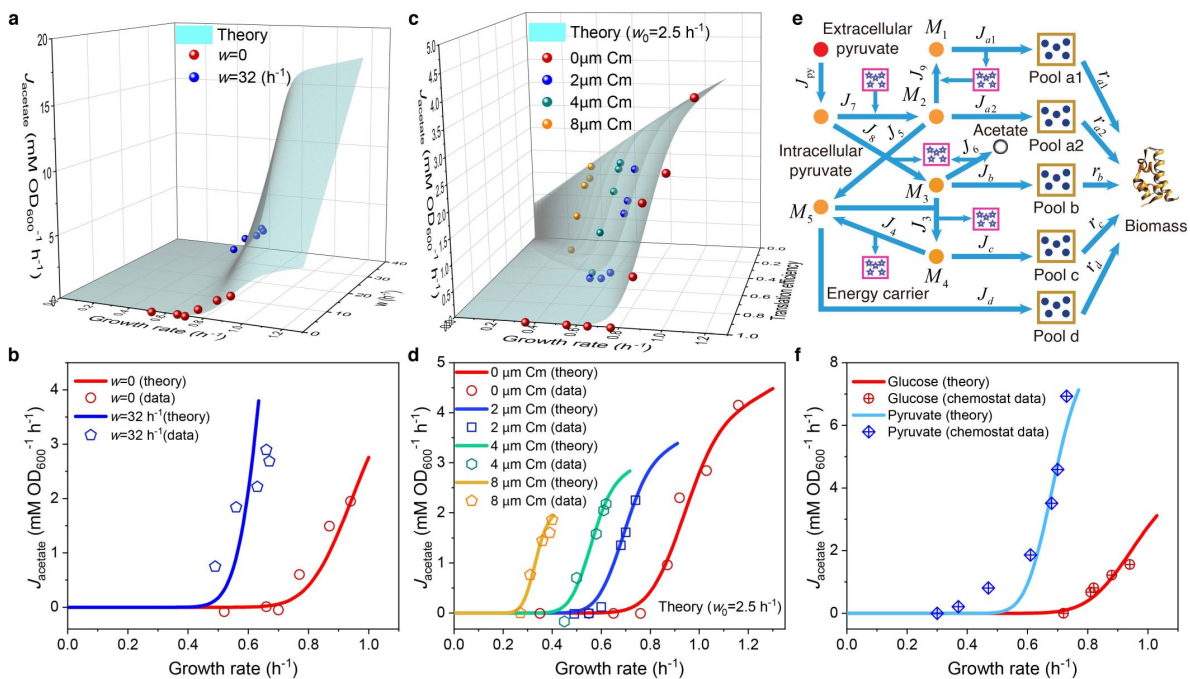


Fig 3.

Influence of energy dissipation, translation inhibition, and carbon source category alteration on overflow metabolism.

(A) A 3D plot of the relations among fermentation flux, growth rate, and the energy dissipation coefficient (Eqs. S70 and S160). (B) Growth rate dependence of the acetate excretion rate as κ_A varies, with each fixed energy dissipation coefficient determined by/fitted from experimental data. (C) A 3D plot of the relations among fermentation flux, growth rate, and the translation efficiency (Eqs. 85 and S160). Here, the translation efficiency is adjusted by the dose of chloramphenicol (Cm). (D) Growth rate dependence of the acetate excretion rate as κ_A varies, with each fixed dose of Cm. (E) Coarse-grained model for pyruvate utilization. (F) The growth rate dependence of fermentation flux in pyruvate (Eqs. 105 and S160) significantly differs from that of the Group A carbon sources (Eqs. 47 and S160).

Testing the model through perturbations

To further test our model, we systematically investigate model predictions under various types of perturbations and compare them with the experimental data from existing studies (Basan et al., 2015; Holms, 1996) (see Appendices 3 and 4.1 for details).

First, we consider the proteomic perturbation by overexpression of useless proteins encoded by the LacZ gene (i.e., ϕ_Z perturbation) in *Escherichia coli*. The net effect of the ϕ_Z perturbation is that the maximum fraction of proteome available for resource allocation changes from $\phi_{\max}$ to $\phi_{\max} - \phi_Z$, where ϕ_Z is the mass fraction of useless proteins. In a cell population, the critical growth rate $\lambda_C(\phi_Z)$ still follows a Gaussian distribution $N(\mu_{\lambda_C}(\phi_Z), \sigma_{\lambda_C}(\phi_Z)^2)$, where the CV of $\lambda_C(\phi_Z)$ remains unchanged. Consequently, the growth rate dependence of fermentation flux changes into

$$J_f^{(N)} = \frac{\phi \cdot \lambda}{\beta_f^{(A)}} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_C}(\phi_Z)}{\sqrt{2} \sigma_{\lambda_C}(\phi_Z)} \right) + 1 \right] \quad (\text{see Appendix 3 for model perturbation results of}$$

respiration flux), where both the growth rate $\lambda(\kappa_A, \phi_Z)$ and the normalized fermentation flux

$J_f^{(N)}(\kappa_A, \phi_Z)$ are bivariate functions of κ_A and ϕ_Z (see Eqs. S49 and S56–S57). For each

degree of LacZ expression (with fixed ϕ_Z), similar to the wild-type strains, the fermentation flux exhibits a threshold-analog response to growth rate as κ_A varies (Fig. 2C), which agrees quantitatively with experimental results (Basan et al., 2015). The shifts in the critical growth rate $\lambda_C(\phi_Z)$ are fully captured by $\mu_{\lambda_C}(\phi_Z) = \mu_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max})$. In contrast, for nutrient conditions with each fixed κ_A , since the growth rate changes with ϕ_Z just like $\lambda_C(\phi_Z)$: $\lambda(\kappa_A, \phi_Z) = \lambda(\kappa_A, 0)(1 - \phi_Z/\phi_{\max})$, the fermentation flux is then proportional to the growth rate for the varying levels of

$$\text{LacZ expression: } J_f^{(N)} = \frac{\phi}{\beta_f^{(A)}} \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_C}(0)}{\sqrt{2} \sigma_{\lambda_C}(0)} \right) + 1 \right] \cdot \lambda, \text{ where the slope is a}$$

monotonically increasing function of the substrate quality κ_A . These scaling relations are well validated by the experimental data (Basan et al., 2015) shown in Fig. 2B. Finally, in the case where both κ_A and ϕ_Z are free to vary, the growth rate dependence of fermentation flux presents a threshold-analog response surface in a 3D plot where ϕ_Z appears explicitly as the y-axis (Fig. 2A). Experimental data points (Basan et al., 2015) lie right on this surface (Fig. 2A), which are highly consistent with the model predictions.

Next, we study the influence of energy dissipation, which introduces an energy dissipation

coefficient “ w ” in Eq. 2: $J_E^{(N)} = \eta_E \cdot \lambda + w$. Similarly, the critical growth rate $\lambda_C(w)$ follows a

Gaussian distribution $N(\mu_{\lambda_C}(w), \sigma_{\lambda_C}(w)^2)$ in a cell population. The relation between the growth rate and fermentation flux can be characterized by (see Appendix 3.2 for details):

$$J_f^{(N)} = \frac{\phi \cdot \lambda + w}{\beta_f^{(A)}} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_C}(w)}{\sqrt{2} \sigma_{\lambda_C}(w)} \right) + 1 \right]. \text{ In Fig. 3A–B, we present a comparison between}$$

the model results and experimental data (Basan et al., 2015) in 3D and 2D plots, which demonstrate good agreement. A notable characteristic of energy dissipation, as distinguished from ϕ_Z perturbation, is that the fermentation flux increases despite a decrease in the growth rate when κ_A is fixed.

We proceed to analyze the impact of translation inhibition with different sub-lethal doses of chloramphenicol on *Escherichia coli*. This type of perturbation introduces an inhibition coefficient “ l ” in the translation rate, thus turning κ_t into $\kappa_t/(l+1)$. Still, the critical growth rate $\lambda_C(l)$ follows a Gaussian distribution $N(\mu_{\lambda_C}(l), \sigma_{\lambda_C}(l)^2)$, and then, the growth rate dependence of fermentation

flux is: $J_f^{(N)} = \frac{\varphi \cdot \lambda}{\beta_f^{(A)}} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)} \right) + 1 \right]$ (see Appendix 3.3 for details). In **Appendix-fig.**

2D-E, we see that the model predictions are generally consistent with the experimental data (Basan et al., 2015). However, there is a noticeable systematic discrepancy when the translation rate is small. Therefore, we turn to maintenance energy, which is tiny and generally negligible for bacteria over the growth rate range of concern (Basan et al., 2015; Neidhardt, 1996). Encouragingly, by assigning a very small value to the maintenance energy coefficient “ w_0 ” ($w_0 = 2.5 \text{ (h}^{-1}\text{)}$), the model results for the growth rate-fermentation flux relation

$$J_f^{(N)} = \frac{\varphi \cdot \lambda + w_0}{\beta_f^{(A)}} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)} \right) + 1 \right] \text{ agree quantitatively with experiments (Basan et al.,$$

2015) (see Fig. 3C-D and Appendix 3.3 for details).

Finally, we consider the alteration of nutrient categories by switching to a non-Group A carbon source: pyruvate, which enters the metabolic network from the endpoint of glycolysis (Neidhardt et al., 1990; Nelson et al., 2008). The coarse-grained model for pyruvate utilization is shown in Fig. 3E (see also Fig. 1A), which shares identical precursor pools as those for Group A carbon sources (Fig. 1B), yet with several differences in the coarse-grained reactions. The growth rate dependencies of both the proteome energy efficiencies (Appendix-fig. 2H) and energy fluxes are qualitatively similar to those of the Group A carbon source utilization, while there are quantitative differences in the coarse-grained parameters (see Appendices 4.1 and 8 for details).

Most notably, the critical growth rate $\lambda_c^{(\text{py})}$ and the ATP production per glucose in the

fermentation pathway $\beta_f^{(\text{py})}$ are noticeably smaller than λ_c and $\beta_f^{(A)}$ (for Group A sources), respectively. Consequently, the growth rate dependence of fermentation flux in pyruvate should present a quite different curve from that of Group A carbon sources (see Eqs. 5 and S105), which is fully validated by experimental results (Holms, 1996) (Fig. 3F).

Enzyme allocation under perturbations

As mentioned above, our coarse-grained model is topologically identical to the central metabolic network (Fig. 1A), and thus it can predict enzyme allocation for each gene in glycolysis and the TCA cycle (see Appendix-fig. 1B and Appendix-table 1) under various types of perturbations. In Fig. 1B, the intermediate nodes M_1, M_2, M_3, M_4 , and M_5 represent G6P, PEP, acetyl-CoA, α -ketoglutarate, and oxaloacetate, respectively. Then, φ_1 and φ_2 correspond to enzymes of glycolysis (or at the junction of glycolysis and the TCA cycle), while φ_3 and φ_4 correspond to enzymes in the TCA cycle.

We first consider enzyme allocation under carbon limitation by varying the nutrient type and concentration of a Group A carbon source (i.e., κ_A perturbation). In fact, this has been extensively studied in more simplified models (Hui et al., 2015; You et al., 2013), where the growth rate dependence of enzyme allocation under κ_A perturbation was generally considered to be a C-line response (Hui et al., 2015; You et al., 2013), i.e., the genes responsible for digesting carbon compounds show a linear increase in gene expression as the growth rate decreases (Hui et al., 2015; You et al., 2013). However, when it comes to enzymes catalyzing reactions between intermediate nodes, we collected experimental data from existing studies (Hui et al., 2015) and found that the enzymes in glycolysis exhibit a completely different response pattern compared to those in the TCA cycle (Appendix-fig. 3A-B). This discrepancy cannot be explained by the C-line response. To address this issue, we apply the coarse-grained model described above (Fig. 1B) to calculate the growth rate dependence of enzyme allocation for each φ_i ($i = 1, 2, 3, 4$) using the

model settings for wild-type strains, where no fitting parameters are involved in determining the shape (see **Eqs. S118**–**S119** and **Appendix 8**). In **Figs. 4A–B** and **Appendix-fig. 3C–D**, we see that the model predictions overall match with the experimental data (Hui et al., 2015) for representative genes from either glycolysis or the TCA cycle, and maintenance energy (with $w_0 = 2.5 \text{ (h}^{-1}\text{)}$) has a negligible effect on this process. Still, there are minor discrepancies that arise from the basal expression of metabolic genes, which may be attributed to the fact that our model deals with relatively stable growth conditions while microbes need to be prepared for fluctuating environments (Basan et al., 2020; Kussell and Leibler, 2005; Mori et al., 2017).

We proceed to analyze the influence of ϕ_Z perturbation and energy dissipation. In both cases, our model predicts a linear response to the growth rate reduction for all genes in either glycolysis or the TCA cycle (see Appendix 5.2–5.3 for details). For ϕ_Z perturbation, all predicted slopes are positive, and there are no fitting parameters involved (Eqs. S120–S121). In **Figs. 4C–D** and **Appendix-fig. 3E–J**, we show that our model quantitatively illustrates the experimental data (Basan et al., 2015) for representative genes in the central metabolic network, and there is a better agreement with experiments (Basan et al., 2015) by incorporating the maintenance energy (with $w_0 = 2.5 \text{ (h}^{-1}\text{)}$). For energy dissipation, however, the predicted slopes of the enzymes corresponding to ϕ_4 are surely negative, and there is a constraint that the slope signs of the enzymes corresponding to the same ϕ_i ($i = 1, 2, 3$) should be the same. In **Appendix-fig. 3K–N**, we see that the model results (**Eqs. S127** and **S123**) are consistent with experiments (Basan et al., 2015).

Discussion

The phenomenon of overflow metabolism, or the Warburg effect, has been a long-standing puzzle in cell metabolism. Although many rationales have been proposed (Basan et al., 2015; Chen and Nielsen, 2019; Majewski and Domach, 1990; Molenaar et al., 2009; Niebel et al., 2019; Pfeiffer et al., 2001; Shlomi et al., 2011; Vander Heiden et al., 2009; Varma and Palsson, 1994; Vazquez et al., 2010; Vazquez and Oltvai, 2016) over the past century, the origin and function of this phenomenon remain unclear (DeBerardinis and Chandel, 2020; Hanahan and Weinberg, 2011; Liberti and Locasale, 2016; Vander Heiden et al., 2009). In this study, we use *Escherichia coli* as a typical example and demonstrate that overflow metabolism can be understood through optimal protein allocation combined with cell heterogeneity. In nutrient-poor conditions, the proteome energy efficiency of respiration is higher than that of fermentation (**Fig. 1E**), and thus the cell uses respiration to optimize growth. In rich media, however, the proteome energy efficiency of fermentation increases faster and is higher than that of respiration (**Fig. 1E**), leading the cell to use fermentation to accelerate growth. In further combination with cell heterogeneity in enzyme catalytic rates (Davidi et al., 2016; García - Contreras et al., 2012), our model quantitatively illustrates the threshold-analog response (Basan et al., 2015; Holms, 1996) in overflow metabolism (**Fig. 1C**).

Cell heterogeneity is crucial for the threshold-analog response in overflow metabolism. In the homogeneous case, the optimal solution is a digital response (**Eq. S44**) that corresponds to an elementary flux mode (Müller et al., 2014; Wortel et al., 2014) and agrees with the numerical study of Molenaar *et al.* (Molenaar et al., 2009). However, this digital response is incompatible with the standard picture of overflow metabolism (Basan et al., 2015; Holms, 1996; Meyer et al., 1984; Nanchen et al., 2006). By incorporating heterogeneity in enzyme catalytic rates (Davidi et al., 2016; García - Contreras et al., 2012), the critical growth rate (i.e., threshold) changes from a single value into a Gaussian distribution (Eq. 45, see Appendix 7 for details; see also **Appendix-fig. 4**) for a cell population, thus turning a digital response into the threshold-analog response in overflow metabolism (**Fig. 1C**). Our model results relying on cell heterogeneity are fully validated by the observed distributions of single-cell growth rate (Wallden

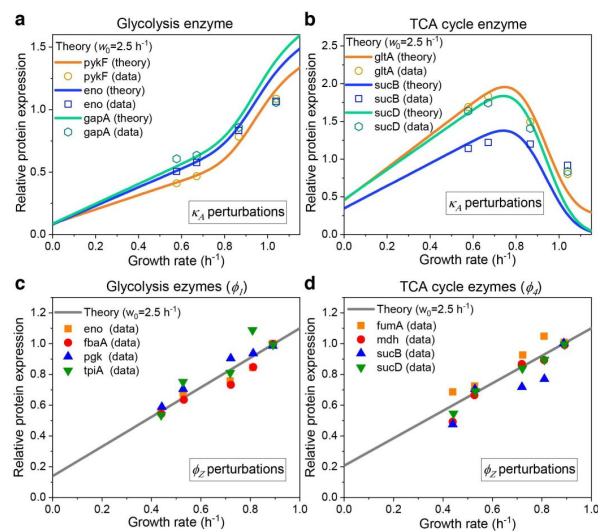


Fig 4.

Relative protein expression of central metabolic enzymes under κ_A and ϕ_Z perturbations.

(A, C) Relative protein expression of representative genes from glycolysis. (B, D) Relative protein expression of representative genes from the TCA cycle. (A, B) Results of κ_A perturbation (Eq. S119). (C, D) Results of ϕ_Z perturbation (Eq. S121).

et al., 2016 [\[1\]](#) (**Appendix-fig. 2B** [\[1\]](#)) and experiments with various types of perturbations (Basan et al., 2015 [\[2\]](#); Holms, 1996 [\[3\]](#); Hui et al., 2015 [\[4\]](#)), both for acetate secretion patterns and gene expression in the central metabolic network (**Figs. 2** [\[1\]](#)–**4** [\[1\]](#) and **Appendix-figs. 2D-E** [\[1\]](#) and **3** [\[1\]](#)).

Finally, our model can be broadly used to address heterogeneity-related challenges in metabolism on a quantitative basis, including the Crabtree effect in yeast (Bagamery et al., 2020 [\[5\]](#); De Deken, 1966 [\[6\]](#)), the Warburg effect in cancer (Duraj et al., 2021 [\[7\]](#); Hanahan and Weinberg, 2011 [\[8\]](#); Liberti and Locasale, 2016 [\[9\]](#); Vander Heiden et al., 2009 [\[10\]](#)) (see Appendix 6.4 [\[1\]](#) for an explanation of the Warburg effect), and the heterogeneous metabolic strategies of cells in various types of environments (Bagamery et al., 2020 [\[5\]](#); Duraj et al., 2021 [\[7\]](#); Escalante-Chong et al., 2015 [\[11\]](#); Hensley et al., 2016 [\[12\]](#); Liu et al., 2015 [\[13\]](#); Solopova et al., 2014 [\[14\]](#); Wang et al., 2019 [\[15\]](#)).

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Author contributions

X.W. conceived and designed the project, developed the model, carried out the analytical/numerical calculations, and wrote the paper.

Data, Materials, and Software Availability

All study data are included in the article and/or appendices.

Appendix 1 Model framework

Appendix 1.1 Proteome partition

Here we adopt the proteome partition framework similar to that introduced by Scott *et al.* (Scott et al., 2010 [\[16\]](#)). All proteins in a cell are classified into three classes: the fixed portion Q-class, the active ribosome-affiliated R-class, and the remaining catabolic/anabolic enzymes C-class. Each

proteome class has a mass $M_i^{(P)}$ ($i = Q, R, C$) and mass fraction ϕ_i , where ϕ_Q is a constant, and

we define $\phi_{\max} \equiv 1 - \phi_Q$. In the exponential growth phase, the ribosome allocation for protein synthesis of each class is f_i , with $f_Q + f_R + f_C = 1$.

To analyze cell growth optimization, we first consider the homogeneous case where all cells share identical biochemical parameters and simplify the mass accumulation of a cell population into a big cell. Essentially, this approximation would not influence the value of growth rate λ . For bacteria, the protein turnover rate is negligible, and thus the mass accumulation of each class follows:

$$dM_i^{(P)}/dt = f_i \cdot k_T \cdot N_R \cdot m_{AA} \quad (i = Q, R, C), \quad (\text{S1})$$

where m_{AA} stands for the average molecular weight of amino acids, k_T is the translation rate, $N_R = M_{rp}^{(P)} / m_R$ is the number of ribosomes, m_R is the protein mass of a single ribosome, and $M_{rp}^{(P)}$ is the protein mass of the ribosomes with $M_R^{(P)} / M_{rp}^{(P)} = \zeta \approx 1.67$ (Neidhardt, 1996; Scott et al., 2010). For a specific stable nutrient environment, f_R and k_T are temporal invariants. Then,

$$M_i^{(P)}(t) = M_i^{(P)}(0) + f_i / f_R \cdot M_R^{(P)}(0) \cdot [\exp(\lambda \cdot t) - 1] \quad (i = Q, R, C), \quad (S2)$$

where $\lambda = f_R \cdot k_T \cdot m_{AA} / (\zeta \cdot m_R)$, and the protein mass of the cell population

$$M_{\text{protein}} \equiv \sum_i^{Q,R,C} M_i^{(P)} \text{ follows:}$$

$$M_{\text{protein}}(t) = M_{\text{protein}}(0) + M_R(0) \cdot [\exp(\lambda \cdot t) - 1] / f_R. \quad (S3)$$

Over a long period in the exponential growth phase ($t \rightarrow +\infty$), $\phi_i = f_i$ ($i = Q, R, C$) and

$$\lambda = \phi_R \cdot \kappa_t, \quad (S4)$$

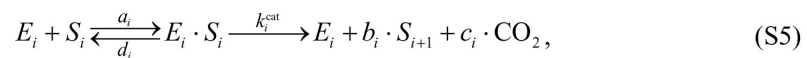
$$\text{where } \kappa_t = k_T \cdot m_{AA} / (\zeta \cdot m_R).$$

Appendix 1.2 Precursor pools

Based on the entry point of the metabolic network, we classify the precursors of biomass components into five pools (Fig. 1A-B): a1 (entry point: G6P/F6P), a2 (entry point: GA3P/3PG/PEP), b (entry point: pyruvate/acetyl-CoA), c (entry point: α -ketoglutarate) and d (entry point: oxaloacetate). For bacteria, these five pools draw roughly $r_{a1} = 24\%$, $r_{a2} = 24\%$, $r_b = 28\%$, $r_c = 12\%$ and $r_d = 12\%$ of the carbon flux (Nelson et al., 2008; Wang et al., 2019). There are overlapping components between Pools a1 and a2 due to the joint synthesis of some precursors, thus we also use Pool a to represent Pools a1-a2 in the descriptions.

Appendix 1.3 Stoichiometric flux

We consider the following biochemical reaction between substrate S_i and enzyme E_i :



where a_i , d_i and k_i^{cat} are the reaction parameters, S_{i+1} is the product, b_i and c_i are the stoichiometric coefficients. For most of the reactions in the central metabolism, $b_i = 1$ and $c_i = 0$. The reaction rate follows Michaelis-Menten kinetics (Nelson et al., 2008):

$$v_i = k_i^{\text{cat}} \cdot [E_i] \cdot \frac{[S_i]}{[S_i] + K_i}, \quad (S6)$$

where $K_i \equiv (d_i + k_i^{\text{cat}}) / a_i$, $[E_i]$ and $[S_i]$ are the concentrations of enzyme E_i and substrate E_i , respectively. For this reaction (Eq. S5), $d[S_{i+1}]/dt = b_i \cdot v_i$ and $d[S_i]/dt = -v_i$. In the cell populations (the “big cell”), suppose that the cell volume is V_{cell} , then the stoichiometric flux of the reaction is:

$$J_i \equiv V_{\text{cell}} \cdot v_i. \quad (\text{S7})$$

The copy number of enzyme E_i is $N_{E_i} = V_{\text{cell}} \cdot [E_i]$ with a total weight of $M_{E_i} = N_{E_i} \cdot m_{E_i}$,

where m_{E_i} is the molecular weight of E_i . By defining the enzyme cost of an E_i molecule as

$$n_{E_i} \equiv m_{E_i} / m_0, \text{ where } m_0 \text{ is a unit mass, then the cost of all } E_i \text{ molecules is } \Phi_i \equiv N_{E_i} \cdot n_{E_i}$$

(Wang et al., 2019). By further defining $\xi_i \equiv \frac{k_i^{\text{cat}}}{n_{E_i}} \cdot \frac{[S_i]}{[S_i] + K_i}$, then,

$$J_i = \Phi_i \cdot \xi_i. \quad (\text{S8})$$

The mass fraction of E_i is $\phi_i = M_{E_i} / M_{\text{protein}}$, and thus

$$\phi_i = \Phi_i \cdot \frac{m_0}{M_{\text{protein}}}. \quad (\text{S9})$$

Appendix 1.4 Carbon flux and cell growth rate

To clarify the relation between the stoichiometric flux J_i and growth rate λ , we consider the carbon flux in the biomass production. The carbon mass of the cell population (the “big cell”) is given by $M_{\text{carbon}} = M_{\text{protein}} \cdot r_{\text{carbon}} / r_{\text{protein}}$, where r_{carbon} and r_{protein} represent the mass fraction of carbon and protein within a cell. In the exponential growth phase, the carbon flux of the biomass production is given by:

$$J_{\text{BM}} = \frac{1}{m_{\text{carbon}}} \cdot \frac{dM_{\text{carbon}}}{dt} = \lambda \cdot \frac{M_{\text{carbon}}}{m_{\text{carbon}}}, \quad (\text{S10})$$

where m_{carbon} is the mass of a carbon atom. In fact, the carbon mass flux per stoichiometry varies depending on the entry point of the precursor pool. Taking Pool b as an example, there are three carbon atoms in a molecule of the entry point metabolite (i.e., pyruvate). Assuming that carbon atoms are conserved from pyruvate to Pool b, then the carbon flux of Pool b is given by

$$J_b^{\text{carbon}} = J_b \cdot N_{\text{py}}^{\text{carbon}}, \text{ where } J_b \text{ is the stoichiometric flux from pyruvate to Pool b (Fig. 1A-}$$

B and $N_{\text{py}}^{\text{carbon}}$ stands for the carbon number of a pyruvate molecule. Combining with **Eq.**

S10 and noting that $J_b^{\text{carbon}} = r_b \cdot J_{\text{BM}}$, we get

$J_b \cdot N_{\text{py}}^{\text{carbon}} \cdot m_{\text{carbon}} = r_b \cdot \lambda \cdot M_{\text{carbon}}$. Similarly, for each precursor pool, we have:

$$J_i \cdot N_{\text{EP}_i}^{\text{carbon}} \cdot m_{\text{carbon}} = r_i \cdot \lambda \cdot M_{\text{carbon}} \quad (i = a1, a2, b, c, d), \quad (\text{S11})$$

where the subscript “EP_{*i*}” represents the entry point of Pool *i*, and $N_{\text{EP}_i}^{\text{carbon}}$ stands for the number of carbon atoms in a molecule of the entry point metabolite.

For each substrate in intermediate steps of the metabolic network, we define κ_i as the substrate quality:

$$\kappa_i \equiv \xi_i \cdot \frac{r_{\text{protein}}}{r_{\text{carbon}}} = \frac{r_{\text{protein}}}{r_{\text{carbon}}} \cdot \frac{k_i^{\text{cat}}}{n_{E_i}} \cdot \frac{[S_i]}{[S_i] + K_i}, \quad (\text{S12})$$

and for each precursor pool, we define:

$$\eta_i \equiv r_i \cdot m_0 / (N_{\text{EP}_i}^{\text{carbon}} \cdot m_{\text{carbon}}) \quad (i = a1, a2, b, c, d). \quad (\text{S13})$$

Combining **Eqs. S8**, **S9** and **S11**, we have

$$\phi_i \cdot \kappa_i = \eta_i \cdot \lambda \quad (i = a1, a2, b, c, d). \quad (\text{S14})$$

Then, we define the normalized flux, which can be regarded as the flux per unit of biomass:

$$J_i^{(\text{N})} \equiv \phi_i \cdot \kappa_i, \quad (\text{S15})$$

where the superscript “(N)” stands for normalized. Combined with **Eqs. S8**, **S9** and **S12**, we have:

$$J_i^{(\text{N})} \equiv J_i \cdot \frac{m_0}{M_{\text{carbon}}}. \quad (\text{S16})$$

Since $\sum_i^{a1, a2, b, c, d} r_i = 1$, by setting

$$m_0 = \left[\sum_i r_i / N_{\text{EP}_i}^{\text{carbon}} \right]^{-1} \cdot m_{\text{carbon}}, \quad (\text{S17})$$

then,

$$\eta_i = \frac{r_i}{N_{EP_i}^{\text{carbon}}} \cdot \left[\sum_j^{a1,a2,b,c,d} \frac{r_j}{N_{EP_j}^{\text{carbon}}} \right]^{-1} \quad (i = a1, a2, b, c, d), \quad (\text{S18})$$

$$\text{and we have } \sum_i^{a1,a2,b,c,d} \eta_i = 1, \text{ and}$$

$$\sum_i^{a1,a2,b,c,d} \phi_i \cdot \kappa_i = \lambda. \quad (\text{S19})$$

Appendix 1.5 Intermediate nodes

In a metabolic network, the metabolites between the carbon source and precursor pools are the intermediate nodes. As specified in Wang *et al.* (Wang *et al.*, 2019), to optimize cell growth rate, the substrate of each intermediate node is nearly saturated, and thus

$$\kappa_i \approx \frac{r_{\text{protein}}}{r_{\text{carbon}}} \cdot \frac{k_i^{\text{cat}}}{n_{E_i}}. \quad (\text{S20})$$

The real cases could be more complicated because of other metabolic regulations. Recent quantitative studies (Bennett *et al.*, 2009; Park *et al.*, 2016) have shown that at least in *E. coli*,

for most of the substrate-enzyme pairs $[S_i] > K_i$, which implies $\kappa_i \approx \frac{r_{\text{protein}}}{r_{\text{carbon}}} \cdot \frac{k_i^{\text{cat}}}{n_{E_i}}$.

Appendix 2 Model and analysis

Appendix 2.1 Coarse-grained model

In the coarse-grained model shown in Fig. 1B, node A represents an arbitrary carbon source of Group A (Wang *et al.*, 2019), which joins at the upper part of glycolysis. Nodes M1, M2, M3, M4, and M5 stand for G6P, PEP, acetyl-CoA, α -ketoglutarate, and oxaloacetate, respectively. In the analysis of carbon supply into precursor pools, we lump sum G6P/F6P as M1, GA3P/3PG/PEP as M2, and pyruvate/acetyl-CoA as M3 for approximation. For the biochemical reactions, each follows Eq. S5 with $b_i = 1$ except that $M1 \rightarrow 2M2$ and $M3+M5 \rightarrow M4$. Basically, there are 3 possible destinies of a Group A carbon source (e.g., glucose, see Appendix-fig. 1C-E): energy contributions in the fermentation and respiration pathways (Appendix-fig. 1C-D), or biomass components accompanied by energy production in the biomass pathway (Appendix-fig. 1E).

By applying flux balance to the stoichiometric fluxes and combining with Eq. S8, we have:

$$\begin{cases} \Phi_A \cdot \xi_A = \Phi_1 \cdot \xi_1 + \Phi_{a1} \cdot \xi_{a1}, \\ 2\Phi_1 \cdot \xi_1 = \Phi_2 \cdot \xi_2 + \Phi_5 \cdot \xi_5 + \Phi_{a2} \cdot \xi_{a2}, \\ \Phi_2 \cdot \xi_2 = \Phi_3 \cdot \xi_3 + \Phi_6 \cdot \xi_6 + \Phi_b \cdot \xi_b, \\ \Phi_5 \cdot \xi_5 + \Phi_4 \cdot \xi_4 = \Phi_3 \cdot \xi_3 + \Phi_d \cdot \xi_d, \\ \Phi_3 \cdot \xi_3 = \Phi_4 \cdot \xi_4 + \Phi_c \cdot \xi_c. \end{cases} \quad (\text{S21})$$

Obviously, the stoichiometric fluxes of respiration J_r and fermentation J_f (**Appendix-fig. 1C-D**) are:

$$\begin{cases} J_r \equiv J_4 = \Phi_4 \cdot \xi_4, \\ J_f \equiv J_6 = \Phi_6 \cdot \xi_6. \end{cases} \quad (\text{S22})$$

We further assume that the carbon atoms are conserved from each entry point metabolite to the precursor pool, and then,

$$\Phi_i \cdot \xi_i \cdot N_{\text{EP}_i}^{\text{carbon}} = r_i \cdot J_{\text{BM}} \quad (i = a1, a2, b, c, d). \quad (\text{S23})$$

In terms of energy production for the relevant reactions, for convenience, we convert all the energy currencies into ATPs, namely, $\text{NADH} \rightarrow 2\text{ATP}$ (Neidhardt et al., 1990), $\text{NADPH} \rightarrow 2\text{ATP}$ (Neidhardt et al., 1990; Sauer et al., 2004), $\text{FADH}_2 \rightarrow 1\text{ATP}$ (Neidhardt et al., 1990). Then, we have

$$\beta_1 \cdot \Phi_1 \cdot \xi_1 + \beta_2 \cdot \Phi_2 \cdot \xi_2 + \beta_3 \cdot \Phi_3 \cdot \xi_3 + \beta_4 \cdot \Phi_4 \cdot \xi_4 + \beta_6 \cdot \Phi_6 \cdot \xi_6 + \beta_{a1} \cdot \Phi_{a1} \cdot \xi_{a1} = J_E, \quad (\text{S24})$$

where J_E represents the stoichiometric flux of ATPs, and β_i is the stoichiometric coefficient with $\beta_1 = 4$, $\beta_2 = 3$, $\beta_3 = 2$, $\beta_4 = 6$, and $\beta_{a1} = 4$ (Neidhardt et al., 1990; Sauer et al., 2004). Generally, the energy demand is proportional to the carbon flux infused into biomass production, thus,

$$J_E = r_E \cdot J_{\text{BM}}, \quad (\text{S25})$$

where r_E is the ratio and also a constant.

By applying the substitutions specified in Eqs. S9, S12, S14–S18, combined with Eqs. S4, S10, S21–S25, and the constraint of proteome resource allocation $\varphi_R + \varphi_C = \varphi_{\text{max}}$, we have:

$$\begin{cases} \phi_A \cdot \kappa_A = \phi_1 \cdot \kappa_1 + \phi_{a1} \cdot \kappa_{a1}, \\ 2\phi_1 \cdot \kappa_1 = \phi_2 \cdot \kappa_2 + \phi_5 \cdot \kappa_5 + \phi_{a2} \cdot \kappa_{a2}, \\ \phi_2 \cdot \kappa_2 = \phi_3 \cdot \kappa_3 + \phi_6 \cdot \kappa_6 + \phi_b \cdot \kappa_b, \\ \phi_5 \cdot \kappa_5 + \phi_4 \cdot \kappa_4 = \phi_3 \cdot \kappa_3 + \phi_d \cdot \kappa_d, \\ \phi_3 \cdot \kappa_3 = \phi_4 \cdot \kappa_4 + \phi_c \cdot \kappa_c, \\ \phi_{a1} \cdot \kappa_{a1} = \eta_{a1} \cdot \lambda, \phi_{a2} \cdot \kappa_{a2} = \eta_{a2} \cdot \lambda, \phi_b \cdot \kappa_b = \eta_b \cdot \lambda, \phi_c \cdot \kappa_c = \eta_c \cdot \lambda, \phi_d \cdot \kappa_d = \eta_d \cdot \lambda, \\ \beta_1 \cdot \phi_1 \cdot \kappa_1 + \beta_2 \cdot \phi_2 \cdot \kappa_2 + \beta_3 \cdot \phi_3 \cdot \kappa_3 + \beta_4 \cdot \phi_4 \cdot \kappa_4 + \beta_6 \cdot \phi_6 \cdot \kappa_6 + \beta_{a1} \cdot \phi_{a1} \cdot \kappa_{a1} = J_E^{(N)}, \\ J_E^{(N)} = \eta_E \cdot \lambda, \lambda = \phi_R \cdot \kappa_t, J_r^{(N)} = \phi_4 \cdot \kappa_4, J_f^{(N)} = \phi_6 \cdot \kappa_6, \\ \phi_R + \phi_A + \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 + \phi_{a1} + \phi_{a2} + \phi_b + \phi_c + \phi_d = \phi_{\text{max}}, \end{cases} \quad (\text{S26})$$

where $\eta_E = r_E \cdot \left[\sum_i r_i / N_{EP_i}^{\text{carbon}} \right]^{-1}$. Here, for each intermediate node, κ_i follows Eq. S20, which can be approximated as a constant. The substrate quality of the Group A carbon source κ_A varies with the identity and concentration of the Group A carbon source:

$$\kappa_A \equiv \frac{r_{\text{protein}}}{r_{\text{carbon}}} \cdot \frac{k_A^{\text{cat}}}{m_{E_A}} \cdot \frac{[A]}{[A] + K_A} \cdot m_0, \quad (\text{S27})$$

which is determined externally by the culture condition. From Eq. S26, all ϕ_i can be expressed by $J_r^{(N)}$, $J_f^{(N)}$, and λ :

$$\begin{cases} \phi_A = \left[J_r^{(N)} + J_f^{(N)} + (2\eta_{a1} + \eta_{a2} + \eta_b + 2\eta_c + \eta_d)\lambda \right] / (2 \cdot \kappa_A), \\ \phi_1 = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d)\lambda \right] / (2 \cdot \kappa_1), \\ \phi_2 = \left[J_r^{(N)} + J_f^{(N)} + (\eta_b + \eta_c)\lambda \right] / \kappa_2, \\ \phi_3 = (J_r^{(N)} + \eta_c \cdot \lambda) / \kappa_3, \phi_4 = J_r^{(N)} / \kappa_4, \\ \phi_5 = (\eta_c + \eta_d)\lambda / \kappa_5, \phi_6 = J_f^{(N)} / \kappa_6, \\ \phi_i = \eta_i \cdot \lambda / \kappa_i \quad (i = a1, a2, b, c, d). \end{cases} \quad (\text{S28})$$

By substituting Eq. S28 into Eq. S26, we have:

$$\begin{cases} J_r^{(E)} + J_f^{(E)} = \varphi \cdot \lambda, \\ \frac{J_r^{(E)}}{\varepsilon_r} + \frac{J_f^{(E)}}{\varepsilon_f} = \phi_{\max} - \psi \cdot \lambda. \end{cases} \quad (\text{S29})$$

Here, $J_r^{(E)}$ and $J_f^{(E)}$ stand for the normalized energy fluxes of respiration and fermentation, with

$$\begin{cases} J_r^{(E)} = (\beta_1/2 + \beta_2 + \beta_3 + \beta_4) \cdot J_r^{(N)}, \\ J_f^{(E)} = (\beta_1/2 + \beta_2 + \beta_6) \cdot J_f^{(N)}. \end{cases} \quad (\text{S30})$$

The coefficients ε_r and ε_f represent the proteome energy efficiencies of the respiration and fermentation pathways (Appendix-fig.1C-D), respectively, with

$$\begin{cases} \varepsilon_r = \frac{\beta_1/2 + \beta_2 + \beta_3 + \beta_4}{1/(2\kappa_A) + 1/(2\kappa_1) + 1/\kappa_2 + 1/\kappa_3 + 1/\kappa_4}, \\ \varepsilon_f = \frac{\beta_1/2 + \beta_2 + \beta_6}{1/(2\kappa_A) + 1/(2\kappa_1) + 1/\kappa_2 + 1/\kappa_6}. \end{cases} \quad (\text{S31})$$

ψ^{-1} is the proteome efficiency of biomass pathway (Appendix-fig. 1E), with

$$\psi = \frac{1}{\kappa_i} + \frac{1 + \eta_{a1} + \eta_c}{2\kappa_A} + \frac{\eta_{a2} + \eta_b + 2\eta_c + \eta_d}{2\kappa_1} + \frac{\eta_b + \eta_c}{\kappa_2} + \frac{\eta_c}{\kappa_3} + \frac{\eta_c + \eta_d}{\kappa_5} + \sum_i^{a1,a2,b,c,d} \frac{\eta_i}{\kappa_i}. \quad (\text{S32})$$

ϕ is the energy demand coefficient (a constant), with

$$\phi \equiv \eta_E - \beta_1 \cdot (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) / 2 - \beta_2 \cdot (\eta_b + \eta_c) - \beta_3 \cdot \eta_c - \beta_{a1} \cdot \eta_{a1}, \quad (\text{S33})$$

and $\phi \cdot \lambda$ stands for the normalized energy demand other than the accompanying energy production from the biomass pathway.

Appendix 2.2 The reason for overflow metabolism

Microbes optimize their growth rate to survive in the evolutionary process (Vander Heiden et al., 2009). Basically, this also applies to tumor cells, which proliferate rapidly ignoring signals of growth restriction (Vander Heiden et al., 2009). To optimize cell growth, we first consider the best strategy for a single cell. The coarse-grained model is summarized in Eq. S26 and further simplified into Eq. S29. Here, ε_r , ε_f and ψ are functions of κ_A (see Eqs. S31, S32), so we also denote them as $\varepsilon_r(\kappa_A)$, $\varepsilon_f(\kappa_A)$, $\psi(\kappa_A)$. Apparently, the fluxes of both respiration and fermentation take non-negative values, i.e., $J_r^{(E)}, J_f^{(E)} \geq 0$, and all the coefficients are positive: $\varepsilon_r(\kappa_A)$, $\varepsilon_f(\kappa_A)$, $\psi(\kappa_A)$, $\phi > 0$.

Thus, if $\varepsilon_r > \varepsilon_f$ then $(\psi + \phi/\varepsilon_r) \cdot \lambda = \phi_{\max} - J_f^{(E)}(1/\varepsilon_f - 1/\varepsilon_r) \leq \phi_{\max}$. Obviously, the optimal solution is:

$$\begin{cases} J_f^{(E)} = 0, \\ J_r^{(E)} = \phi \cdot \lambda. \end{cases} \quad \varepsilon_r > \varepsilon_f. \quad (\text{S34})$$

Similarly, if $\varepsilon_f > \varepsilon_r$ then the optimal solution is:

$$\begin{cases} J_f^{(E)} = \phi \cdot \lambda, \\ J_r^{(E)} = 0. \end{cases} \quad \varepsilon_r < \varepsilon_f. \quad (\text{S35})$$

In both cases, the growth rate λ takes the maximum value for a given nutrient condition (i.e., given κ_A):

$$\lambda = \begin{cases} \lambda_r \equiv \frac{\phi_{\max}}{\phi/\varepsilon_r(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \lambda_f \equiv \frac{\phi_{\max}}{\phi/\varepsilon_f(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A). \end{cases} \quad (\text{S36})$$

So, why do microbes use the wasteful fermentation pathway when the growth rate is large under aerobic conditions? An intuitive speculation is that the fermentation pathway is more efficient in terms of the proteome energy efficiency, i.e., $\varepsilon_f > \varepsilon_r$. If so, then why do microbes still use the

normal respiration pathway when the growth rate is small? The answer lies in that both $\varepsilon_r(\kappa_A)$ and $\varepsilon_f(\kappa_A)$ are not constants, but are dependent on nutrient conditions. In Eq. S31, when κ_A is small, just consider the extreme case of $\kappa_A \rightarrow 0$, and then

$$\begin{cases} \varepsilon_r(\kappa_A \rightarrow 0) \approx [\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)] \cdot \kappa_A, \\ \varepsilon_f(\kappa_A \rightarrow 0) \approx [\beta_1 + 2(\beta_2 + \beta_6)] \cdot \kappa_A. \end{cases} \quad (\text{S37})$$

Since $\beta_3 + \beta_4 \gg \beta_6$, clearly,

$$\varepsilon_r(\kappa_A \rightarrow 0) > \varepsilon_f(\kappa_A \rightarrow 0). \quad (\text{S38})$$

Combined with Eq. S36, thus cells would certainly use the respiration pathway when the growth rate is very small. Meanwhile, suppose that $\kappa_A^{\max}$ is the maximum value of κ_A available across different Group A carbon sources, and if there exists a κ_A (with $\kappa_A \leq \kappa_A^{\max}$) satisfying $\varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A)$, specifically,

$$\frac{\beta_3 + \beta_4 - \beta_6}{2\kappa_A} < \left(\frac{\beta_1}{2} + \beta_2 + \beta_6 \right) \left(\frac{1}{2\kappa_1} + \frac{1}{\kappa_2} + \frac{1}{\kappa_3} + \frac{1}{\kappa_4} \right) - \left(\frac{\beta_1}{2} + \beta_2 + \beta_3 + \beta_4 \right) \cdot \left(\frac{1}{2\kappa_1} + \frac{1}{\kappa_2} + \frac{1}{\kappa_6} \right), \quad (\text{S39})$$

then $\Delta(\kappa_A) \equiv \varepsilon_f(\kappa_A)/\varepsilon_r(\kappa_A)$ is a monotonously increasing function of κ_A . Thus,

$$\varepsilon_r(\kappa_A^{\max}) < \varepsilon_f(\kappa_A^{\max}), \quad (\text{S40})$$

and cells would use the fermentation pathway when the growth rate is large.

In practice, experimental studies (Basan et al., 2015) in *E. coli* have reported that the proteome energy efficiency in fermentation is higher than that in respiration when the Group A carbon source is lactose at saturated concentration (Molenaar et al., 2009), i.e.,

$$\varepsilon_r(\kappa_{\text{lactose}}^{(\text{ST})}) < \varepsilon_f(\kappa_{\text{lactose}}^{(\text{ST})}). \text{ Here, } \kappa_{\text{lactose}}^{(\text{ST})} \text{ represents the substrate quality of lactose and}$$

the superscript “(ST)” signifies saturated concentration. In fact, *E. coli* grows much faster in G6p than lactose (Basan et al., 2015), thus, $\kappa_A^{\max} > \kappa_{\text{lactose}}^{(\text{ST})}$. The above example verifies that Eq.

S40 holds for *E. coli*. In fact, more recent studies (Chen and Nielsen, 2019) supported that Eq. S40 holds generically for many microbial species. From the theoretical side, we can verify Eq. S39 and thus Eq. S40 using Eq. S20 combined with the in vivo/in vitro biochemical parameters obtained from experimental data (see Appendix-tables 1–2). For example, it is easy to check that $\varepsilon_r(\kappa_{\text{glucose}}^{(\text{ST})}) < \varepsilon_f(\kappa_{\text{glucose}}^{(\text{ST})})$ with this method (see Appendix 8.2), which also confirms the validity of Eqs. S39–S40.

Now that **Eqs. S38–S40** are all valid, then there exists a critical value of κ_A (denoted as

$\kappa_A^{(C)}$) satisfying $\Delta(\kappa_A^{(C)}) = 1$. Thus,

$$\begin{cases} \varepsilon_f(\kappa_A) > \varepsilon_r(\kappa_A), & \kappa_A > \kappa_A^{(C)}; \\ \varepsilon_f(\kappa_A) = \varepsilon_r(\kappa_A), & \kappa_A = \kappa_A^{(C)}; \\ \varepsilon_f(\kappa_A) < \varepsilon_r(\kappa_A), & \kappa_A < \kappa_A^{(C)}. \end{cases} \quad (\text{S41})$$

Combined with **Eq. S31**, we have:

$$\kappa_A^{(C)} = \frac{\beta_3 + \beta_4 - \beta_6}{(\beta_1/2 + \beta_2 + \beta_6)(1/\kappa_1 + 2/\kappa_2 + 2/\kappa_3 + 2/\kappa_4) - (\beta_1/2 + \beta_2 + \beta_3 + \beta_4)(1/\kappa_1 + 2/\kappa_2 + 2/\kappa_6)}. \quad (\text{S42})$$

By substituting **Eq. S42** into **Eqs. S31, S32** and **S36**, we obtain the expressions for

$\varepsilon_r(\kappa_A^{(C)})$, $\varepsilon_f(\kappa_A^{(C)})$ and the critical growth rate at the transition point (i.e., $\lambda_C \equiv \lambda(\kappa_A^{(C)})$):

$$\begin{cases} \varepsilon_r(\kappa_A^{(C)}) = \varepsilon_f(\kappa_A^{(C)}) = \frac{\beta_3 + \beta_4 - \beta_6}{1/\kappa_3 + 1/\kappa_4 - 1/\kappa_6}, \\ \lambda_C = \frac{\phi_{\max}}{\varphi/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi(\kappa_A^{(C)})}, \end{cases} \quad (\text{S43})$$

where $\varepsilon_{r/f}$ represents either ε_r or ε_f . In **Fig. 1E**, we show the dependencies of $\varepsilon_r(\kappa_A)$, $\varepsilon_f(\kappa_A)$ and $\lambda(\kappa_A)$ on κ_A in a 3-dimensional form, as κ_A changes.

Appendix 2.3 The relation between respiration/fermentation fluxes and growth rate

We proceed to study the relation between the respiration/fermentation flux and the cell growth rate. From **Eqs. S16** and **S30**, we see that the stoichiometric fluxes J_r, J_f the normalized fluxes

$J_r^{(N)}$, $J_f^{(N)}$ and the normalized energy fluxes $J_r^{(E)}$, $J_f^{(E)}$ are all interconvertible. For

convenience, we first analyze the relations between $J_r^{(E)}$, $J_f^{(E)}$, and λ under growth rate

optimization. In fact, all these terms are merely functions of κ_A (see **Eqs. S34–S36**), which is determined by the nutrient condition (**Eq. S27**).

In the homogeneous case, i.e., all microbes share identical biochemical parameters, as $\lambda(\kappa_A)$

increases with κ_A , $J_f^{(E)}$ show up abruptly and $J_r^{(E)}$ vanish simultaneously as κ_A right exceeds

$\kappa_A^{(C)}$ (**Fig. 1E**, see also **Eqs. S34–S35, S41**). Combining **Eqs. S34–S36** and **S43**, we have:

$$\begin{cases} J_f^{(E)} = \varphi \cdot \lambda \cdot \theta(\lambda - \lambda_C), \\ J_r^{(E)} = \varphi \cdot \lambda \cdot [1 - \theta(\lambda - \lambda_C)], \end{cases} \quad (\text{S44})$$

where “ θ ” stands for the Heaviside step function. Defining $\lambda_{\max} = \lambda(\kappa_A^{\max})$, and then, $[0, \lambda_{\max}]$ is the relevant range of the x axis. In fact, the digital responses in **Eq. S44** are consistent with the numerical simulation results in Molenaar *et al.* (Molenaar *et al.*, 2009). However, these results are incompatible with the threshold-analog response in the standard picture of overflow metabolism (Basan *et al.*, 2015; Holms, 1996).

In practice, the values of k_i^{cat} can be greatly influenced by the concentrations of potassium and phosphate (García-Contreras *et al.*, 2012), which vary from cell to cell. Then, there is a distribution of the values for k_i^{cat} among the cell populations, commonly we call it extrinsic noise (Elowitz *et al.*, 2002). For convenience, we assume that each k_i^{cat} (and thus κ_i) follows a Gaussian distribution with the coefficient of variation (CV) taken as 25%. Then, the distribution of λ_c can be approximated by a Gaussian distribution (see Appendix 7.1):

$$\lambda_c \sim \mathcal{N}(\mu_{\lambda_c}, \sigma_{\lambda_c}^2), \quad (\text{S45})$$

where μ_{λ_c} and σ_{λ_c} stand for the mean and standard deviation of λ_c , with the CV

$\sigma_{\lambda_c} / \mu_{\lambda_c}$ calculated to be 12% (see Appendix 8.2 for details). Note that λ is κ_A dependent, while λ_c is independent of κ_A . Thus, given the growth rate of microbes in a culturing medium (e.g., in a chemostat), the normalized energy fluxes are:

$$\begin{cases} J_f^{(\text{E})}(\lambda) = \frac{1}{2} \varphi \cdot \lambda \cdot \left[\text{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2} \sigma_{\lambda_c}} \right) + 1 \right], \\ J_r^{(\text{E})}(\lambda) = \frac{1}{2} \varphi \cdot \lambda \cdot \left[1 - \text{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2} \sigma_{\lambda_c}} \right) \right], \end{cases} \quad (\text{S46})$$

where “erf” represents the error function. In practice, given a culturing medium, there is also a probability distribution for the growth rate (**Appendix-fig.2B**, see also **Eq. S157**). For approximation, in plotting the flux-growth rate relations, we use the deterministic (noise-free) value of the growth rate as the proxy. To compare with experiments, basically, we are comparing the normalized fluxes $J_r^{(\text{N})}$, $J_f^{(\text{N})}$ (see **Appendix 8.1** for details). Combining **Eqs. S30** and **S46**, we get:

$$\begin{cases} J_f^{(\text{N})}(\lambda) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \lambda \cdot \left[\text{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2} \sigma_{\lambda_c}} \right) + 1 \right], \\ J_r^{(\text{N})}(\lambda) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \lambda \cdot \left[1 - \text{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2} \sigma_{\lambda_c}} \right) \right]. \end{cases} \quad (\text{S47})$$

In **Fig. 1C-D**, we see that **Eq. S47** quantitatively illustrates the experimental data (Basan et al., 2015), where the model parameters were obtained using the biochemical data for the catalytic enzymes (see **Appendix-table1** for details).

Appendix 3 Model perturbations

Appendix 3.1 Overexpression of useless proteins

Here we consider the case of overexpression of the protein encoded by LacZ gene in *E. coli*. Effectively, this limits the proteome by altering $\phi_{\max}$:

$$\phi_{\max} \xrightarrow{\text{LacZ overexpression}} \phi_{\max} - \phi_Z, \quad (\text{S48})$$

where ϕ_Z stands for the fraction of useless proteins, which is controllable in experiments. Then, the growth rate changes into a bivariate function of κ_A and ϕ_Z :

$$\lambda(\kappa_A, \phi_Z) = \begin{cases} \frac{\phi_{\max} - \phi_Z}{\varphi/\varepsilon_r(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \frac{\phi_{\max} - \phi_Z}{\varphi/\varepsilon_f(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A), \end{cases} \quad (\text{S49})$$

and thus,

$$\lambda(\kappa_A, \phi_Z) = \lambda(\kappa_A, 0)(1 - \phi_Z/\phi_{\max}). \quad (\text{S50})$$

Obviously, $\kappa_A^{(C)}$ is still a constant (following **Eq. S42**), while $\lambda_c(\phi_Z) \equiv \lambda(\kappa_A^{(C)}, \phi_Z)$ and

$\lambda_{\max}(\phi_Z) \equiv \lambda(\kappa_A^{\max}, \phi_Z)$ become functions of ϕ_Z :

$$\begin{cases} \lambda_c(\phi_Z) = \lambda_c(0)(1 - \phi_Z/\phi_{\max}), \\ \lambda_{\max}(\phi_Z) = \lambda_{\max}(0)(1 - \phi_Z/\phi_{\max}). \end{cases} \quad (\text{S51})$$

In the homogeneous case, $J_f^{(E)}$ and $J_r^{(E)}$ follow:

$$\begin{cases} J_f^{(E)}(\kappa_A, \phi_Z) = \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot \theta(\lambda(\kappa_A, \phi_Z) - \lambda_c(\phi_Z)), \\ J_r^{(E)}(\kappa_A, \phi_Z) = \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot [1 - \theta(\lambda(\kappa_A, \phi_Z) - \lambda_c(\phi_Z))]. \end{cases} \quad (\text{S52})$$

Combined with **Eqs. S50–S51**, we have:

$$\begin{cases} J_f^{(E)}(\kappa_A, \phi_Z) = \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot \theta(\lambda(\kappa_A, 0) - \lambda_c(0)), \\ J_r^{(E)}(\kappa_A, \phi_Z) = \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot [1 - \theta(\lambda(\kappa_A, 0) - \lambda_c(0))]. \end{cases} \quad (\text{S53})$$

To compare with experiments, we assume that each k_i^{cat} and κ_i follow the extrinsic noise with a CV of 25% specified in **Appendix 2.3**, and neglect the noise on φ_Z and $\varphi_{\max}$. Combining **Eqs. S45** and **S51**, we see that $\lambda_C(\varphi_Z)$ approximately follows a Gaussian distribution:

$$\lambda_C(\phi_Z) \sim \mathcal{N}(\mu_{\lambda_C}(\phi_Z), \sigma_{\lambda_C}(\phi_Z)^2), \quad (\text{S54})$$

where $\mu_{\lambda_C}(\phi_Z)$ and $\sigma_{\lambda_C}(\phi_Z)$ represent the mean and standard deviation of $\lambda_C(\varphi_Z)$, with

$$\begin{cases} \mu_{\lambda_C}(\phi_Z) = \mu_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max}), \\ \sigma_{\lambda_C}(\phi_Z) = \sigma_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max}). \end{cases} \quad (\text{S55})$$

Here, $\mu_{\lambda_C}(0)$, $\sigma_{\lambda_C}(0)$, $\lambda_C(0)$, $\lambda_{\max}(0)$ and $\lambda(\kappa_A, 0)$ represent the parameters or variables free from LacZ perturbation, just as those in **Appendix 2.3**. Since the noise on the multiplier term (i.e., $1 - \phi_Z/\phi_{\max}$) is negligible, the CV of $\lambda_C(\varphi_Z)$ (i.e., $\sigma_{\lambda_C}(\phi_Z)/\mu_{\lambda_C}(\phi_Z)$) is unaffected by φ_Z . By combining **Eqs. S46** and **S48**, we obtain the relations between the normalized energy fluxes and growth rate:

$$\begin{cases} J_f^{(E)}(\lambda(\kappa_A, \phi_Z), \phi_Z) = \frac{1}{2} \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot \left[\text{erf}\left(\frac{\lambda(\kappa_A, \phi_Z) - \mu_{\lambda_C}(\phi_Z)}{\sqrt{2}\sigma_{\lambda_C}(\phi_Z)}\right) + 1 \right], \\ J_r^{(E)}(\lambda(\kappa_A, \phi_Z), \phi_Z) = \frac{1}{2} \varphi \cdot \lambda(\kappa_A, \phi_Z) \cdot \left[1 - \text{erf}\left(\frac{\lambda(\kappa_A, \phi_Z) - \mu_{\lambda_C}(\phi_Z)}{\sqrt{2}\sigma_{\lambda_C}(\phi_Z)}\right) \right], \end{cases} \quad (\text{S56})$$

where $\lambda(\kappa_A, \varphi_Z)$, $\mu_{\lambda_C}(\phi_Z)$ and $\sigma_{\lambda_C}(\phi_Z)$ follow **Eqs. S50** and **S55** accordingly. For a given value of φ_Z , i.e., φ_Z is fixed, then, $\lambda(\kappa_A, \varphi_Z)$ changes monotonously with κ_A . Combining **Eqs. S55**–**S56** and **S30**, we have the relation between the normalized fluxes $J_r^{(N)}$, $J_f^{(N)}$ and growth rate (here φ_Z is a parameter):

$$\begin{cases} J_f^{(N)}(\lambda, \phi_Z) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \lambda \cdot \left[\text{erf}\left(\frac{\lambda - \mu_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max})}{\sqrt{2}\sigma_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max})}\right) + 1 \right], \\ J_r^{(N)}(\lambda, \phi_Z) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \lambda \cdot \left[1 - \text{erf}\left(\frac{\lambda - \mu_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max})}{\sqrt{2}\sigma_{\lambda_C}(0)(1 - \phi_Z/\phi_{\max})}\right) \right]. \end{cases} \quad (\text{S57})$$

In **Fig. 2C**, we show that the model predictions (**Eq. S57**) quantitatively agree with the experiments (Basan et al., 2015).

Meanwhile, we can also perturb the growth rate by tuning ϕ_Z in a stable culturing environment with fixed concentration of a Group A carbon source (i.e., given $[A]$). In fact, for this case there is a distribution of κ_A values due to the extrinsic noise in k_A^{cat} , yet this distribution is fixed. For convenience of description, we still referred to it as fixed κ_A . Then, combining Eqs. S30, S50, S55 and S56, we get:

$$\begin{cases} J_f^{(N)}(\lambda, \phi_Z) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\text{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] \cdot \lambda, \\ J_r^{(N)}(\lambda, \phi_Z) = \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \text{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \cdot \lambda. \end{cases} \quad (\text{S58})$$

Here, $\lambda(\kappa_A, 0)$ remains unaltered as κ_A is fixed. Therefore, in this case, $J_f^{(N)}$ and $J_r^{(N)}$ are proportional to λ , where the slopes are both functions of κ_A . More specifically, the slope of $J_f^{(N)}$ is a monotonically increasing function of κ_A , while that of $J_r^{(N)}$ is a monotonically decreasing function of κ_A . In Fig. 2B, we see that the model predictions (Eq. S58) agree quantitatively with the experiments (Basan et al., 2015).

In fact, the growth rate can be altered by tuning ϕ_Z and κ_A simultaneously. Then, the relations among the energy fluxes, growth rate and ϕ_Z still follow Eq. S57 (here ϕ_Z is a variable). In a 3-D representation, these relations correspond to a surface. In Fig. 2A, we show that the model predictions (Eq. S57) match well with the experimental data (Basan et al., 2015).

Appendix 3.2 Energy dissipation

In practice, energy dissipation breaks the proportional relationship between energy demand and biomass production. Thus, Eq. S25 changes to:

$$J_E = r_E \cdot J_{\text{BM}} + w \cdot \frac{M_{\text{carbon}}}{m_0}, \quad (\text{S59})$$

where W represents the dissipation coefficient. In fact, maintenance energy contributes to energy dissipation, and we define the maintenance energy coefficient as w_0 . In bacteria, the impact of w_0 is often negligible, particularly for all the analysis in the sections above. While in tumor cell, w_0 plays a much more significant role.

The introduction of energy dissipation leads to a modification to Eq. S26: combining Eq. S59 and Eq. S16, we have:

$$J_E^{(N)} = \eta_E \cdot \lambda + w. \quad (\text{S60})$$

Then, Eq. S29 [changes to](#):

$$\begin{cases} J_r^{(E)} + J_f^{(E)} = \varphi \cdot \lambda + w, \\ \frac{J_r^{(E)}}{\varepsilon_r} + \frac{J_f^{(E)}}{\varepsilon_f} = \phi_{\max} - \psi \cdot \lambda. \end{cases} \quad (\text{S61})$$

Consequently, if $\varepsilon_r > \varepsilon_f$ the best strategy for the cell is:

$$\begin{cases} J_f^{(E)} = 0, \\ J_r^{(E)} = \varphi \cdot \lambda + w, \end{cases} \quad \varepsilon_r > \varepsilon_f, \quad (\text{S62})$$

and if $\varepsilon_f > \varepsilon_r$, the best strategy is:

$$\begin{cases} J_f^{(E)} = \varphi \cdot \lambda + w, \\ J_r^{(E)} = 0. \end{cases} \quad \varepsilon_r < \varepsilon_f. \quad (\text{S63})$$

Then, the growth rate turns into a bivariate function of both κ_A and w :

$$\lambda(\kappa_A, w) = \begin{cases} \frac{\phi_{\max} - w/\varepsilon_r(\kappa_A)}{\varphi/\varepsilon_r(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \frac{\phi_{\max} - w/\varepsilon_f(\kappa_A)}{\varphi/\varepsilon_f(\kappa_A) + \psi(\kappa_A)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A). \end{cases} \quad (\text{S64})$$

Clearly, $\kappa_A^{(C)}$ is still a constant, while $\lambda_C(w) \equiv \lambda(\kappa_A^{(C)}, w)$ and

$\lambda_{\max}(w) \equiv \lambda(\kappa_A^{\max}, w)$ changes into functions of w :

$$\begin{cases} \lambda_C(w) = \lambda_C(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}, \\ \lambda_{\max}(w) = \lambda_{\max}(0) \left\{ 1 - w / \left[\varepsilon_f(\kappa_A^{\max}) \phi_{\max} \right] \right\}. \end{cases} \quad (\text{S65})$$

For a cell population, in the homogeneous case, $J_f^{(E)}$ and $J_r^{(E)}$ follow:

$$\begin{cases} J_f^{(E)}(\kappa_A, w) = [\varphi \cdot \lambda(\kappa_A, w) + w] \cdot \theta(\lambda(\kappa_A, w) - \lambda_C(w)), \\ J_r^{(E)}(\kappa_A, w) = [\varphi \cdot \lambda(\kappa_A, w) + w] \cdot [1 - \theta(\lambda(\kappa_A, w) - \lambda_C(w))]. \end{cases} \quad (\text{S66})$$

To compare with experiments, we assume the same extent of extrinsic noise in k_i^{cat} (and thus κ_i) as that specified in Appendix 2.3. Combining Eqs. S45 and S65, then, $\lambda_C(w)$ approximately follows a Gaussian distribution:

$$\lambda_C(w) \sim \mathcal{N}(\mu_{\lambda_C}(w), \sigma_{\lambda_C}(w)^2), \quad (\text{S67})$$

where $\mu_{\lambda_C}(w)$ and $\sigma_{\lambda_C}(w)$ represent the mean and standard deviation of $\lambda_C(w)$, and

$$\begin{cases} \mu_{\lambda_C}(w) = \mu_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}, \\ \sigma_{\lambda_C}(w) \approx \sigma_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}. \end{cases} \quad (\text{S68})$$

Here, $\mu_{\lambda_C}(0)$, $\sigma_{\lambda_C}(0)$, $\lambda_C(0)$, $\lambda_{\max}(0)$, and $\lambda(\kappa_A, 0)$ represent the parameters or variables free from energy dissipation. In fact, there is a distribution of values for $\varepsilon_{r/f}(\kappa_A^{(C)})$. For approximation, we use the deterministic value of $\varepsilon_{r/f}(\kappa_A^{(C)})$ in Eq. S68, and then the CV of $\lambda_C(w)$ is roughly unperturbed by w . Combining Eqs. S46, S66 and S67, we have:

$$\begin{cases} J_f^{(E)}(\lambda(\kappa_A, w), w) = \frac{1}{2}(\varphi \cdot \lambda(\kappa_A, w) + w) \cdot \left[\text{erf} \left(\frac{\lambda(\kappa_A, w) - \mu_{\lambda_C}(w)}{\sqrt{2}\sigma_{\lambda_C}(w)} \right) + 1 \right], \\ J_r^{(E)}(\lambda(\kappa_A, w), w) = \frac{1}{2}(\varphi \cdot \lambda(\kappa_A, w) + w) \cdot \left[1 - \text{erf} \left(\frac{\lambda(\kappa_A, w) - \mu_{\lambda_C}(w)}{\sqrt{2}\sigma_{\lambda_C}(w)} \right) \right]. \end{cases} \quad (\text{S69})$$

Since the dissipation coefficient w is tunable in experiments, for a given value of w , $\lambda(\kappa_A, w)$ changes monotonously with κ_A . Combining Eqs. S68–S69 and S30, we have (here w is a parameter):

$$\begin{cases} J_f^{(N)}(\lambda, w) = \frac{\varphi \cdot \lambda + w}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\text{erf} \left(\frac{\lambda - \mu_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}}{\sqrt{2}\sigma_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}} \right) + 1 \right], \\ J_r^{(N)}(\lambda, w) = \frac{\varphi \cdot \lambda + w}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \text{erf} \left(\frac{\lambda - \mu_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}}{\sqrt{2}\sigma_{\lambda_C}(0) \left\{ 1 - w / \left[\varepsilon_{r/f}(\kappa_A^{(C)}) \phi_{\max} \right] \right\}} \right) \right]. \end{cases} \quad (\text{S70})$$

The comparison between model predictions (Eq. S70) and experimental results (Basan et al., 2015) is shown in Fig. 3B, which agrees quantitatively. Meanwhile, the growth rate can also be perturbed by changing κ_A and w simultaneously. Then, the relations among the energy fluxes,

growth rate and follow [Eq. S70](#) (here w is a variable). In a 3D representation, these relations correspond to a surface. As shown in [Fig. 3A](#), the model predictions ([Eq. S70](#)) agree quantitatively with the experimental results ([Basan et al., 2015](#)).

Appendix 3.3 Translation inhibition

In *E. coli*, the translation rate can be modified by adding different concentrations of translation inhibitors, e.g., chloramphenicol (Cm). The net effect of this perturbation is represented as:

$$\kappa_t \xrightarrow{\text{Translation inhibition}} \kappa_t / (1 + \iota), \quad (\text{S71})$$

where ι stands for the inhibition coefficient with $\iota > 0$, and $(1 + \iota)^{-1}$ represents the translation efficiency. Thus, [Eq. S32](#) changes to:

$$\psi(\kappa_A, \iota) = \frac{\iota + 1}{\kappa_t} + \frac{1 + \eta_{a1} + \eta_c}{2\kappa_A} + \frac{\eta_{a2} + \eta_b + 2\eta_c + \eta_d}{2\kappa_1} + \frac{\eta_b + \eta_c}{\kappa_2} + \frac{\eta_c}{\kappa_3} + \frac{\eta_c + \eta_d}{\kappa_5} + \sum_i^{a1, a2, b, c, d} \frac{\eta_i}{\kappa_i}. \quad (\text{S72})$$

First, we consider the case of neglecting the maintenance energy, i.e., $w_0 = 0$. Then, the growth rate takes the following form:

$$\lambda(\kappa_A, \iota) = \begin{cases} \frac{\phi_{\max}}{\varphi / \varepsilon_r(\kappa_A) + \psi(\kappa_A, \iota)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \frac{\phi_{\max}}{\varphi / \varepsilon_f(\kappa_A) + \psi(\kappa_A, \iota)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A), \end{cases} \quad (\text{S73})$$

where $\lambda(\kappa_A, 0)$ and $\psi(\kappa_A, 0)$ represent the terms free from translation inhibition. Thus,

$$\lambda_c(\iota) \equiv \lambda(\kappa_A^{(c)}, \iota) \quad \text{and} \quad \lambda_{\max}(\iota) \equiv \lambda(\kappa_A^{\max}, \iota) \quad \text{change into functions of } \iota:$$

$$\begin{cases} \lambda_c(\iota) = \lambda_c(0) \frac{\varphi / \varepsilon_{r/f}(\kappa_A^{(c)}) + \psi(\kappa_A^{(c)}, 0)}{\varphi / \varepsilon_{r/f}(\kappa_A^{(c)}) + \psi(\kappa_A^{(c)}, \iota)}, \\ \lambda_{\max}(\iota) = \lambda_{\max}(0) \frac{\varphi / \varepsilon_f(\kappa_A^{\max}) + \psi(\kappa_A^{\max}, 0)}{\varphi / \varepsilon_f(\kappa_A^{\max}) + \psi(\kappa_A^{\max}, \iota)}. \end{cases} \quad (\text{S74})$$

In the homogeneous case, $J_f^{(E)}$ and $J_r^{(E)}$ follow:

$$\begin{cases} J_f^{(E)}(\kappa_A, \iota) = \varphi \cdot \lambda(\kappa_A, \iota) \cdot \theta(\lambda(\kappa_A, \iota) - \lambda_c(\iota)), \\ J_r^{(E)}(\kappa_A, \iota) = \varphi \cdot \lambda(\kappa_A, \iota) \cdot [1 - \theta(\lambda(\kappa_A, \iota) - \lambda_c(\iota))]. \end{cases} \quad (\text{S75})$$

To compare with experiments, we assume that there exists extrinsic noise in k_i^{cat} and κ_i as specified in Appendix 2.3. Combining Eqs. S45 and S74, $\lambda_C(\iota)$ can be approximated by a Gaussian distribution:

$$\lambda_C(\iota) \sim \mathcal{N}(\mu_{\lambda_C}(\iota), \sigma_{\lambda_C}(\iota)^2), \quad (\text{S76})$$

where $\mu_{\lambda_C}(\iota)$ and $\sigma_{\lambda_C}(\iota)$ represent the mean and standard deviation of $\lambda_C(\iota)$, with

$$\begin{cases} \mu_{\lambda_C}(\iota) = \mu_{\lambda_C}(0) \frac{\varphi/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi(\kappa_A^{(C)}, 0)}{\varphi/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi(\kappa_A^{(C)}, \iota)}, \\ \sigma_{\lambda_C}(\iota) \approx \sigma_{\lambda_C}(0) \frac{\varphi/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi(\kappa_A^{(C)}, 0)}{\varphi/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi(\kappa_A^{(C)}, \iota)}. \end{cases} \quad (\text{S77})$$

Here, $\mu_{\lambda_C}(0)$, $\sigma_{\lambda_C}(0)$, $\psi(\kappa_A^{(C)}, 0)$, $\lambda_C(0)$ and $\lambda_{\max}(0)$ stand for the terms free from translation inhibition. Basically, there are distributions of values for $\varepsilon_{r/f}(\kappa_A^{(C)})$, $\psi(\kappa_A^{(C)}, 0)$ and $\psi(\kappa_A^{(C)}, \iota)$. For approximation, we use the deterministic values of these terms in Eq. S77, and then the CV of $\lambda_C(\iota)$ can be approximated by $\lambda_C(0)$. Combining Eqs. S46, S75 and S76, we have:

$$\begin{cases} J_f^{(E)}(\lambda(\kappa_A, \iota), \iota) = \frac{1}{2} \varphi \cdot \lambda(\kappa_A, \iota) \cdot \left[\text{erf} \left(\frac{\lambda(\kappa_A, \iota) - \mu_{\lambda_C}(\iota)}{\sqrt{2} \sigma_{\lambda_C}(\iota)} \right) + 1 \right], \\ J_r^{(E)}(\lambda(\kappa_A, \iota), \iota) = \frac{1}{2} \varphi \cdot \lambda(\kappa_A, \iota) \cdot \left[1 - \text{erf} \left(\frac{\lambda(\kappa_A, \iota) - \mu_{\lambda_C}(\iota)}{\sqrt{2} \sigma_{\lambda_C}(\iota)} \right) \right]. \end{cases} \quad (\text{S78})$$

In the experiments, the inhibition coefficient ι is controllable by tuning the concentration of translation inhibitor. For a given value of ι , $\lambda(\kappa_A, \iota)$ changes monotonously with κ_A . Combining Eqs. S30 and S78, we have (ι is a parameter here):

$$\begin{cases} J_f^{(N)}(\lambda, \iota) = \frac{\varphi \cdot \lambda}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\text{erf} \left(\frac{\lambda - \mu_{\lambda_C}(\iota)}{\sqrt{2} \sigma_{\lambda_C}(\iota)} \right) + 1 \right], \\ J_r^{(N)}(\lambda, \iota) = \frac{\varphi \cdot \lambda}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \text{erf} \left(\frac{\lambda - \mu_{\lambda_C}(\iota)}{\sqrt{2} \sigma_{\lambda_C}(\iota)} \right) \right], \end{cases} \quad (\text{S79})$$

where $\mu_{\lambda_c}(\iota)$ and $\sigma_{\lambda_c}(\iota)$ follow Eq. S77. The growth rate can also be perturbed by altering κ_A and ι simultaneously. Then, the relations among the energy fluxes, growth rate and ι still follow Eq. S79 (here ι is a variable). The comparison between Eq. S79 and experimental data (Basan et al., 2015) is shown in **Appendix-fig. 2D** (3-D) and **2E** (2-D). Overall, it shows good consistency, however, there is still a noticeable discrepancy when ι is large (i.e., with high concentration of translation inhibitor). Then, it led us to consider the maintenance energy w_0 , which is small but may account for this discrepancy. Then, $\lambda(\kappa_A, \iota)$ changes into:

$$\lambda(\kappa_A, \iota) = \begin{cases} \frac{\phi_{\max} - w_0/\varepsilon_r(\kappa_A)}{\varphi/\varepsilon_r(\kappa_A) + \psi(\kappa_A, \iota)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \frac{\phi_{\max} - w_0/\varepsilon_f(\kappa_A)}{\varphi/\varepsilon_f(\kappa_A) + \psi(\kappa_A, \iota)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A), \end{cases} \quad (\text{S80})$$

while $\lambda_c(\iota) \equiv \lambda(\kappa_A^{(c)}, \iota)$ and $\lambda_{\max}(\iota) \equiv \lambda(\kappa_A^{\max}, \iota)$ still follow Eq. S74, although the forms of $\lambda_c(0)$ and $\lambda_{\max}(0)$ change into:

$$\begin{cases} \lambda_c(0) = \frac{\phi_{\max} - w_0/\varepsilon_{r/f}(\kappa_A^{(c)})}{\varphi/\varepsilon_{r/f}(\kappa_A^{(c)}) + \psi(\kappa_A^{(c)}, 0)}, \\ \lambda_{\max}(0) = \frac{\phi_{\max} - w_0/\varepsilon_f(\kappa_A^{\max})}{\varphi/\varepsilon_f(\kappa_A^{\max}) + \psi(\kappa_A^{\max}, 0)}. \end{cases} \quad (\text{S81})$$

In the homogeneous case, $J_f^{(E)}$ and $J_r^{(E)}$ follow:

$$\begin{cases} J_f^{(E)}(\kappa_A, \iota) = [\varphi \cdot \lambda(\kappa_A, \iota) + w_0] \cdot \theta(\lambda(\kappa_A, \iota) - \lambda_c(\iota)), \\ J_r^{(E)}(\kappa_A, \iota) = [\varphi \cdot \lambda(\kappa_A, \iota) + w_0] \cdot [1 - \theta(\lambda(\kappa_A, \iota) - \lambda_c(\iota))]. \end{cases} \quad (\text{S82})$$

To compare with experiments, we assume that the extrinsic noise follows that specified in **Appendix 2.3**. Combining Eqs. S45, S74 and S81, then, $\lambda_c(\iota)$ approximately follows a Gaussian distribution:

$$\lambda_c(\iota) \sim \mathcal{N}(\mu_{\lambda_c}(\iota), \sigma_{\lambda_c}(\iota)^2). \quad (\text{S83})$$

Here $\mu_{\lambda_c}(\iota)$ and $\sigma_{\lambda_c}(\iota)$ still follow [Eq. S77](#), while $\mu_{\lambda_c}(0)$ and $\sigma_{\lambda_c}(0)$ change accordingly with [Eq. S81](#). For approximation, we use the deterministic values of the relevant terms in [Eq. S77](#), and then the CV of $\lambda_c(\iota)$ is roughly the same as $\lambda_c(0)$. Combining [Eqs. S46](#), [S82](#) and [S83](#), we have:

$$\begin{cases} J_f^{(E)}(\lambda(\kappa_A, \iota), \iota) = \frac{1}{2}(\varphi \cdot \lambda(\kappa_A, \iota) + w_0) \cdot \left[\operatorname{erf}\left(\frac{\lambda(\kappa_A, \iota) - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)}\right) + 1 \right], \\ J_r^{(E)}(\lambda(\kappa_A, \iota), \iota) = \frac{1}{2}(\varphi \cdot \lambda(\kappa_A, \iota) + w_0) \cdot \left[1 - \operatorname{erf}\left(\frac{\lambda(\kappa_A, \iota) - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)}\right) \right]. \end{cases} \quad (\text{S84})$$

Thus, for a given ι , $\lambda(\kappa_A, \iota)$ changes monotonously with κ_A . Combining [Eqs. S30](#) and [S84](#), we have (here ι is a parameter):

$$\begin{cases} J_f^{(N)}(\lambda, \iota) = \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)}\right) + 1 \right], \\ J_r^{(N)}(\lambda, \iota) = \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}(\iota)}{\sqrt{2}\sigma_{\lambda_c}(\iota)}\right) \right]. \end{cases} \quad (\text{S85})$$

The growth rate and fluxes can also be perturbed by altering κ_A and ι simultaneously. The relations among the energy fluxes, growth rate and ι would still follow [Eq. S85](#) other than that now ι is regarded as a variable. Assuming that there is a tiny amount of maintenance energy. Basically, we assign $w_0 = 2.5(h^{-1})$. Then, we see that the experimental results ([Basan et al., 2015](#)) agree quantitatively well with the model predictions ([Fig.3C-D](#)).

Appendix 4 Overflow metabolism in substrates other than Group A carbon sources

Due to the topology of metabolic network, Group A carbon sources follow the equation ([Eq. S47](#)) of overflow metabolism upon κ_A perturbation (i.e., varying the type or concentration of a Group A carbon source). This has been demonstrated clearly in the above analysis, which agrees quantitatively with experiments. However, further analysis is required for substrates other than Group A sources due to the topological differences in carbon utilization ([Wang et al., 2019](#)). Basically, substrates entering from glycolysis or the points before acetyl-CoA are potentially involved in overflow metabolism, while those join from the TCA cycle are not relevant to this behavior. Still, mixed carbon sources are likely to induce a different profile of overflow metabolism, so long as there is a carbon source coming from glycolysis.

Appendix 4.1 Pyruvate

The coarse-grained model for pyruvate utilization is shown in **Fig. 3E**. Here, nodes M_1, M_2, M_3, M_4, M_5 follow everything depicted in **Appendix 2.1**. Each biochemical reaction follows **Eq. S5** with $b_i = 1$ except that $2M_2 \rightarrow M_1$ and $M_3 + M_5 \rightarrow M_4$. By applying flux balance to the stoichiometric fluxes, combining with **Eq. S8**, we have:

$$\begin{cases} \Phi_{\text{py}} \cdot \xi_{\text{py}} = \Phi_7 \cdot \xi_7 + \Phi_8 \cdot \xi_8, \\ \Phi_7 \cdot \xi_7 = 2\Phi_9 \cdot \xi_9 + \Phi_5 \cdot \xi_5 + \Phi_{a2} \cdot \xi_{a2}, \\ \Phi_9 \cdot \xi_9 = \Phi_{a1} \cdot \xi_{a1}, \\ \Phi_8 \cdot \xi_8 = \Phi_3 \cdot \xi_3 + \Phi_6 \cdot \xi_6 + \Phi_b \cdot \xi_b, \\ \Phi_5 \cdot \xi_5 + \Phi_4 \cdot \xi_4 = \Phi_3 \cdot \xi_3 + \Phi_d \cdot \xi_d, \\ \Phi_3 \cdot \xi_3 = \Phi_4 \cdot \xi_4 + \Phi_c \cdot \xi_c. \end{cases} \quad (\text{S86})$$

For energy production, we convert all the energy currencies into ATPs, and then,

$$\beta_8 \cdot \Phi_8 \cdot \xi_8 + \beta_3 \cdot \Phi_3 \cdot \xi_3 + \beta_4 \cdot \Phi_4 \cdot \xi_4 + \beta_6 \cdot \Phi_6 \cdot \xi_6 + \beta_{a1} \cdot \Phi_{a1} \cdot \xi_{a1} - \beta_7 \cdot \Phi_7 \cdot \xi_7 - \beta_9 \cdot \Phi_9 \cdot \xi_9 = J_E, \quad (\text{S87})$$

where $\beta_7 = 1, \beta_8 = 2, \beta_3 = 2, \beta_4 = 6, \beta_6 = 1, \beta_9 = 6, \beta_{a1} = 4$ (Neidhardt et al., 1990; Sauer et al., 2004), and J_E follows **Eq. S25**. By applying the substitutions specified in **Eqs. S9, S12, S14–S18**, combined with **Eqs. S4, S10, S22, S23, S25, S86–S87**, and the constraint of proteome resource allocation, we have:

$$\begin{cases} \phi_{\text{py}} \cdot \kappa_{\text{py}} = \phi_7 \cdot \kappa_7 + \phi_8 \cdot \kappa_8, \\ \phi_7 \cdot \kappa_7 = 2\phi_9 \cdot \kappa_9 + \phi_5 \cdot \kappa_5 + \phi_{a2} \cdot \kappa_{a2}, \\ \phi_9 \cdot \kappa_9 = \phi_{a1} \cdot \kappa_{a1} \\ \phi_8 \cdot \kappa_8 = \phi_3 \cdot \kappa_3 + \phi_6 \cdot \kappa_6 + \phi_b \cdot \kappa_b \\ \phi_3 \cdot \kappa_3 = \phi_4 \cdot \kappa_4 + \phi_c \cdot \kappa_c \\ \phi_5 \cdot \kappa_5 + \phi_4 \cdot \kappa_4 = \phi_3 \cdot \kappa_3 + \phi_d \cdot \kappa_d \\ \phi_{a1} \cdot \kappa_{a1} = \eta_{a1} \cdot \lambda, \phi_{a2} \cdot \kappa_{a2} = \eta_{a2} \cdot \lambda, \phi_b \cdot \kappa_b = \eta_b \cdot \lambda, \phi_c \cdot \kappa_c = \eta_c \cdot \lambda, \phi_d \cdot \kappa_d = \eta_d \cdot \lambda, \\ \beta_8 \cdot \phi_8 \cdot \kappa_8 + \beta_3 \cdot \phi_3 \cdot \kappa_3 + \beta_4 \cdot \phi_4 \cdot \kappa_4 + \beta_6 \cdot \phi_6 \cdot \kappa_6 + \beta_{a1} \cdot \phi_{a1} \cdot \kappa_{a1} - \beta_7 \cdot \phi_7 \cdot \kappa_7 - \beta_9 \cdot \phi_9 \cdot \kappa_9 = J_E^{(N)}, \\ J_E^{(N)} = \eta_E \cdot \lambda, \lambda = \phi_R \cdot \kappa_r, J_r^{(N)} = \phi_4 \cdot \kappa_4, J_f^{(N)} = \phi_6 \cdot \kappa_6, \\ \phi_R + \phi_{\text{py}} + \phi_3 + \phi_4 + \phi_5 + \phi_6 + \phi_7 + \phi_8 + \phi_9 + \phi_{a1} + \phi_{a2} + \phi_b + \phi_c + \phi_d = \phi_{\text{max}}, \end{cases} \quad (\text{S88})$$

where $\eta_E = r_E \cdot \left[\sum_i r_i / N_{EP_i}^{\text{carbon}} \right]^{-1}$. κ_i is approximately a constant which follows Eq. S20 for each of the intermediate node. The substrate quality of κ_{py} varies with the external concentration of pyruvate ([py]),

$$\kappa_{py} \equiv \frac{r_{\text{protein}}}{r_{\text{carbon}}} \cdot \frac{k_{py}^{\text{cat}}}{m_{E_{py}}} \cdot \frac{[py]}{[py] + K_{py}} \cdot m_0. \quad (\text{S89})$$

From Eq. S88, all ϕ_i can be expressed by $J_r^{(N)}$, $J_f^{(N)}$, and λ :

$$\left\{ \begin{array}{l} \phi_{py} = \left[(2\eta_{a1} + \eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda + J_r^{(N)} + J_f^{(N)} \right] / \kappa_{py}, \\ \phi_7 = (2\eta_{a1} + \eta_{a2} + \eta_c + \eta_d) \lambda / \kappa_7, \phi_9 = \eta_{a1} \cdot \lambda / \kappa_9 \\ \phi_8 = \left[J_r^{(N)} + J_f^{(N)} + (\eta_b + \eta_c) \lambda \right] / \kappa_8 \\ \phi_3 = (J_r^{(N)} + \eta_c \cdot \lambda) / \kappa_3, \phi_4 = J_r^{(N)} / \kappa_4, \\ \phi_5 = (\eta_c + \eta_d) \lambda / \kappa_5, \phi_6 = J_f^{(N)} / \kappa_6, \\ \phi_i = \eta_i \cdot \lambda / \kappa_i \quad (i = a1, a2, b, c, d). \end{array} \right. \quad (\text{S90})$$

By substituting Eq. S90 into Eq. S88, we have:

$$\left\{ \begin{array}{l} J_r^{(E,py)} + J_f^{(E,py)} = \phi_{py} \cdot \lambda, \\ \frac{J_r^{(E,py)}}{\mathcal{E}_r^{(py)}} + \frac{J_f^{(E,py)}}{\mathcal{E}_f^{(py)}} = \phi_{\text{max}} - \psi_{py} \cdot \lambda. \end{array} \right. \quad (\text{S91})$$

Here, $J_r^{(E,py)}$ and $J_f^{(E,py)}$ stand for the normalized energy fluxes of respiration and fermentation, respectively, with

$$\left\{ \begin{array}{l} J_r^{(E,py)} = (\beta_3 + \beta_4 + \beta_8) \cdot J_r^{(N)}, \\ J_f^{(E,py)} = (\beta_6 + \beta_8) \cdot J_f^{(N)}. \end{array} \right. \quad (\text{S92})$$

The coefficients $\mathcal{E}_r^{(py)}$ and $\mathcal{E}_f^{(py)}$ represent the proteome energy efficiencies of respiration and fermentation, respectively (Appendix-fig. 1C-D), with

$$\left\{ \begin{array}{l} \mathcal{E}_r^{(py)} = \frac{\beta_3 + \beta_4 + \beta_8}{1/\kappa_{py} + 1/\kappa_8 + 1/\kappa_3 + 1/\kappa_4}, \\ \mathcal{E}_f^{(py)} = \frac{\beta_6 + \beta_8}{1/\kappa_{py} + 1/\kappa_8 + 1/\kappa_6}. \end{array} \right. \quad (\text{S93})$$

ψ_{py}^{-1} is the proteome efficiency of biomass pathway (**Appendix-fig. 1E**), with

$$\psi_{py} = \frac{1}{\kappa_t} + \frac{1 + \eta_{a1} + \eta_c}{\kappa_{py}} + \frac{1 - \eta_b + \eta_{a1}}{\kappa_7} + \frac{\eta_b + \eta_c}{\kappa_8} + \frac{\eta_{a1}}{\kappa_9} + \frac{\eta_c}{\kappa_3} + \frac{\eta_c + \eta_d}{\kappa_5} + \sum_i^{a1,a2,b,c,d} \frac{\eta_i}{\kappa_i}. \quad (S94)$$

ϕ_{py} is the energy demand coefficient (a constant), with

$$\phi_{py} \equiv \eta_E + \beta_7 \cdot (1 - \eta_b + \eta_{a1}) + \beta_9 \cdot \eta_{a1} - \beta_8 \cdot (\eta_c + \eta_b) - \beta_3 \cdot \eta_c - \beta_{a1} \cdot \eta_{a1}, \quad (S95)$$

Evidently, **Eq. S91** is identical in form with **Eq. S29**. The growth rate changes into κ_{py} dependent:

$$\lambda(\kappa_{py}) = \begin{cases} \frac{\phi_{\max}}{\phi_{py}/\varepsilon_r^{(py)}(\kappa_{py}) + \psi_{py}(\kappa_{py})} & \varepsilon_r^{(py)}(\kappa_{py}) > \varepsilon_f^{(py)}(\kappa_{py}), \\ \frac{\phi_{\max}}{\phi_{py}/\varepsilon_f^{(py)}(\kappa_{py}) + \psi_{py}(\kappa_{py})} & \varepsilon_r^{(py)}(\kappa_{py}) < \varepsilon_f^{(py)}(\kappa_{py}). \end{cases} \quad (S96)$$

When κ_{py} is very small, combined with **Eq. S93**, then,

$$\begin{cases} \varepsilon_r^{(py)}(\kappa_{py} \rightarrow 0) \approx (\beta_3 + \beta_4 + \beta_8) \cdot \kappa_{py}, \\ \varepsilon_f^{(py)}(\kappa_{py} \rightarrow 0) \approx (\beta_6 + \beta_8) \cdot \kappa_{py}. \end{cases} \quad (S97)$$

Obviously,

$$\varepsilon_r^{(py)}(\kappa_{py} \rightarrow 0) > \varepsilon_f^{(py)}(\kappa_{py} \rightarrow 0). \quad (S98)$$

As long as

$$\frac{\beta_3 + \beta_4 - \beta_6}{\kappa_{py}^{(ST)}} < (\beta_6 + \beta_8) \left(\frac{1}{\kappa_8} + \frac{1}{\kappa_3} + \frac{1}{\kappa_4} \right) - (\beta_3 + \beta_4 + \beta_8) \cdot \left(\frac{1}{\kappa_8} + \frac{1}{\kappa_6} \right), \quad (S99)$$

where the superscript “(ST)” stands for the saturated concentration, then,

$$\varepsilon_r^{(py)}(\kappa_{py}^{(ST)}) < \varepsilon_f^{(py)}(\kappa_{py}^{(ST)}), \quad (S100)$$

and there exists a critical value of κ_{py} , denoted as $\kappa_{py}^{(C)}$, with

$$\begin{cases} \varepsilon_r^{(py)}(\kappa_{py}^{(C)}) = \varepsilon_f^{(py)}(\kappa_{py}^{(C)}) = \frac{\beta_3 + \beta_4 - \beta_6}{1/\kappa_3 + 1/\kappa_4 - 1/\kappa_6}, \\ \lambda_C^{(py)} \equiv \lambda(\kappa_{py}^{(C)}) = \frac{\phi_{\max}}{\phi_{py}/\varepsilon_r^{(py)}(\kappa_{py}^{(C)}) + \psi_{py}(\kappa_{py}^{(C)})}. \end{cases} \quad (S101)$$

Here, $\lambda_C^{(py)}$ is the growth rate at the transition point, and $\mathcal{E}_{r/f}^{(py)}$ stands for either $\mathcal{E}_r^{(py)}$ or $\mathcal{E}_f^{(py)}$. In **Appendix-fig. 2H**, we show the dependencies of $\mathcal{E}_r^{(py)}(\kappa_{py})$, $\mathcal{E}_f^{(py)}(\kappa_{py})$ and $\lambda(\kappa_{py})$ on κ_{py} in a 3-dimensional form. In the homogeneous case, $J_f^{(E,py)}$ and $J_r^{(E,py)}$ follow:

$$\begin{cases} J_f^{(E,py)} = \varphi_{py} \cdot \lambda \cdot \theta(\lambda - \lambda_C^{(py)}), \\ J_r^{(E,py)} = \varphi_{py} \cdot \lambda \cdot [1 - \theta(\lambda - \lambda_C^{(py)})]. \end{cases} \quad (S102)$$

Defining $\lambda_{\max}^{(py)} = \lambda(\kappa_{py}^{(ST)})$, and then, $[0, \lambda_{\max}^{(py)}]$ is the relevant range of the x axis. To compare with experiments, we assume the same extent of extrinsic noise in k_i^{cat} as specified in **Appendix 2.3**. Then, $\lambda_C^{(py)}$ approximately follows a Gaussian distribution:

$$\lambda_C^{(py)} \sim \mathcal{N}(\mu_{\lambda_C^{(py)}}, \sigma_{\lambda_C^{(py)}}^2), \quad (S103)$$

where $\mu_{\lambda_C^{(py)}}$ and $\sigma_{\lambda_C^{(py)}}$ stand for the mean and standard deviation of $\lambda_C^{(py)}$. Then, the relations between the normalized energy fluxes and growth rate are:

$$\begin{cases} J_f^{(E,py)}(\lambda) = \frac{1}{2} \varphi_{py} \cdot \lambda \cdot \left[\text{erf}\left(\frac{\lambda - \mu_{\lambda_C^{(py)}}}{\sqrt{2}\sigma_{\lambda_C^{(py)}}}\right) + 1 \right], \\ J_r^{(E,py)}(\lambda) = \frac{1}{2} \varphi_{py} \cdot \lambda \cdot \left[1 - \text{erf}\left(\frac{\lambda - \mu_{\lambda_C^{(py)}}}{\sqrt{2}\sigma_{\lambda_C^{(py)}}}\right) \right]. \end{cases} \quad (S104)$$

Combined with **Eq. S92**, we have:

$$\begin{cases} J_f^{(N)}(\lambda) = \frac{\varphi_{py}}{2(\beta_6 + \beta_8)} \cdot \lambda \cdot \left[\text{erf}\left(\frac{\lambda - \mu_{\lambda_C^{(py)}}}{\sqrt{2}\sigma_{\lambda_C^{(py)}}}\right) + 1 \right], \\ J_r^{(N)}(\lambda) = \frac{\varphi_{py}}{2(\beta_3 + \beta_4 + \beta_8)} \cdot \lambda \cdot \left[1 - \text{erf}\left(\frac{\lambda - \mu_{\lambda_C^{(py)}}}{\sqrt{2}\sigma_{\lambda_C^{(py)}}}\right) \right]. \end{cases} \quad (S105)$$

In **Fig. 3F**, we show that the model predictions (**Eq. S105**) agree quantitatively with the experimental results (Holms, 1996).

Appendix 4.2 Mixture of a Group A carbon source with extracellular amino acids

In the case of a Group A carbon source mixed with amino acids, the coarse-grained model is shown in **Appendix-fig. 2A**. In fact, this model can be used to analyze mixtures with one or multiple types of extracellular amino acids. Here, **Eqs. S21**, **S22**, **S24** and **S25** still apply, but **Eq. S23** changes to (the case of i remains the same as **Eq. S23**):

$$\Phi_i \cdot \xi_i \cdot N_{EP_i}^{\text{carbon}} + \Phi'_i \cdot \xi'_i \cdot N_{P_i}^{\text{carbon}} = r_i \cdot J_{\text{BM}} \quad (i = a2, b, c, d). \quad (\text{S106})$$

Here, $N_{P_i}^{\text{carbon}}$ represents the number of carbon atoms in a molecule of Pool i . For simplicity, we assume:

$$N_{P_i}^{\text{carbon}} \approx N_{EP_i}^{\text{carbon}}. \quad (\text{S107})$$

In the case where all 21 types of amino acids are present and each in saturated concentration (denoted as “21AA”), we have:

$$\left\{ \begin{array}{l} \phi_A \cdot \kappa_A = \phi_1 \cdot \kappa_1 + \phi_{a1} \cdot \kappa_{a1}, \\ 2\phi_1 \cdot \kappa_1 = \phi_2 \cdot \kappa_2 + \phi_5 \cdot \kappa_5 + \phi_{a2} \cdot \kappa_{a2}, \\ \phi_2 \cdot \kappa_2 = \phi_3 \cdot \kappa_3 + \phi_6 \cdot \kappa_6 + \phi_b \cdot \kappa_b, \\ \phi_5 \cdot \kappa_5 + \phi_4 \cdot \kappa_4 = \phi_3 \cdot \kappa_3 + \phi_d \cdot \kappa_d, \\ \phi_3 \cdot \kappa_3 = \phi_4 \cdot \kappa_4 + \phi_c \cdot \kappa_c, \\ \phi_{a1} \cdot \kappa_{a1} = \eta_{a1} \cdot \lambda, \phi_{a2} \cdot \kappa_{a2} + \phi_{a2}^{(21AA)} \cdot \kappa_{a2}^{(21AA)} = \eta_{a2} \cdot \lambda, \phi_b \cdot \kappa_b + \phi_b^{(21AA)} \cdot \kappa_b^{(21AA)} = \eta_b \cdot \lambda, \\ \phi_c \cdot \kappa_c + \phi_c^{(21AA)} \cdot \kappa_c^{(21AA)} = \eta_c \cdot \lambda, \phi_d \cdot \kappa_d + \phi_d^{(21AA)} \cdot \kappa_d^{(21AA)} = \eta_d \cdot \lambda, \\ \beta_1 \cdot \phi_1 \cdot \kappa_1 + \beta_2 \cdot \phi_2 \cdot \kappa_2 + \beta_3 \cdot \phi_3 \cdot \kappa_3 + \beta_4 \cdot \phi_4 \cdot \kappa_4 + \beta_6 \cdot \phi_6 \cdot \kappa_6 + \beta_{a1} \cdot \phi_{a1} \cdot \kappa_{a1} = J_E^{(N)}, \\ J_E^{(N)} = \eta_E \cdot \lambda, \lambda = \phi_R \cdot \kappa_r, J_r^{(N)} = \phi_4 \cdot \kappa_4, J_f^{(N)} = \phi_6 \cdot \kappa_6, \\ \phi_R + \phi_A + \sum_i \phi_i + \sum_j^{a1, a2, b, c, d} \phi_j + \phi_{a2}^{(21AA)} + \phi_b^{(21AA)} + \phi_c^{(21AA)} + \phi_d^{(21AA)} = \phi_{\max}, \end{array} \right. \quad (\text{S108})$$

where ϕ_i and κ_i are defined following **Eqs. S9** and **S12**. Since the cell growth rate elevates significantly with the mixture of amino acids, we deduce that Pools a2-d are supplied by amino acids in growth optimization, with

$$\phi_i = 0 \quad (i = a2, b, c, d). \quad (\text{S109})$$

Basically, amino acids should be more efficient in the supply of biomass production than the Group A carbon source for Pools a2-d, i.e.,

$$\left\{ \begin{array}{l} 1/\kappa_{a2}^{(21AA)} < 1/\kappa_{a2} + 1/(2\kappa_1) + 1/(2\kappa_A), \\ 1/\kappa_b^{(21AA)} < 1/\kappa_b + 1/\kappa_2 + 1/(2\kappa_1) + 1/(2\kappa_A), \\ 1/\kappa_c^{(21AA)} < 1/\kappa_c + 1/\kappa_5 + 1/\kappa_3 + 1/\kappa_2 + 1/\kappa_1 + 1/\kappa_A, \\ 1/\kappa_d^{(21AA)} < 1/\kappa_d + 1/\kappa_5 + 1/(2\kappa_1) + 1/(2\kappa_A). \end{array} \right. \quad (\text{S110})$$

In practice, the requirement is even higher for proteome efficiency using amino acids, since the biomass production pathway is accompanied by energy production in the case of Group A carbon sources, yet not for amino acids. Combining **Eqs. S108** and **S109**, we have:

$$\begin{cases} J_r^{(E)} + J_f^{(E)} = \varphi_{21AA} \cdot \lambda, \\ \frac{J_r^{(E)}}{\varepsilon_r} + \frac{J_f^{(E)}}{\varepsilon_f} = \phi_{\max} - \psi_{21AA} \cdot \lambda, \end{cases} \quad (\text{S111})$$

where $J_r^{(E)}$, $J_f^{(E)}$ follow **Eq. S30**, while ε_r and ε_f follow **Eq. S31**. ψ_{21AA}^{-1} is the proteome efficiency of biomass pathway under this nutrient condition, with

$$\psi_{21AA} = \frac{1}{\kappa_t} + \frac{\eta_{a1}}{\kappa_A} + \frac{\eta_{a1}}{\kappa_{a1}} + \frac{\eta_{a2}}{\kappa_{a2}^{(21AA)}} + \frac{\eta_b}{\kappa_b^{(21AA)}} + \frac{\eta_c}{\kappa_c^{(21AA)}} + \frac{\eta_d}{\kappa_d^{(21AA)}}. \quad (\text{S112})$$

φ_{21AA} is the energy demand coefficient, with

$$\varphi_{21AA} \equiv \eta_E - \beta_{a1} \cdot \eta_{a1}. \quad (\text{S113})$$

Combining **Eqs. S111** and **S31**, it is easy to obtain the formula for the growth rate:

$$\lambda(\kappa_A) = \begin{cases} \lambda_r^{(21AA)} \equiv \frac{\phi_{\max}}{\varphi_{21AA}/\varepsilon_r(\kappa_A) + \psi_{21AA}(\kappa_A)} & \varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A), \\ \lambda_f^{(21AA)} \equiv \frac{\phi_{\max}}{\varphi_{21AA}/\varepsilon_f(\kappa_A) + \psi_{21AA}(\kappa_A)} & \varepsilon_r(\kappa_A) < \varepsilon_f(\kappa_A). \end{cases} \quad (\text{S114})$$

In fact, **Eqs. S37–S42** still apply. $\varepsilon_{r/f}(\kappa_A^{(C)})$ satisfies **Eq. S43**, while

$\lambda_C^{(21AA)} \equiv \lambda(\kappa_A^{(C)})$ and $\lambda_{\max}^{(21AA)} \equiv \lambda(\kappa_A^{\max})$ are:

$$\begin{cases} \lambda_C^{(21AA)} = \frac{\phi_{\max}}{\varphi_{21AA}/\varepsilon_{r/f}(\kappa_A^{(C)}) + \psi_{21AA}(\kappa_A^{(C)})}, \\ \lambda_{\max}^{(21AA)} = \frac{\phi_{\max}}{\varphi_{21AA}/\varepsilon_f(\kappa_A^{\max}) + \psi_{21AA}(\kappa_A^{\max})}. \end{cases} \quad (\text{S115})$$

When extrinsic noise is taken into account, $\lambda_C^{(21AA)}$ approximately follows a Gaussian distribution:

$$\lambda_C^{(21AA)} \sim \mathcal{N}\left(\mu_{\lambda_C^{(21AA)}}, \sigma_{\lambda_C^{(21AA)}}^2\right), \quad (\text{S116})$$

and the normalized fluxes $J_r^{(N)}$, $J_f^{(N)}$ change into:

$$\begin{cases} J_f^{(N)}(\lambda) = \frac{\varphi_{21AA}}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \lambda \cdot \left[\operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c^{(21AA)}}}{\sqrt{2}\sigma_{\lambda_c^{(21AA)}}}\right) + 1 \right], \\ J_r^{(N)}(\lambda) = \frac{\varphi_{21AA}}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \lambda \cdot \left[1 - \operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c^{(21AA)}}}{\sqrt{2}\sigma_{\lambda_c^{(21AA)}}}\right) \right]. \end{cases} \quad (\text{S117})$$

In fact, the above analysis can be extended to cases where a Group A carbon source is mixed with arbitrary combinations of amino acids. **Eqs. S111**, **S114**–**S117** would remain in a similar form, while **Eqs. S112**–**S113** would change depending on the amino acid combinations. In **Appendix-fig.2B-C**, we show the comparisons between model predictions (see also **Appendix7.2** and **Eq. S157**) and experimental data (Basan et al., 2015; Wallden et al., 2016) in mixtures of 21 or 7 types of amino acids together with a Group A carbon source, which agree quantitatively.

Appendix 5 Enzyme allocation upon perturbations

Appendix 5.1 Carbon limitation within Group A carbon sources

In **Eq. S28**, we present the model predictions of the dependencies of enzyme protein fractions on growth rate and energy fluxes. To compare with experiments, we assume the same extent of extrinsic noise in k_i^{cat} as specified in **Appendix 2.3**. In fact, relative protein expression data for enzymes within glycolysis and the TCA cycle are available from existing studies, which are comparable to the φ_1 – φ_4 enzymes of our model (**Fig. 1B**). Upon κ_A perturbation, κ_A is a variable while $w_0 = 0$ is fixed (see **Appendix 1.5**). Combining **Eqs. S28** and **S47** (with $w_0 = 0$), we obtain:

$$\begin{cases} \phi_1 = \frac{\lambda}{\kappa_1} \left\{ \frac{\varphi \cdot (\beta_3 + \beta_4 - \beta_6)}{[\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)] \cdot [\beta_1 + 2(\beta_2 + \beta_6)]} \cdot \left[\operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) + 1 \right] + \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + \frac{\eta_{a2} + \eta_b + 2\eta_c + \eta_d}{2} \right\}, \\ \phi_2 = \frac{\lambda}{\kappa_2} \left\{ \frac{2\varphi \cdot (\beta_3 + \beta_4 - \beta_6)}{[\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)] \cdot [\beta_1 + 2(\beta_2 + \beta_6)]} \cdot \left[\operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) + 1 \right] + \frac{2\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + \eta_b + \eta_c \right\}, \\ \phi_3 = \frac{\lambda}{\kappa_3} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) \right] + \eta_c \right\}, \\ \phi_4 = \frac{\lambda}{\kappa_4} \cdot \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf}\left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}}\right) \right]. \end{cases} \quad (\text{S118})$$

In **Appendix-fig.3C-D**, we show the comparisons between model predictions (Eq. S118, $w_0 = 0$) and experimental data (Hui et al., 2015), which are consistent overall. We proceed to consider the influence of maintenance energy as specified in Appendix 3.2. Here, we still choose $w_0 = 2.5$ (h^{-1}) as previously adopted in Appendix 3.3. Then, Eq. S28 still holds, combined with Eq. S85 in the condition that $\iota = 0$, we have:

$$\begin{cases} \phi_1 = \frac{1}{2 \cdot \kappa_1} \left\{ \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) + 1 \right] + \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) \right] + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right\}, \\ \phi_2 = \frac{1}{\kappa_2} \left\{ \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) + 1 \right] + \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) \right] + (\eta_b + \eta_c) \lambda \right\}, \\ \phi_3 = \frac{1}{\kappa_3} \left\{ \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) \right] + \eta_c \cdot \lambda \right\}, \\ \phi_4 = \frac{1}{\kappa_4} \cdot \frac{\varphi \cdot \lambda + w_0}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}}{\sqrt{2}\sigma_{\lambda_c}} \right) \right]. \end{cases} \quad (S119)$$

In **Fig. 4A-B**, we show that the model predictions (Eq. S119, $w_0 = 2.5$ (h^{-1})) generally agree with the experiments (Hui et al., 2015). However, there are different basal expressions of these enzymes, which are probably due to living demands other than cell proliferation, such as preparation for starvation (Mori et al., 2017) or alteration in the type of the nutrient (Basan et al., 2020; Kussell and Leibler, 2005).

Appendix 5.2 Overexpression of useless proteins

In the case of φ_Z perturbation under each nutrient condition with fixed κ_A (see Appendix 3.1), we consider the same extent of extrinsic noise in k_i^{cat} as specified in Appendix 2.3. The relation between enzyme allocation and growth rate can be obtained by combining Eqs. S28 and S58 (with $w_0 = 0$):

$$\begin{cases} \phi_1 = \frac{\lambda}{2 \cdot \kappa_1} \left\{ \frac{\varphi \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + \frac{\varphi \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right]}{\beta_1 + 2(\beta_2 + \beta_6)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \right\}, \\ \phi_2 = \frac{\lambda}{\kappa_2} \left\{ \frac{\varphi \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + \frac{\varphi \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right]}{\beta_1 + 2(\beta_2 + \beta_6)} + (\eta_b + \eta_c) \right\}, \\ \phi_3 = \frac{\lambda}{\kappa_3} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] + \eta_c \right\}, \\ \phi_4 = \frac{\lambda}{\kappa_4} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right\}. \end{cases} \quad (S120)$$

Here $\lambda(\kappa_A, 0)$ is the growth rate for $\varphi_Z = 0$ and thus is a parameter rather than a variable. The growth rate is defined as $\lambda(\kappa_A, \varphi_Z)$, which follows Eq. S50. Thus, φ_i is proportional to the growth rate λ . In **Appendix-fig. 3E-F**, we see that the model predictions (Eq. S120) agree with the experiments (Basan et al., 2015) overall. Next, we consider the influence of maintenance energy with $w_0 = 2.5$ (h^{-1}). Combining Eqs. S28, S58 and S85 (with $\iota = 0$), we get:

$$\begin{aligned} \phi_1 &= \frac{w_0}{2 \cdot \kappa_1} \left\{ \frac{1}{\beta_1 + 2(\beta_2 + \beta_6)} \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{1}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right\} \\ &+ \frac{\lambda}{2 \cdot \kappa_1} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_6)} \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \right\}, \\ \phi_2 &= \frac{w_0}{\kappa_2} \left\{ \frac{1}{\beta_1 + 2(\beta_2 + \beta_6)} \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{1}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right\} \\ &+ \frac{\lambda}{\kappa_2} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_6)} \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] + (\eta_b + \eta_c) \right\}, \\ \phi_3 &= \frac{\lambda}{\kappa_3} \left\{ \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] + \eta_c \right\} + \frac{w_0}{\kappa_3} \cdot \frac{1}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right], \\ \phi_4 &= \frac{\lambda}{\kappa_4} \cdot \frac{\varphi}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] + \frac{w_0}{\kappa_4} \cdot \frac{1}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right]. \end{aligned} \quad (S121)$$

Here, the growth rate is defined as $\lambda(\kappa_A, \varphi_Z)$, and $\lambda(\kappa_A, 0)$ is a parameter rather than a variable. Thus, φ_i is a linear function of the growth rate λ , with a positive slope and a positive y-intercept. In **Fig. 4C-D** and **Appendix-fig. 3I-J**, we show that the model predictions (Eq. S121) agree quantitatively with the experimental data (Basan et al., 2015).

Appendix 5.3 Energy dissipation

In the case of energy dissipation under each nutrient condition, w is perturbed while κ_A is fixed. The relation between protein allocation and growth rate can be obtained by combining Eqs. S28 and S70. However, since w is explicitly present in Eq. S70, we need to reduce this variable to obtain the growth rate dependence of enzyme allocation. In fact, from Eq. S64, we have:

$$\lambda(\kappa_A, w) = \lambda(\kappa_A, 0) \left\{ 1 - \frac{w}{\phi_{\max}} \cdot \left[\frac{1}{\varepsilon_r(\kappa_A)} - \theta(\varepsilon_f(\kappa_A) - \varepsilon_r(\kappa_A)) \cdot \left(\frac{1}{\varepsilon_r(\kappa_A)} - \frac{1}{\varepsilon_f(\kappa_A)} \right) \right] \right\}. \quad (S122)$$

Here, $\lambda(\kappa_A, 0) \equiv \lambda(\kappa_A, w = 0)$ (satisfying Eq. S64), which is a parameter rather than a variable. “ θ ” stands for the Heaviside step function. Thus, we have:

$$w(\lambda) = \frac{\phi_{\max} \cdot [1 - \lambda/\lambda(\kappa_A, 0)]}{[1/\varepsilon_r(\kappa_A) - \theta(\varepsilon_f(\kappa_A) - \varepsilon_r(\kappa_A)) \cdot (1/\varepsilon_r(\kappa_A) - 1/\varepsilon_f(\kappa_A))]}, \quad (S123)$$

where the energy dissipation coefficient w is regarded as a function of the growth rate.

Combining Eqs. S28, S70 and S123, we get:

$$\begin{cases} \phi_1 = \frac{1}{2\kappa_1} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_6)} - \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \right] \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(0) \left\{ 1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right]} \right) + 1 \right] + \frac{2[\varphi \cdot \lambda + w(\lambda)]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \cdot \lambda, \\ \phi_2 = \frac{1}{\kappa_2} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_6)} - \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \right] \cdot \left[\operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(0) \left\{ 1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right]} \right) + 1 \right] + \frac{2[\varphi \cdot \lambda + w(\lambda)]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + (\eta_b + \eta_c) \cdot \lambda, \\ \phi_3 = \frac{1}{\kappa_3} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(0) \left\{ 1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right] + \eta_c \cdot \lambda, \\ \phi_4 = \frac{1}{\kappa_4} \cdot \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda - \mu_{\lambda_c}(0) \left\{ 1 - w \left[\frac{\varepsilon_{r/f}(\kappa_A^{(c)}) \phi_{\max}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right\}}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right], \end{cases} \quad (S124)$$

where $w(\lambda)$ follows Eq. S123. When κ_A lies in the vicinity of $\kappa_A^{(c)}$ or w is small so that

$$\left(1 - \frac{w}{\varepsilon_{r/f}(\kappa_A) \cdot \phi_{\max}} \right) / \left(1 - \frac{w}{\varepsilon_{r/f}(\kappa_A^{(c)}) \cdot \phi_{\max}} \right) \approx 1, \quad (S125)$$

then we have:

$$\begin{cases} J_f^{(N)}(\lambda, w) = \frac{\varphi \cdot \lambda + w}{\beta_1 + 2(\beta_2 + \beta_6)} \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right], \\ J_r^{(N)}(\lambda, w) = \frac{\varphi \cdot \lambda + w}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right], \end{cases} \quad (S126)$$

and thus,

$$\begin{cases} \phi_1 = \frac{1}{2\kappa_1} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_6)} - \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \right] \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{2[\varphi \cdot \lambda + w(\lambda)]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \cdot \lambda, \\ \phi_2 = \frac{1}{\kappa_2} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_6)} - \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \right] \cdot \left[\operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) + 1 \right] + \frac{2[\varphi \cdot \lambda + w(\lambda)]}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} + (\eta_b + \eta_c) \cdot \lambda, \\ \phi_3 = \frac{1}{\kappa_3} \left[\frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right] \right] + \eta_c \cdot \lambda, \\ \phi_4 = \frac{1}{\kappa_4} \cdot \frac{\varphi \cdot \lambda + w(\lambda)}{\beta_1 + 2(\beta_2 + \beta_3 + \beta_4)} \cdot \left[1 - \operatorname{erf} \left(\frac{\lambda(\kappa_A, 0) - \mu_{\lambda_c}(0)}{\sqrt{2}\sigma_{\lambda_c}(0)} \right) \right], \end{cases} \quad (S127)$$

Note that in Eq. S123, w is a linear function of λ with a negative slope. Thus φ_i exhibits a linear relation with λ when Eq. S125 is satisfied (see Eq. S127). In fact, the slope of φ_4 is surely negative (combining Eqs. S64, S123 and S127), while the slope sign of other φ_i depends on parameters. For a given nutrient, the enzymes corresponding to the same φ_i should exhibit the same slope sign. Another restriction is that if the slope sign of φ_1 is negative, then the slope sign of φ_2 is surely negative. In **Appendix-fig. 3K-N**, we show that our model results agree well with the experimental data (Basan et al., 2015) (Eq. S127).

Appendix 6 Other aspects of the model

Appendix 6.1 A coarse-grained model with more details

To compare with experiments, we consider a coarse-grained model with more details as shown in **Appendix-fig. 2F**. Here, nodes M_6, M_7 represent GA3P and DHAP, respectively. Other nodes follow the descriptions specified in **Appendix 2.1**. Each biochemical reaction follows **Eq. S5** with $b_i = 1$ except that $M_1 \rightarrow M_6 + M_7$ and $M_3 + M_5 \rightarrow M_4$. By applying flux balance to the stoichiometric fluxes, combined with **Eq. S8**, we obtain:

$$\begin{cases} \Phi_A \cdot \xi_A = \Phi_1 \cdot \xi_1 + \Phi_{a1} \cdot \xi_{a1}, \\ \Phi_{11} \cdot \xi_{11} = \Phi_{10} \cdot \xi_{10} + \Phi_1 \cdot \xi_1, \Phi_{10} \cdot \xi_{10} = \Phi_1 \cdot \xi_1, \\ \Phi_{11} \cdot \xi_{11} = \Phi_2 \cdot \xi_2 + \Phi_5 \cdot \xi_5 + \Phi_{a2} \cdot \xi_{a2}, \\ \Phi_2 \cdot \xi_2 = \Phi_3 \cdot \xi_3 + \Phi_6 \cdot \xi_6 + \Phi_b \cdot \xi_b, \\ \Phi_5 \cdot \xi_5 + \Phi_4 \cdot \xi_4 = \Phi_3 \cdot \xi_3 + \Phi_d \cdot \xi_d, \\ \Phi_3 \cdot \xi_3 = \Phi_4 \cdot \xi_4 + \Phi_c \cdot \xi_c. \end{cases} \quad (\text{S128})$$

While **Eqs. S22–S25** still hold. By applying the substitutions specified in **Eqs. S9, S12, S14–S18**, combined with **Eqs. S4, S10, S22–S25, S128**, and the constraint of proteome resource allocation, we get:

$$\begin{cases} \phi_A \cdot \kappa_A = \phi_1 \cdot \kappa_1 + \phi_{a1} \cdot \kappa_{a1}, \\ \phi_{11} \cdot \kappa_{11} = \phi_{10} \cdot \kappa_{10} + \phi_1 \cdot \kappa_1, \phi_{10} \cdot \kappa_{10} = \phi_1 \cdot \kappa_1, \\ \phi_{11} \cdot \kappa_{11} = \phi_2 \cdot \kappa_2 + \phi_5 \cdot \kappa_5 + \phi_{a2} \cdot \kappa_{a2}, \\ \phi_2 \cdot \kappa_2 = \phi_3 \cdot \kappa_3 + \phi_6 \cdot \kappa_6 + \phi_b \cdot \kappa_b, \\ \phi_5 \cdot \kappa_5 + \phi_4 \cdot \kappa_4 = \phi_3 \cdot \kappa_3 + \phi_d \cdot \kappa_d, \\ \phi_3 \cdot \kappa_3 = \phi_4 \cdot \kappa_4 + \phi_c \cdot \kappa_c, \\ \phi_{a1} \cdot \kappa_{a1} = \eta_{a1} \cdot \lambda, \phi_{a2} \cdot \kappa_{a2} = \eta_{a2} \cdot \lambda, \phi_b \cdot \kappa_b = \eta_b \cdot \lambda, \phi_c \cdot \kappa_c = \eta_c \cdot \lambda, \phi_d \cdot \kappa_d = \eta_d \cdot \lambda, \\ \beta_1 \cdot \phi_1 \cdot \kappa_1 + \beta_2 \cdot \phi_2 \cdot \kappa_2 + \beta_3 \cdot \phi_3 \cdot \kappa_3 + \beta_4 \cdot \phi_4 \cdot \kappa_4 + \beta_6 \cdot \phi_6 \cdot \kappa_6 + \beta_{a1} \cdot \phi_{a1} \cdot \kappa_{a1} = J_E^{(N)}, \\ J_E^{(N)} = \eta_E \cdot \lambda, \lambda = \phi_R \cdot \kappa_t, J_r^{(N)} = \phi_4 \cdot \kappa_4, J_f^{(N)} = \phi_6 \cdot \kappa_6, \\ \phi_R + \phi_A + \phi_1 + \phi_2 + \phi_3 + \phi_4 + \phi_5 + \phi_6 + \phi_7 + \phi_8 + \phi_{a1} + \phi_{a2} + \phi_b + \phi_c + \phi_d = \phi_{\max}. \end{cases} \quad (\text{S129})$$

Then, **Eq. S28** still hold, while ϕ_{10} and ϕ_{11} are:

$$\begin{cases} \phi_{10} = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right] / (2 \cdot \kappa_{10}), \\ \phi_{11} = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right] / \kappa_{11}. \end{cases} \quad (\text{S130})$$

By substituting **Eqs. S28** and **S130** into **Eq. S129**, we get:

$$\begin{cases} J_r^{(E)} + J_f^{(E)} = \varphi \cdot \lambda, \\ \frac{J_r^{(E)}}{\mathcal{E}_r^{(dt)}} + \frac{J_f^{(E)}}{\mathcal{E}_f^{(dt)}} = \phi_{\max} - \psi_{dt} \cdot \lambda, \end{cases} \quad (\text{S131})$$

where “dt” stands for details. **Eqs. S30** and **S33** still hold. $\mathcal{E}_r^{(dt)}$ and $\mathcal{E}_f^{(dt)}$ represent the proteome energy efficiencies of respiration and fermentation, respectively, with

$$\begin{cases} \mathcal{E}_r^{(dt)} = \frac{\beta_1/2 + \beta_2 + \beta_3 + \beta_4}{1/(2\kappa_A) + 1/(2\kappa_1) + 1/(2\kappa_{10}) + 1/\kappa_{11} + 1/\kappa_2 + 1/\kappa_3 + 1/\kappa_4}, \\ \mathcal{E}_f^{(dt)} = \frac{\beta_1/2 + \beta_2 + \beta_6}{1/(2\kappa_A) + 1/(2\kappa_1) + 1/(2\kappa_{10}) + 1/\kappa_{11} + 1/\kappa_2 + 1/\kappa_6}. \end{cases} \quad (\text{S132})$$

ψ_{dt}^{-1} is the proteome efficiency of biomass pathway, with

$$\psi_{dt} = \frac{1}{\kappa_i} + \frac{1 + \eta_{a1} + \eta_c}{2\kappa_A} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \left(\frac{1}{2\kappa_1} + \frac{1}{2\kappa_{10}} + \frac{1}{\kappa_{11}} \right) + \frac{\eta_b + \eta_c}{\kappa_2} + \frac{\eta_c}{\kappa_3} + \frac{\eta_c + \eta_d}{\kappa_5} + \sum_i^{a1,a2,b,c,d} \frac{\eta_i}{\kappa_i}. \quad (\text{S133})$$

Appendix 6.2 Estimation of the in vivo enzyme catalytic rates

We use the method introduced by Davidi *et al.* (Davidi *et al.*, 2016), combined with proteome experimental data (Basan *et al.*, 2015) (**Appendix-table2**), to estimate the in vivo enzyme catalytic rates. Combining **Eqs. S28** and **S130**, we have:

$$\begin{cases} \kappa_1 = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right] / (2 \cdot \phi_1), \\ \kappa_2 = \left[J_r^{(N)} + J_f^{(N)} + (\eta_b + \eta_c) \lambda \right] / \phi_2, \\ \kappa_3 = (J_r^{(N)} + \eta_c \cdot \lambda) / \phi_3, \kappa_4 = J_r^{(N)} / \phi_4, \\ \kappa_5 = (\eta_c + \eta_d) \lambda / \phi_5, \kappa_6 = J_f^{(N)} / \phi_6, \\ \kappa_{10} = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right] / (2 \cdot \phi_{10}), \\ \kappa_{11} = \left[J_r^{(N)} + J_f^{(N)} + (\eta_{a2} + \eta_b + 2\eta_c + \eta_d) \lambda \right] / \phi_{11}. \end{cases} \quad (\text{S134})$$

Here, $J_r^{(N)}$, $J_f^{(N)}$, λ and ϕ_i ($i = 1-6, 10-11$) are measurable from experiments (see **Appendix**

8.1 and **Appendix-table2**). Thus, we can obtain the in vivo values of κ_i from **Eq. S134**. Combined with **Eqs. S17** and **S20**, we have

$$k_i^{\text{cat}} = \frac{r_{\text{carbon}}}{r_{\text{protein}}} \cdot \frac{m_{E_i}}{m_{\text{carbon}}} \cdot \kappa_i \cdot \left[\sum_i r_i / N_{\text{EP}_i}^{\text{carbon}} \right]. \quad (\text{S135})$$

Eq. S135 is the in vivo result for the enzyme catalytic rate. In **Appendix-fig. 2G**, we show a comparison between in vivo and in intro results for k_{cat} values of enzymes within glycolysis and the TCA cycle, which are roughly consistent. In the applications, we prioritized the use of in vivo results for enzyme catalytic rates, and use in intro data as a substitute when there were vacancies.

Appendix 6.3 Comparison with existing models that illustrate experimental results

For the coarse-grained model described in [Appendix 2](#), the normalized stoichiometric influx of a Group A carbon source is given by:

$$J_{in}^{(N)} \equiv J_A^{(N)} = \phi_A \cdot \kappa_A. \quad (S136)$$

Combined with the first equation in [Eq. S28](#), we obtain:

$$J_{in}^{(N)} - \vartheta \cdot \lambda = \frac{J_r^{(E)}}{e_r} + \frac{J_f^{(E)}}{e_f}, \quad (S137)$$

where $e_r = \beta_1 + 2(\beta_2 + \beta_3 + \beta_4)$, $e_f = \beta_1 + 2(\beta_2 + \beta_6)$, and $\vartheta = \eta_{a1} + \eta_c + (\eta_{a2} + \eta_b + \eta_d)/2$.

Evidently, e_r , e_f and ϑ are constant parameters. In this subsection, we highlight the major differences between our model presented in [Appendix 2](#) and existing models that may illustrate the growth rate dependence of fermentation flux in the standard picture ([Basan et al., 2015](#); [Holms, 1996](#); [Meyer et al., 1984](#); [Nanchen et al., 2006](#)) of overflow metabolism.

Based on the modeling principles rather than the detailed mechanisms, there are two major classes of existing models that can illustrate experimental results. In fact, both classes of models regard the proteome energy efficiencies ε_r and ε_f as constants, with $\varepsilon_f > \varepsilon_r$ if used, or follow functionally equivalent propositions. In our model, however, ε_r and ε_f are both functions of κ_A , which vary significantly upon nutrient perturbation, where $\varepsilon_r(\kappa_A \rightarrow 0) > \varepsilon_f(\kappa_A \rightarrow 0)$

and $\varepsilon_r(\kappa_A^{\max}) < \varepsilon_f(\kappa_A^{\max})$ (see [Eqs. S38](#), [S40](#)–[S41](#)). Furthermore, there are significant differences in the modeling/optimization principles, as listed below.

The first class of models ([Chen and Nielsen, 2019](#); [Majewski and Domach, 1990](#); [Niebel et al., 2019](#); [Shlomi et al., 2011](#); [Varma and Palsson, 1994](#); [Vazquez et al., 2010](#); [Vazquez and Oltvai, 2016](#)) optimize the ratio of biomass outflow to carbon influx: $\lambda/J_{in}^{(N)}$. Either to

optimize the growth rate for a given carbon influx, or to minimize the carbon influx for a given growth rate. Since respiration is far more efficient than fermentation in terms of the energy production per unit carbon, to optimize the ratio $\lambda/J_{in}^{(N)}$, cells would preferentially use

respiration when the carbon influx is small. As carbon influx increases above a certain threshold, factors such as proteome allocation direct cells towards fermentation in an analog way since they consider $\varepsilon_f > \varepsilon_r$. Our model is significantly different from this class of models in the optimization principle, where we purely optimize the cell growth rate for a given nutrient condition, without any further constraints on the carbon influx.

The second class of models, represented by Basan *et al.* ([Basan et al., 2015](#)), also adopt the optimization of $\lambda/J_{in}^{(N)}$ in the interpretation of their model results. However, the growth rate dependence of fermentation flux was derived prior to the application of growth rate optimization.

In fact, Eqs. S29 and S137 in our model are very similar form to Basan *et al.* (Basan *et al.*, 2015), yet there are critical differences as we list below. In Eq. S29, by regarding $J_r^{(E)}$ and $J_f^{(E)}$ as the two variables in a system of linear equations, we obtain the following expressions:

$$\begin{cases} J_r^{(E)} = \frac{\phi_{\max} - (\psi + \varphi/\varepsilon_f) \cdot \lambda}{1/\varepsilon_r - 1/\varepsilon_f}, \\ J_f^{(E)} = \frac{(\psi + \varphi/\varepsilon_r) \cdot \lambda - \phi_{\max}}{1/\varepsilon_r - 1/\varepsilon_f}. \end{cases} \quad (\text{S138})$$

In Basan *et al.* (Basan *et al.*, 2015), Eq. S138 is considered to be the relation between $J_{r/f}^{(E)}$ and λ upon nutrient (and thus $J_{in}^{(N)}$) perturbation, while ε_r and ε_f are regarded as constants through the perturbation. By contrast, in our model, Eq. S138 is the constraint under a given nutrient condition with fixed κ_A , yet not relevant to nutrient perturbation. For wild-type strains, if $\varepsilon_r(\kappa_A) > \varepsilon_f(\kappa_A)$ (or vice versa), then the optimal solution is $J_r^{(E)}(\kappa_A) = \varphi \cdot \lambda(\kappa_A)$ and

$$J_f^{(E)}(\kappa_A) = 0, \text{ with } \lambda(\kappa_A) = \frac{\varepsilon_r(\kappa_A) \cdot \phi_{\max}}{\varphi + \varepsilon_r(\kappa_A) \cdot \psi(\kappa_A)}.$$

This solution (which satisfies Eq.

S138) corresponds to a point rather than a line in the relation between $J_{r/f}^{(E)}$ and λ upon κ_A perturbation.

Appendix 6.4 Explanation of the Warburg effect in tumor cells

Our model and analysis shown in Appendix 2 can be naturally extended to explain the Warburg effect in tumor cell metabolism with the following modifications in the model settings. In the applications for tumor cell metabolism, the fermentation flux changes from acetate secretion rate into the lactate secretion rate, and thus the stoichiometric coefficients (β_i) for ATP production change accordingly. Consequently, in the coarse-grained model shown in Fig. 1B, M_3 stands for pyruvate, and β_2, β_6 change into $\beta_2 = 1, \beta_6 = -2$ (Nelson *et al.*, 2008).

Evidently, Eqs. S37–S38 still hold. As long as $\varepsilon_r(\kappa_A^{\max}) < \varepsilon_f(\kappa_A^{\max})$ (Eq. S40), then there exists a critical switching point for κ_A (denoted as $\kappa_A^{(C)}$, see Eq. S41), below which the respiration pathway is more efficient, while above $\kappa_A^{(C)}$, the fermentation pathway is more efficient in the proteome energy production. To maximize proliferation rate, the tumor cells would preferentially use the respiration pathway upon starvation while use the aerobic glycolysis pathway with abundant nutrient (Warburg effect). Based on the facts that the rate of glucose metabolism is 10–100 times faster in aerobic glycolysis than that of respiration (Liberti and Locasale, 2016), and oncogenes induce Warburg effect intrinsically to promote cell

proliferation (Hanahan and Weinberg, 2011; Liberti and Locasale, 2016), it is suggested that $\varepsilon_r(\kappa_A^{\max}) < \varepsilon_f(\kappa_A^{\max})$. Thus, the aerobic glycolysis with abundant nutrients provides a fitness advantage for tumor cell proliferation.

Appendix 7 Probability density functions of variables and parameters

Appendix 7.1 Probability density function of k_i

Enzyme catalysis is crucial for the survival of living organisms, as it can significantly accelerate the rate of a biochemical reaction by moderating the energy barrier between the substrate and product (Nelson et al., 2008). However, the maximal turnover rate of enzymes, k_{cat} values, vary notably between the in vivo and in vitro measurements (Davidi et al., 2016). Recent studies suggest that differences in the aquatic medium should be the causes (Davidi et al., 2016; García-Contreras et al., 2012). In particular, the concentrations of potassium and phosphate have a big influence on k_{cat} (García-Contreras et al., 2012), which possess a certain degree of variation among the cell populations under intracellular conditions (García-Contreras et al., 2012). For

simplicity, we assume that the turnover rate of each E_i enzyme k_i^{cat} follows a Gaussian

distribution $\mathcal{N}(\mu_{k_i^{\text{cat}}}, \sigma_{k_i^{\text{cat}}}^2)$ with $k_i^{\text{cat}} > 0$ among cells (extrinsic noise (Elowitz et al., 2002)), denoted as η_{ext} , and then the probability density function of k_i^{cat} is given by:

$$k_i^{\text{cat}} \sim \mathcal{N}(x, \mu_{k_i^{\text{cat}}}, \sigma_{k_i^{\text{cat}}}^2) = \begin{cases} \frac{1}{\sigma_{k_i^{\text{cat}}} \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu_{k_i^{\text{cat}}}}{\sigma_{k_i^{\text{cat}}}} \right)^2}, & x \geq 0. \\ 0, & x < 0. \end{cases} \quad (\text{S139})$$

When the CV of the k_i^{cat} distribution (i.e., $\sigma_{k_i^{\text{cat}}} / \mu_{k_i^{\text{cat}}}$) is less than 1/3, $\mathcal{N}(x, \mu_{k_i^{\text{cat}}}, \sigma_{k_i^{\text{cat}}}^2)$ is

almost identical to $\mathcal{N}(\mu_{k_i^{\text{cat}}}, \sigma_{k_i^{\text{cat}}}^2)$. In this regard, $1/k_i^{\text{cat}}$ follows the positive inverse of Gaussian distribution (IOG), and the probability density function is:

$$\text{IOG}(x, \mu_{1/k_i^{\text{cat}}}, \zeta_{1/k_i^{\text{cat}}}) = \begin{cases} \sqrt{\frac{\zeta_{1/k_i^{\text{cat}}}}{2\pi x^4}} \exp\left(-\frac{1}{2} \frac{\zeta_{1/k_i^{\text{cat}}} (x - \mu_{1/k_i^{\text{cat}}})^2}{x^2 \mu_{1/k_i^{\text{cat}}}^2}\right), & x \geq 0, \\ 0, & x < 0, \end{cases} \quad (\text{S140})$$

where $\zeta_{1/k_i^{\text{cat}}} = 1/\sigma_{k_i^{\text{cat}}}^2$ and $\mu_{1/k_i^{\text{cat}}} = 1/\mu_{k_i^{\text{cat}}}$.

Meanwhile, due to the stochastic nature of biochemical reactions, we apply Gillespie's chemical Langevin equation (Gillespie, 2000) to account for the intrinsic noise (Elowitz et al., 2002) (denoted as η_{int}). For cell size regulation of *E. coli* within a cell cycle, the cell mass at initiation of DNA per chromosome origin remains constant (Donachie, 1968). Thus, the duration of enzyme E_i to finish a catalytic job (with a timescale of $1/k_i^{\text{cat}}$) can be approximated as the first passage time of a stochastic process, with

$$\begin{cases} X_i(t=0) = 0, \\ dX_i/dt = \alpha_i + \sqrt{\alpha_i} \Gamma_i(t), \\ T_\Theta = \inf \{t > 0 \mid X_i(t) = \Theta\}. \end{cases} \quad (\text{S141})$$

Here $\alpha_i \equiv k_i^{\text{cat}} \cdot \Theta$, Θ is proportional to the cell volume, and $\Gamma_i(t)$ is the independent temporally uncorrelated Gaussian white noise. Then, for a given value of k_i^{cat} , the first passage time T_i follows an Inverse Gaussian distribution (IG) (Folks and Chhikara, 1978):

$$\text{IG}\left(x, \mu'_{1/k_i^{\text{cat}}}, \zeta'_{1/k_i^{\text{cat}}}\right) = \begin{cases} \sqrt{\frac{\zeta'_{1/k_i^{\text{cat}}}}{2\pi x^3}} \exp\left(-\frac{1}{2} \frac{\zeta'_{1/k_i^{\text{cat}}} \left(x - \mu'_{1/k_i^{\text{cat}}}\right)^2}{x \mu'^2_{1/k_i^{\text{cat}}}}\right), & x \geq 0, \\ 0, & x < 0, \end{cases} \quad (\text{S142})$$

where $\mu'_{1/k_i^{\text{cat}}} = 1/k_i^{\text{cat}}$, $\zeta'_{1/k_i^{\text{cat}}} = \Theta/k_i^{\text{cat}}$, and the variance is

$$\sigma'^2_{1/k_i^{\text{cat}}} \equiv \mu'^3_{1/k_i^{\text{cat}}} / \zeta'_{1/k_i^{\text{cat}}} = 1 / \left[\Theta \cdot \left(k_i^{\text{cat}}\right)^2 \right]. \text{ Thus, we can obtain the CV:}$$

$$\sigma'_{1/k_i^{\text{cat}}} / \mu'_{1/k_i^{\text{cat}}} = \Theta^{-\frac{1}{2}}, \quad (\text{S143})$$

which is inversely scaled with the square root of cell volume. Evidently, the intrinsic and extrinsic noise make orthogonal contributions to the total noise (Elowitz et al., 2002) (denoted as θ_{tot}):

$$\eta_{\text{tot}}^2 = \eta_{\text{int}}^2 + \eta_{\text{ext}}^2. \quad (\text{S144})$$

In fact, when the CV is small (i.e., $CV \ll 1$), both IOG and IG distributions converge into Gaussian distributions (**Appendix-fig.4**). In the back-of-the-envelope calculations, we approximate x in all the denominator terms of $\text{IOG}(x, \mu, \zeta)$ and $\text{IG}(x, \mu, \zeta)$ as μ (since $CV \ll 1$). Then, both IOG and IG distributions can be approximated as follows:

$$\text{IOG}\left(x, \mu_{1/k_i^{\text{cat}}}, \zeta_{1/k_i^{\text{cat}}}\right) \xrightarrow{CV \ll 1} \mathcal{N}\left(\mu_{1/k_i^{\text{cat}}}, \sigma_{1/k_i^{\text{cat}}}^2\right), \quad (\text{S145})$$

with $\sigma_{1/k_i^{\text{cat}}}^2 = \mu_{1/k_i^{\text{cat}}}^4 / \zeta_{1/k_i^{\text{cat}}}$, and

$$\text{IG}\left(x, \mu'_{1/k_i^{\text{cat}}}, \zeta'_{1/k_i^{\text{cat}}}\right) \xrightarrow{CV \ll 1} \mathcal{N}\left(\mu'_{1/k_i^{\text{cat}}}, \sigma'^2_{1/k_i^{\text{cat}}}\right), \quad (\text{S146})$$

with $\sigma'^2_{1/k_i^{\text{cat}}} = \mu'^3_{1/k_i^{\text{cat}}} / \zeta'_{1/k_i^{\text{cat}}}$. Rigorously, we show below that $\text{IG}(x, \mu, \zeta)$ shrinks to be $N(\mu, \mu^3/\zeta)$

when the CV is small. Here, the characteristic function of the variable x is (Folks and Chhikara, 1978; Van Kampen, 1992):

$$G(k) = \int_{-\infty}^{\infty} e^{ikx} \cdot \text{IG}(x, \mu, \zeta) dx = \exp\left\{\frac{\zeta}{\mu} \left[1 - \sqrt{1 - \frac{2i\mu^2 k}{\zeta}}\right]\right\}, \quad (\text{S147})$$

and thus,

$$\text{IG}(x, \mu, \zeta) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} \cdot G(k) dk. \quad (\text{S148})$$

When the variance $\sigma^2 \equiv \mu^3/\zeta$ is very small, basically, we require $2\mu^2 k/\zeta = 2\sigma^2 k/\mu \ll 1$, and then

$$\sqrt{1 - \frac{2i\mu^2 k}{\zeta}} \approx 1 - \frac{\mu^2}{\zeta} ki + \frac{\mu^4}{2\zeta^2} k^2. \text{ Thus,}$$

$$\begin{cases} G(k) \approx \exp\left(\mu ki - \frac{\mu^3}{2\zeta} k^2\right), \\ \text{IG}(x, \mu, \zeta) \approx \sqrt{\frac{\zeta}{2\pi\mu^3}} e^{-\frac{\zeta(x-\mu)^2}{2\mu^3}} = N\left(\mu, \mu^3/\zeta\right). \end{cases} \quad (\text{S149})$$

Then, we have:

$$\lim_{\sigma \rightarrow 0} \text{IG}(x, \mu, \zeta) = N\left(\mu, \mu^3/\zeta\right). \quad (\text{S150})$$

In fact, although intrinsic noise affects the short-term measurement of enzyme catalytic rate and growth rate at the single-cell level, its contribution in the long-term is negligible. Thus, we approximate $\eta_{\text{tot}} \approx \eta_{\text{ext}}$. Combined with Eqs. S145–S146, it is easy to check that $1/k_i^{\text{cat}}$

shares roughly identical CV as k_i^{cat} :

$$\sigma_{1/k_i^{\text{cat}}} / \mu_{1/k_i^{\text{cat}}} = \sigma_{k_i^{\text{cat}}} / \mu_{k_i^{\text{cat}}} . \quad (\text{S151})$$

For convenience, in the model analysis, we approximate both IOG and IG distributions as Gaussian distributions. Then, all $1/k_i^{\text{cat}}$ are normally distributed independent random variables following Gaussian distributions:

$$1/k_i^{\text{cat}} \sim \mathcal{N}\left(\mu_{1/k_i^{\text{cat}}}, \sigma_{1/k_i^{\text{cat}}}^2\right). \quad (\text{S152})$$

Using the properties of Gaussian distributions, for a series of constant real numbers γ_i , the

summation of $\gamma_i / k_i^{\text{cat}}$, where we defined it as $\Xi \equiv \sum_{i=1}^n \gamma_i / k_i^{\text{cat}}$, follows a Gaussian

distribution (Van Kampen, 1992):

$$\Xi \sim \mathcal{N}\left(\mu_{\Xi}, \sigma_{\Xi}^2\right), \quad (\text{S153})$$

with $\mu_{\Xi} = \sum_{i=1}^n \gamma_i \mu_{1/k_i^{\text{cat}}}$ and $\sigma_{\Xi}^2 = \sum_{i=1}^n \left(\gamma_i \sigma_{1/k_i^{\text{cat}}}\right)^2$. The relation between κ_i and k_i^{cat} is shown in

Eq. S12. To optimize growth rate, each κ_i of the intermediate nodes satisfies Eq. S20, while κ_A

satisfies Eq. S27. Thus, for a given nutrient condition ($[A]$ is fixed), all the ratios $k_i^{\text{cat}} / \kappa_i$ are

constants. Combined with Eqs. S139, S145–S146, S152, then, the distributions of all κ_i and $1/\kappa_i$ can be approximated as Gaussian distributions:

$$\begin{cases} \kappa_i \sim \mathcal{N}\left(\mu_{\kappa_i}, \sigma_{\kappa_i}^2\right), \\ 1/\kappa_i \sim \mathcal{N}\left(\mu_{1/\kappa_i}, \sigma_{1/\kappa_i}^2\right), \end{cases} \quad (\text{S154})$$

where μ_{κ_i} and μ_{1/κ_i} are the means of κ_i and $1/\kappa_i$, while σ_{κ_i} and σ_{1/κ_i} are their standard

deviations. Using the properties of Gaussian distributions combined with Eq. S31, S32, S36,

S42–S43, S145–S146 and S153, then ε_r , ε_f , ψ , λ_r , λ_f , $K_A^{(C)}$ and λ_C also roughly follow

Gaussian distributions.

Appendix 7.2 Probability density function of the growth rate λ

From Appendix 7.1 [\[1\]](#), we note that λ_r and λ_f (see Eq. S36 [\[1\]](#)) roughly follow Gaussian distributions, with

$$\begin{cases} \lambda_r \sim \mathcal{N}(\mu_{\lambda_r}, \sigma_{\lambda_r}^2), \\ \lambda_f \sim \mathcal{N}(\mu_{\lambda_f}, \sigma_{\lambda_f}^2), \end{cases} \quad (\text{S155})$$

where $\mu_{\lambda_{r/f}}$ and $\sigma_{\lambda_{r/f}}$ represent the mean and standard deviation, respectively. We further assume that the correlation between λ_r and λ_f is ρ_{rf} . From Eq. S36 [\[1\]](#), we see that the growth rate λ takes the maximum of λ_r and λ_f i.e.,

$$\lambda = \max(\lambda_r, \lambda_f). \quad (\text{S156})$$

Then, the cumulative distribution function of λ is $P(\lambda \leq x) = \int_{-\infty}^x \int_{-\infty}^x f(x_1, x_2) dx_1 dx_2$, where

$$f(x_1, x_2) = \frac{(1 - \rho_{rf}^2)^{-\frac{1}{2}}}{2\pi\sigma_{\lambda_r}\sigma_{\lambda_f}} \exp\left(-\frac{1}{2(1 - \rho_{rf}^2)} \left[\left(\frac{x_1 - \mu_{\lambda_r}}{\sigma_{\lambda_r}}\right)^2 - 2\rho_{rf} \left(\frac{x_1 - \mu_{\lambda_r}}{\sigma_{\lambda_r}}\right) \left(\frac{x_2 - \mu_{\lambda_f}}{\sigma_{\lambda_f}}\right) + \left(\frac{x_2 - \mu_{\lambda_f}}{\sigma_{\lambda_f}}\right)^2 \right]\right)$$

. Thus, the probability density function of the growth rate λ is given by:

$$f_{\lambda}(x) = \frac{1}{2\sqrt{2\pi}\sigma_{\lambda_r}} e^{-\frac{1}{2}\left(\frac{x - \mu_{\lambda_r}}{\sigma_{\lambda_r}}\right)^2} \left[\operatorname{erf}\left(\frac{(x - \mu_{\lambda_f})\sigma_{\lambda_r} - \rho_{rf}\sigma_{\lambda_f}(x - \mu_{\lambda_r})}{\sigma_{\lambda_r}\sigma_{\lambda_f}\sqrt{2(1 - \rho_{rf}^2)}}\right) + 1 \right] + \frac{1}{2\sqrt{2\pi}\sigma_{\lambda_f}} e^{-\frac{1}{2}\left(\frac{x - \mu_{\lambda_f}}{\sigma_{\lambda_f}}\right)^2} \left[\operatorname{erf}\left(\frac{(x - \mu_{\lambda_r})\sigma_{\lambda_f} - \rho_{rf}\sigma_{\lambda_r}(x - \mu_{\lambda_f})}{\sigma_{\lambda_r}\sigma_{\lambda_f}\sqrt{2(1 - \rho_{rf}^2)}}\right) + 1 \right]. \quad (\text{S157})$$

In Appendix-fig. 2B [\[1\]](#), we show that Eq. S157 [\[1\]](#) quantitatively illustrates the experimental data of *E. coli* under the relevant conditions.

Appendix 8 Model comparison with experiments

Appendix 8.1 Flux comparison with experiments

In Appendix 6.2 [\[1\]](#), we see that the values of $J_f^{(N)}$ and $J_r^{(N)}$ are required to calculate the in

vivo enzyme catalytic rates of the intermediate nodes. Here, we use J_{acetate} and $J_{\text{CO}_2, r}$ to

represent the stoichiometric fluxes of acetate from the fermentation pathway and CO_2 from the respiration pathway, respectively. Combined with the stoichiometric coefficients of both pathways, we have:

$$\begin{cases} J_{\text{acetate}} = J_f, \\ J_{\text{CO}_2, r} = 3 \cdot J_r. \end{cases} \quad (\text{S158})$$

By further combining with **Eqs. S16–S17**, we get:

$$\begin{cases} J_f^{(N)} = J_{\text{acetate}} \cdot \frac{m_{\text{carbon}}}{M_{\text{carbon}}} \cdot \left[\sum_i r_i / N_{\text{EP}_i}^{\text{carbon}} \right]^{-1}, \\ J_r^{(N)} = \frac{1}{3} \cdot J_{\text{CO}_2, r} \cdot \frac{m_{\text{carbon}}}{M_{\text{carbon}}} \cdot \left[\sum_i r_i / N_{\text{EP}_i}^{\text{carbon}} \right]^{-1}. \end{cases} \quad (\text{S159})$$

In fact, the values of J_{acetate} and $J_{\text{CO}_2, r}$ scale with the mass of the “big cell”, which increases over time. In experiments, the measurable fluxes are typically expressed in the unit of mM/OD₆₀₀/h (Basan et al., 2015). Thus, we define $J_{\text{acetate}}^{(M)}$ and $J_{\text{CO}_2, r}^{(M)}$ as the fluxes of J_{acetate} and $J_{\text{CO}_2, r}$ in the unit of mM/OD₆₀₀/h, respectively. The superscript “(M)” represents the measurable flux in this unit. For *E. coli*, we use the following biochemical data collected from published literature: 1 OD₆₀₀ roughly corresponds to 6×10^8 cells/mL (Stevenson et al., 2016), the average mass of a cell is 1pg (Milo and Phillips, 2015), the biomass percentage of the cell weight is 30% (Neidhardt et al., 1990), the molar mass of carbon is 12g (Nelson et al., 2008), $r_{\text{carbon}} = 0.48$ (Neidhardt et al., 1990) and $r_{\text{protein}} = 0.55$ (Neidhardt et al., 1990). Combined with the values of r_i (see Appendix 1.2) and $N_{\text{EP}_i}^{\text{carbon}}$, where $N_{\text{EP}_{a1}}^{\text{carbon}} = 6$, $N_{\text{EP}_{a2}}^{\text{carbon}} = 3$, $N_{\text{EP}_b}^{\text{carbon}} = 3$, $N_{\text{EP}_c}^{\text{carbon}} = 5$, and $N_{\text{EP}_d}^{\text{carbon}} = 4$ (Nelson et al., 2008), we have:

$$\begin{cases} J_f^{(N)} \approx J_{\text{acetate}}^{(M)} / 2, \\ J_r^{(N)} \approx J_{\text{CO}_2, r}^{(M)} / 6. \end{cases} \quad (\text{S160})$$

From Eq. S18, we obtain the values of η_i for each precursor pool: $\eta_{a1} = 0.15$, $\eta_{a2} = 0.30$, $\eta_b = 0.35$, $\eta_c = 0.09$, $\eta_d = 0.11$. Still, the value of η_E is required to compare the growth rate dependence of fermentation/respiration fluxes between model results and experiments, which we will specify in Appendix 8.2.

Appendix 8.2 Model parameter settings using experimental data

We have collected biochemical data of *E. coli* shown in **Appendix-tables 1–2** to set the model parameters. This includes the molecular weight (MW) and in vitro k_{cat} values of the catalytic enzymes, as well as the proteome and flux data used to calculate the in vivo turnover numbers. To reduce measurement noise, we take the average rather than the maximum value of in vivo k_{cat} from calculations with data from four cultures (see **Appendix-table 2**). Here, we prioritize the use of in vivo k_{cat} wherever applicable unless there is a vacancy (see **Appendix-table 1**).

Note that our models are coarse grained. For example, the flux J_3 shown in **Fig. 1B** actually corresponds to three different reactions in the metabolic network (see **Fig. 1A** and **Appendix-table 1**), which we label as $J_3^{(i)}$ ($i=1, 2, 3$). For each $J_3^{(i)}$, there are corresponding

variables/parameters of $\Phi_3^{(i)}$, $\xi_3^{(i)}$, $\phi_3^{(i)}$, $\kappa_3^{(i)}$ satisfying [Eqs. S8](#), [S9](#) and [S12](#),

Evidently, $J_3^{(i)} = J_3$ ($i=1, 2, 3$), and it is straightforward to derive the following relation

between $\kappa_3^{(i)}$ and κ_3 :

$$1/\kappa_3 = \sum_{i=1}^3 1/\kappa_3^{(i)}. \quad (\text{S161})$$

In fact, [Eq. S161](#) can be generalized to determine the values of other κ_i in the coarse-grained models combined with the biochemical data. For the coarse-grained model of Group A carbon source utilization shown in [Fig. 1B](#), we have the values for parameters κ_i ($i=1, \dots, 6$), and then

$$\mathcal{E}_{r/f}(\kappa_A^{(C)}) = 122 \text{ (h}^{-1}\text{)}. \text{ Evidently, } \mathcal{E}_r(\kappa_{\text{glucose}}^{(\text{ST})}) < \mathcal{E}_f(\kappa_{\text{glucose}}^{(\text{ST})}),$$

$$\mathcal{E}_r(\kappa_{\text{lactose}}^{(\text{ST})}) < \mathcal{E}_f(\kappa_{\text{lactose}}^{(\text{ST})}), \text{ and thus } \mathcal{E}_r(\kappa_A^{\text{max}}) < \mathcal{E}_f(\kappa_A^{\text{max}}). \text{ For pyruvate, we have}$$

$$\mathcal{E}_{r/f}^{(\text{py})}(\kappa_{\text{py}}^{(C)}) = \mathcal{E}_{r/f}(\kappa_A^{(C)}) = 122 \text{ (h}^{-1}\text{)} \text{ (see [Eqs. S43](#) and [S101](#))}, \text{ and it is easy to check}$$

$$\text{that } \mathcal{E}_r(\kappa_{\text{py}}^{(\text{ST})}) < \mathcal{E}_f(\kappa_{\text{py}}^{(\text{ST})}).$$

For the remaining model parameters, note that we have classified the inactive ribosomal-affiliated proteins into the Q-class, and then $\phi_{\text{max}} = 48\%$ ([Scott et al., 2010](#)). The values of κ_t is obtainable from experiments: the translational speed is 20.1aa/s ([Scott et al., 2010](#)), with 7336 amino acids per ribosome ([Neidhardt, 1996](#)) and $\zeta \approx 1.67$ ([Neidhardt, 1996](#); [Scott et al., 2010](#)) (see [Appendix 1.1](#)), hence $\kappa_t = 1/610 \text{ (s}^{-1}\text{)}$. However, there are insufficient data to determine the values of κ_i ($i=a1, a2, b, c, d$) from the entry point metabolites to the precursor pools. Basically, it involves many steps, and thus these values should be considerably large. Here, we lump sum the contributions of κ_t and κ_i ($i=a1, a2, b, c, d$) by defining a composite parameter:

$$\Omega \equiv 1/\kappa_t + \sum_i^{a1, a2, b, c, d} \eta_i/\kappa_i. \quad (\text{S162})$$

We proceed to estimate the values Ω of ϕ and using experimental data ([Basan et al., 2015](#)) for wild-type strain on the $J_{\text{acetate}}^{(M)} - \lambda$ relation ([Fig. 1C](#)), and then all the rest of model parameters are set accordingly.

For the case of $w_0 = 0$, where all k_{cat} values follow a Gaussian distribution with an extrinsic noise of 25% CV, which is the general setting we use unless otherwise specified, then $\phi = 10.8$ and $\Omega = 1345 \text{ (s)}$. Accordingly, we have $\eta_E = 14.78$, $\mu_{\lambda_c} = 0.92 \text{ (h}^{-1}\text{)}$, and $\sigma_{\lambda_c} = 0.12\mu_{\lambda_c}$, where the CV of the extrinsic noise for Ω is estimated using the averaged CV of other κ_i . For the

translation inhibition effect of Cm, we estimate the values for ι as $\iota_{w_0=0}^{(2\mu\text{M Cm})} = 1.15$,

$\iota_{w_0=0}^{(4\mu\text{M Cm})} = 2.33$, $\iota_{w_0=0}^{(8\mu\text{M Cm})} = 6.25$, where the superscript stands for the concentration of Cm, and the subscript represents the choice of w_0 .

For pyruvate, with the value of η_E , we get $\phi_{\text{py}} = 14.82$. However, there is still a lack of proteome data to determine the value of κ_9 , which actually involves many steps in the metabolic network and thus can be considerably large. Here we define another composite parameter

$\Omega'_{\text{Gg}} \equiv (\eta_b + \eta_c)/\kappa_8 + \eta_{a1}/\kappa_9$ and estimate its value as $\Omega'_{\text{Gg}} = 690$ (s) from the growth rate data for *E. coli* measured under the relevant nutrient conditions (Basan et al., 2015), where the subscript “Gg” stands for glucogenesis. Then, $\mu_{\lambda_c^{(\text{py})}} = 0.67$ (h^{-1}), and

$\sigma_{\lambda_c^{(\text{py})}} = 0.10\mu_{\lambda_c^{(\text{py})}}$, where the same CV of extrinsic noise for Ω applies to Ω'_{Gg} .

For the case of a Group A carbon source mixed with 21 amino acids (21AA, with saturated concentrations), then $\phi_{21\text{AA}} = 14.2$. Comparing Eq. S32 with Eq. S112, the parameter Ω should

change to $\Omega_{21\text{AA}} \equiv 1/\kappa_t + \eta_{a1}/\kappa_{a1} + \sum_i^{a2,b,c,d} \eta_i/\kappa_i^{(21\text{AA})}$. Obviously, $1/\kappa_t < \Omega_{21\text{AA}} < \Omega$, and we

estimate $\Omega_{21\text{AA}} = 1000$ (s) from the growth rate data for *E. coli* measured under the relevant

nutrient conditions (Wallden et al., 2016). Then, we have $\mu_{\lambda_c^{21\text{AA}}} = 1.13$ (h^{-1}), and

$\sigma_{\lambda_c^{21\text{AA}}} = 0.12\mu_{\lambda_c^{21\text{AA}}}$.

For the case of a Group A carbon source mixed with 7 amino acids (7AA: His, Iso, Leu, Lys, Met, Phe, and Val), similar to the roles of $\phi_{21\text{AA}}$ and $\Omega_{21\text{AA}}$, we define $\phi_{7\text{AA}}$ and $\Omega_{7\text{AA}}$. Using the mass fraction of the 7AA combined with Eq. S18, we have $\phi_{7\text{AA}} = 11.6$. For the value of $\Omega_{7\text{AA}}$, evidently, $\Omega_{21\text{AA}} < \Omega_{7\text{AA}} < \Omega$, and we estimate $\phi_{7\text{AA}} = 1215$ (s) from the growth rate data for *E. coli* measured under the relevant culture media (Basan et al., 2015). Then, $\mu_{\lambda_c^{(7\text{AA})}} = 0.98$ (h^{-1}),

and $\sigma_{\lambda_c^{(7\text{AA})}} = 0.12\mu_{\lambda_c^{(7\text{AA})}}$.

For the case of $w_0 = 2.5$ (h^{-1}), we have $\phi = 8.3$, and thus $\eta_E = 12.28$, while other parameters such as Ω , μ_{λ_c} and σ_{λ_c} remain the same values as for $w_0 = 0$. Nevertheless, the values for ι upon translation inhibition with Cm are influenced by the choice of $w_0 = 0$, where the values of ι change into $\iota_{w_0=2.5}^{(2\mu\text{M Cm})} = 1.05$, $\iota_{w_0=2.5}^{(4\mu\text{M Cm})} = 2.00$, $\iota_{w_0=2.5}^{(8\mu\text{M Cm})} = 5.40$.

From [Appendix 7.1](#)–[7.2](#), combined with [Eq. S114](#), the distributions of $\lambda_r^{(21AA)}$ and $\lambda_f^{(21AA)}$ can be approximated by Gaussian distributions:

$$\begin{cases} \lambda_r^{(21AA)} \sim \mathcal{N}\left(\mu_{\lambda_r^{(21AA)}}, \sigma_{\lambda_r^{(21AA)}}^2\right), \\ \lambda_f^{(21AA)} \sim \mathcal{N}\left(\mu_{\lambda_f^{(21AA)}}, \sigma_{\lambda_f^{(21AA)}}^2\right), \end{cases} \quad (\text{S163})$$

where $\mu_{\lambda_r^{(21AA)}}$ and $\mu_{\lambda_f^{(21AA)}}$ stand for the mean values, while $\sigma_{\lambda_r^{(21AA)}}$ and $\sigma_{\lambda_f^{(21AA)}}$ represent the standard deviations. For the case of glucose mixed with 21AA (labeled as “Glucose+21AA”), the distribution of the growth rate $\lambda_{\text{glucose}}^{(21AA)}$ follows [Eq. S157](#). With $\Omega_{21AA} = 1000$ (s), we have $\mu_{\lambda_{\text{glucose},r}^{(21AA)}} = 1.34 \text{ (h}^{-1}\text{)}$, $\mu_{\lambda_{\text{glucose},f}^{(21AA)}} = 1.46 \text{ (h}^{-1}\text{)}$ (both definitions follow [Eq. S163](#)), and $\rho_{rf} \approx 1.0$ (obtained with numerical results).

For the case of succinate mixed with 21AA (labeled as “Succinate+21AA”), the respiration pathway is always more efficient since succinate lies within the TCA cycle, then the cell growth rate (defined as $\lambda_{\text{succinate}}^{(21AA)}$) would just take the value of the respiration one and thus follows a Gaussian distribution:

$$\lambda_{\text{succinate}}^{(21AA)} \sim \mathcal{N}\left(\mu_{\lambda_{\text{succinate}}^{(21AA)}}, \sigma_{\lambda_{\text{succinate}}^{(21AA)}}^2\right). \quad (\text{S164})$$

For the case that acetate is the sole carbon source, evidently, the cells only use the respiration pathway, and thus the growth rate (defined as λ_{acetate}) follows a Gaussian distribution:

$$\lambda_{\text{acetate}} \sim \mathcal{N}\left(\mu_{\lambda_{\text{acetate}}}, \sigma_{\lambda_{\text{acetate}}}^2\right). \quad (\text{S165})$$

With the measured growth rate data (Wallden et al., 2016), we estimate

$\mu_{\lambda_{\text{succinate}}^{(21AA)}} = 0.67 \text{ (h}^{-1}\text{)}$ and $\mu_{\lambda_{\text{acetate}}} = 0.253 \text{ (h}^{-1}\text{)}$. To illustrate the distribution of growth rates $\lambda_{\text{glucose}}^{(21AA)}$, $\lambda_{\text{succinate}}^{(21AA)}$ and λ_{acetate} shown in [Appendix-fig. 2B](#), if there was no other source of noise, extrinsic noise with a CV of 40% would be required for each k_{cat} value. Then, $\sigma_{\lambda_{\text{glucose},r}^{(21AA)}} \approx 0.21\mu_{\lambda_{\text{glucose},r}^{(21AA)}}$, $\sigma_{\lambda_{\text{glucose},f}^{(21AA)}} \approx 0.23\mu_{\lambda_{\text{glucose},f}^{(21AA)}}$, $\sigma_{\lambda_{\text{succinate}}^{(21AA)}} = 0.22\mu_{\lambda_{\text{succinate}}^{(21AA)}}$, and $\sigma_{\lambda_{\text{acetate}}} = 0.22\mu_{\lambda_{\text{acetate}}}$. Allowing for that intrinsic noise may also play a non-negligible role for the observed single-cell growth rate (which is not a long-term average), we still use extrinsic noise with a CV of 25% other than those shown in [Appendix-fig. 2B](#).

Appendix 8.3 Notes on the application of reference data

Data calibration

Throughout our manuscript, we use experimental data from the original references except for two calibrations. The first calibration is shown in the footnote to **Appendix-table2**. With this

calibration, the $J_{\text{acetate}}^{(M)} - \lambda$ data (Basan et al., 2015) in **Appendix-table2** align with the curve

shown in **Fig. 1C**, which includes experimental data from other sources. The second calibration is for the data shown in **Figs. 3F** and **1C** (chemostat data). The unit in the original reference (Holms, 1996) is mmol/(dry mass)g/h. To convert this into the unit mM/OD₆₀₀/h applied in our text, the conversion factor should be 0.18. Here, we deduce that only 60% of the measured dry biomass in centrifuged material is effective to calibrate with other experimental results. Therefore, there is a calibration factor of 0.6, and the conversion factor changes into 0.3.

Data of the inducible strains

We note that part of the experiment data in the original references (Basan et al., 2015; Hui et al., 2015) were obtained using strains with titratable systems (e.g. titratable ptsG, LacY). Basically,

the $J_{\text{acetate}}^{(M)} - \lambda$ relation of these inducible strains align with the same curve as that of the wild-

type *E. coli* (**Fig. 1C**). Given that evolution treatment is not involved for the inducible strain, we regard the titration perturbation as a technique to fool the strains. That is, these inducible strains behave as if they were cultured in a less efficient Group A carbon source.

Experimental data sources

The batch culture data shown in **Fig. 1C** (labeled with minimum/rich media or inducible strains) and **Appendix-fig.2C** were taken from the source data of the reference's **figure 1** (Basan et al., 2015). The chemostat data shown in **Fig. 1C** were taken from the reference's table 7 (Holms, 1996). The data shown in **Fig. 1D** were taken from the reference's extended data figure 3a (Basan et al., 2015) with the calibration specified in the footnote to **Appendix-table2**.

The data shown in **Fig. 2A** were adopted from the reference's extended data figure 4a-b (Basan et al., 2015). The data shown in **Fig. 2B** were taken from the source data of the reference's **figure 2a** (Basan et al., 2015). The data shown in **Fig. 2C** were taken from the source data of the reference's **figure 3a** (Basan et al., 2015). The data shown in **Fig. 3A-B** were taken from the source data of the reference's **figure 3d** (Basan et al., 2015).

The data shown in **Fig. 3C-D** and **Appendix-fig.2D-E** were taken from the source data of the reference's **figure 3c** (Basan et al., 2015). The data shown in **Fig. 3F** were taken from the reference's table 7 (Holms, 1996), with a calibration factor specified in the above paragraph ("Data calibration").

The data shown in **Fig.4A-B** and **Appendix-fig.3A-D** were taken from the reference's table S2 with the label "C-lim" (Hui et al., 2015). We excluded the reference's data with $\lambda=0.45205$ (h⁻¹) as there are other unconsidered factors involved during slow growth (Dai et al., 2016) (for $\lambda < 0.5$ h⁻¹), and we suspect that there may be unknown calibration factors. The data shown in **Fig. 4C-D** and **Appendix-fig.3E-N** were adopted from the reference's extended data figure 6-7 (Basan et al., 2015).

The gene names depicted in **Appendix-fig. 1B** were identified using KEGG database. The data shown in **Appendix-fig. 2G** were drawn from **Appendix-table1**, which includes the original references themselves. The flux data presented in **Appendix-table2** were obtained from the

reference's extended data figure 3a(Basan et al., 2015 [link](#)), with the calibration specified in the footnote. The proteome data shown in **Appendix-table2** [link](#) were taken from the reference's supplementary Table N5(Basan et al., 2015 [link](#)).

No.*	Reaction	Enzyme	Gene name	EC	MW (kDa)	In vitro k_{cat} (s ⁻¹)	References	In vivo k_{cat} (s ⁻¹)	Chosen k_{cat} (s ⁻¹)
J_1	Glucose-6P ↔ Fructose-6P	Glucose-6-phosphate isomerase	pgi	EC:5.3.1.9	1.2×10 ²	2.6×10 ²	PMID: 7004378; DOI:10.1016/j.ijms.2004.09.017	8.7×10 ²	8.7×10 ²
	Fructose-6P → Fructose-1,6P	Phosphofructokinase	pfkA [‡]	EC:2.7.1.11	1.4×10 ²	4.4×10 ²	PMID: 6218375; 70226	1.7×10 ³	1.7×10 ³
	Fructose-1,6P ↔ Glyceraldehyde 3-phosphate+Dihydroxyacetone phosphate	Fructose-bisphosphate aldolase	fbaA [‡]	EC:4.1.2.13	7.8×10	1.4×10	PMID: 8939754;15531627	1.6×10 ²	1.6×10 ²
	Dihydroxyacetone phosphate ↔ Glyceraldehyde 3-phosphate	Triosephosphate Isomerase	tpiA	EC:5.3.1.1	5.4×10	4.3×10 ²	PMID: 3887397; 6092857	2.7×10 ²	2.7×10 ²
	Glyceraldehyde 3-phosphate ↔ 1,3-Bisphosphoglycerate	Glyceraldehyde-3-phosphate dehydrogenase	gapA	EC:1.2.1.12	1.4×10 ²	9.5×10	PMID: 4932978; 2200929	1.5×10 ²	1.5×10 ²
	1,3-Bisphosphoglycerate ↔ 3-Phosphoglycerate	Phosphoglycerate kinase	pgk	EC:2.7.2.3	4.4×10	3.5×10 ²	PMID: 367367;166274	1.9×10 ²	1.9×10 ²
	3-Phosphoglycerate ↔ 2-Phosphoglycerate	Phosphoglycerate mutase	gpmA [‡]	EC:5.4.2.11	4.9×10	3.3×10 ²	PMID: 10437801	4.5×10 ²	4.5×10 ²
	2-Phosphoglycerate ↔ Phosphoenolpyruvate	Enolase	eno	EC:4.2.1.11	9.0×10	2.2×10 ²	PMID: 1094232;4942326	1.7×10 ²	1.7×10 ²
J_2	Phosphoenolpyruvate → Pyruvate	Pyruvate kinase	pykF [‡]	EC:2.7.1.40	2.4×10 ²	5.0×10 ²	PMID: 6759852	1.6×10 ³	1.6×10 ³
	Pyruvate → Acetyl-CoA	Pyruvate dehydrogenase	aceE [‡]	EC:1.2.4.1	1.0×10 ²	1.2×10 ²	PMID: 23088422	3.4×10 ²	3.4×10 ²
J_3	Oxaloacetate +Acetyl-CoA → Citrate	Citrate synthase	gltA	EC:2.3.3.1	9.7×10	2.4×10 ²	PMID:4900996;23954305	7.1×10	7.1×10
	Citrate ↔ Isocitrate	Aconitate hydratase	acnB [‡]	EC:4.2.1.3	9.4×10	7.0×10	PMID:15963579;12473114	6.3×10	6.3×10
	Isocitrate → α-Ketoglutarate	Isocitrate dehydrogenase	icd	EC:1.1.1.42	9.5×10	2.0×10 ²	PMID: 8141;36923;2200929	3.3×10	3.3×10
J_4	α-Ketoglutarate → Succinyl-CoA	α-Ketoglutarate dehydrogenase complex E1 component	suc A suc B [‡]	EC:1.2.4.2, EC:2.3.1.61	1.9×10 ²	1.5×10 ²	PMID: 6380583; 4588679	1.3×10 ²	1.3×10 ²
	Succinyl-CoA ↔ Succinate	Succinyl-CoA synthetase	suc C suc D	EC:6.2.1.5	1.6×10 ²	9.1×10	PMID: 5338130	1.0×10 ²	1.0×10 ²
	Succinate → Fumarate	Succinate dehydrogenase	sdh A sdh B [‡]	EC:1.3.5.1	1.0×10 ²	1.1×10 ²	PMID: 4334990; 16484232	1.1×10 ²	1.1×10 ²
	Fumarate ↔ Malate	Fumarase	fumA [‡]	EC:4.2.1.2	2.0×10 ²	1.2×10 ³	PMID: 3282546;12021453	4.9×10 ²	4.9×10 ²
	Malate ↔ Oxaloacetate	Malate dehydrogenase	mdh	EC:1.1.1.37	6.1×10	5.5×10 ²	doi:10.1016/0076-6879 (69)13029-3	6.6×10	6.6×10
J_5	Phosphoenolpyruvate →Oxaloacetate	Phosphoenolpyruvate carboxylase	ppc	EC:4.1.1.31	4.0×10 ²	1.5×10 ²	PMID: 9927652; 4932977	/	1.5×10 ²
J_6	Acetyl-CoA ↔ Acetyl phosphate	Phosphate acetyltransferase	pta	EC:2.3.1.8	7.7×10	3.0×10	PMID:20236319	3.7×10 ²	3.7×10 ²
	Acetyl phosphate ↔ Acetate	Acetate kinase	ackA	EC:2.7.2.1	4.3×10	3.6×10 ³	EcoCyc: EG10027; PMID: 24801996	3.3×10 ²	3.3×10 ²
	Acetate (intracellular) ↔ Acetate (extracellular)	Acetate transporter	actP	/	2×10	4.7×10 ²	PMID: 31405984 (Estimated)	/	4.7×10 ²

Appendix-table 1.

Molecular weight (MW) and in vivo/in vitro k_{cat} data for *E. coli*

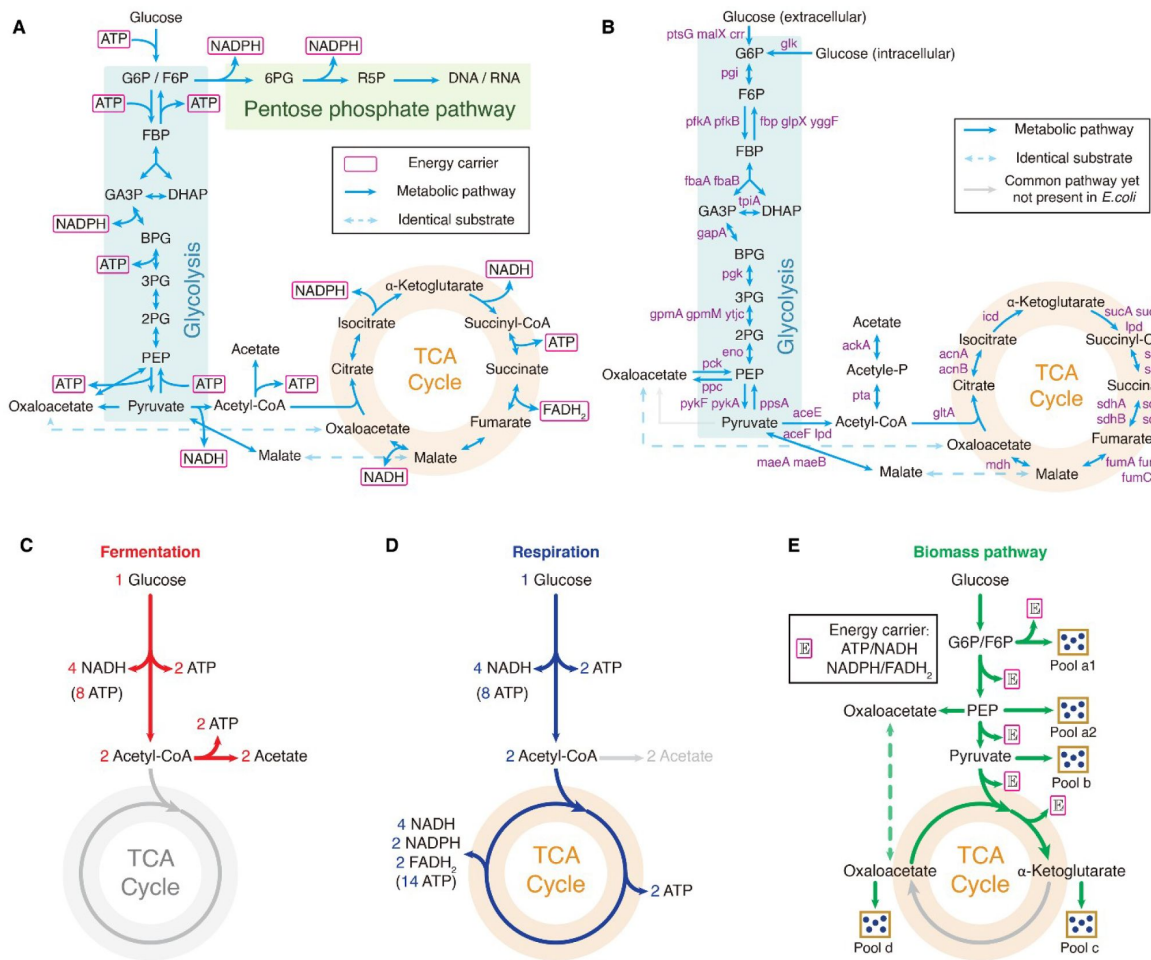
J_7	Pyruvate → Phosphoenolpyruvate	Pyruvate, water dikinase	ppsA	EC:2.7.9.2	2.5×10^2	3.5×10	PMID: 4319237	/	3.5×10
J_A	Glucose-6P (extracellular) → Glucose-6P (intracellular)	Glucose-6-phosphate transporter	UhpT	/	5×10	2×10^2	PMID: 3283129;2197272; 20018695 (Estimated)	/	2×10^2
	Glucose (extracellular) → Glucose-6P	Glucose-specific PTS enzyme	ptsG	EC: 2.7.1.199	5×10	1×10^2	PMID: 9575173; 20018695; 12146972	/	1×10^2
	Lactose (extracellular) → Lactose (intracellular)	Lactose transporter	lacY	/	4.6×10	6×10	PMID: 6444453; 20018695	/	6×10
	Lactose → Glucose + Galactose	β-galactosidase	lacZ	EC:3.2.1.23	4.6×10^2	6.4×10^2	PMID: 8008071; 23011886 (Estimated)	/	6.4×10^2
J_{py}	Pyruvate (extracellular) → Pyruvate (intracellular)	Pyruvate transporter	btsT CstA	/	8×10	6×10	PMID:20018695; 33260635; EcoCyc: G7942;EG10167 (Estimated)	/	6×10

Appendix-table 1. (continued)

	Culture1	Culture2	Culture3	Culture4
Growth rate λ (h ⁻¹) *	0.82	0.87	0.97	1.03
J_{acetate} (mM OD ₆₀₀ ⁻¹ h ⁻¹) †	0.39	1.18	2.68	2.84
J_{CO_2} (mM OD ₆₀₀ ⁻¹ h ⁻¹) †	7.44	6.05	4.30	3.04
Gene name	Protein fractions obtained using absolute abundance (ϕ)			
pgi	0.09%	0.09%	0.10%	0.11%
pfkA	0.06%	0.06%	0.06%	0.06%
fbaA	0.32%	0.35%	0.35%	0.39%
tpiA	0.12%	0.15%	0.13%	0.18%
gapA	1.19%	1.29%	1.33%	1.47%
pgk	0.30%	0.31%	0.32%	0.36%
gpmA	0.15%	0.15%	0.15%	0.16%
eno	0.63%	0.70%	0.75%	0.83%
pykF	0.15%	0.15%	0.18%	0.21%
aceE	0.30%	0.32%	0.34%	0.41%
gltA	0.88%	0.80%	0.61%	0.48%
acnB	0.92%	0.84%	0.66%	0.57%
icd	1.55%	1.55%	1.31%	1.39%
suc A	0.71%	0.75%	0.64%	0.55%
suc B				
suc C	0.88%	0.84%	0.66%	0.52%
suc D				
sdh A	0.49%	0.45%	0.42%	0.35%
sdh B				
fumA	0.24%	0.21%	0.17%	0.13%
mdh	0.45%	0.45%	0.41%	0.39%
pta	0.10%	0.10%	0.10%	0.10%
ackA	0.06%	0.07%	0.06%	0.06%

Appendix-table 2.

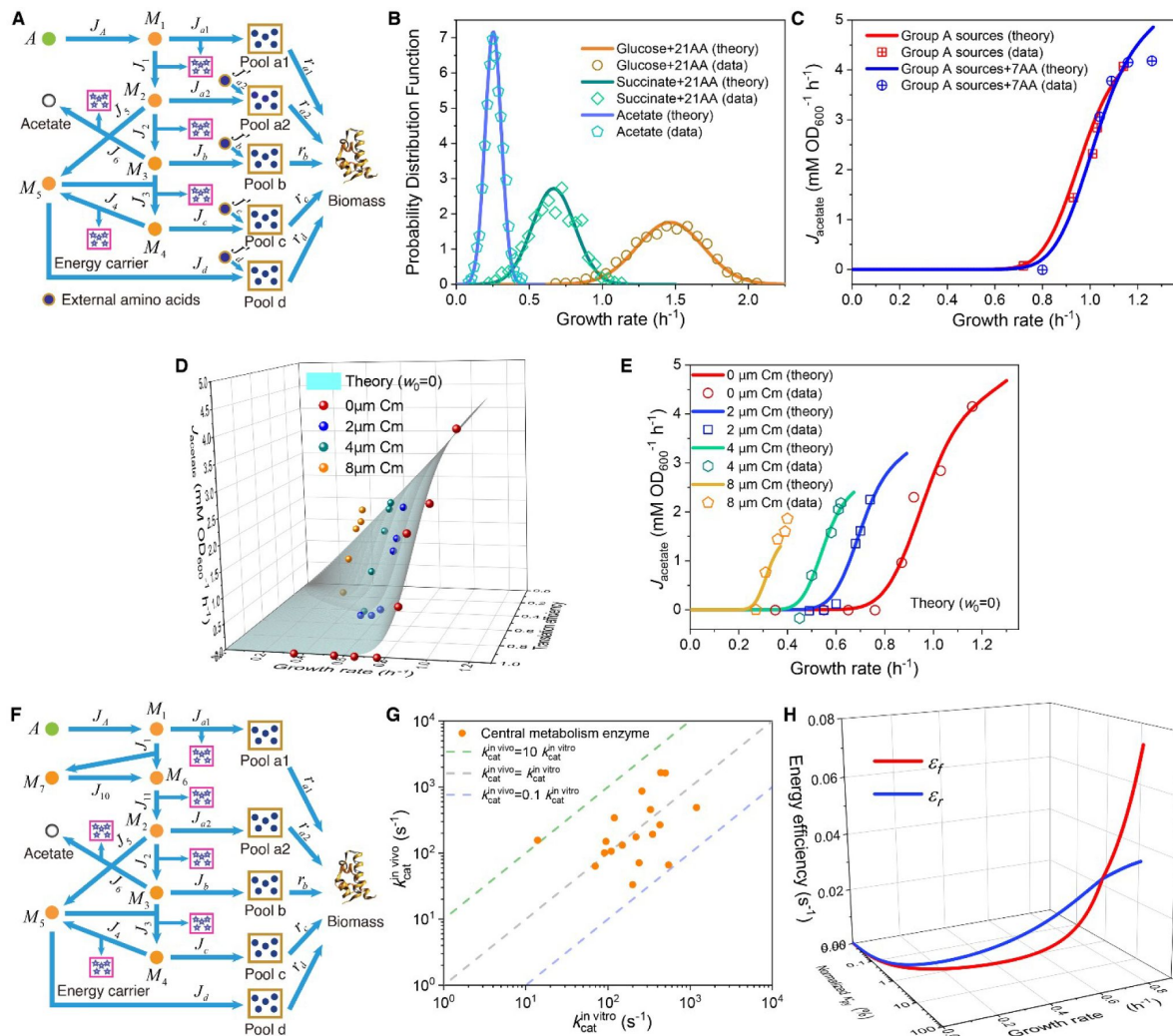
Proteome and flux data([Basan et al., 2015](#) [↗](#))used to calculate the in vivo k_{cat}



Appendix-figure 1.

Central metabolic network and carbon utilization pathways

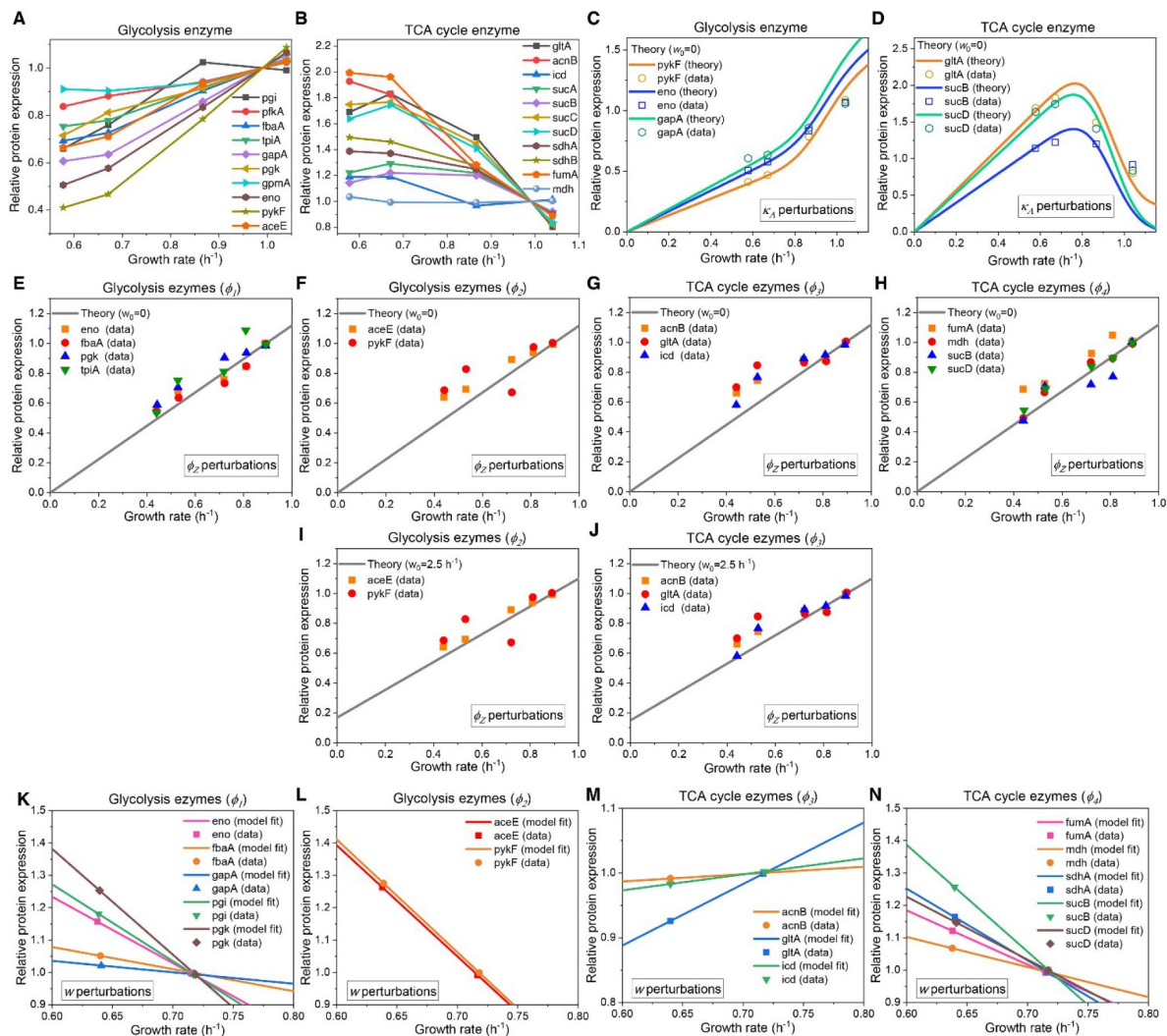
(A) Energy production details of the central metabolic network. In *E. coli*, NADPH and NADH are interconvertible (Sauer et al., 2004), and all energy carriers can be converted to ATP with ADP. The conversion factors are: NADH=2ATP, NADPH=2ATP, FADH₂=1ATP (Neidhardt et al., 1990). (B) Relevant genes for enzymes in the central metabolic network. (C-E) Three destinies of glucose metabolism. (C) Fermentation pathway, where a molecule of glucose generates 12 ATPs in *E. coli*. (D) Respiration pathway, where a molecule of glucose generates 26 ATPs. (E) Biomass pathway, where glucose turns into precursors of biomass. Note that the process of biomass generation is accompanied by ATPs production (see Appendix 2.1).



Appendix-figure 2.

Model and results for experimental comparison

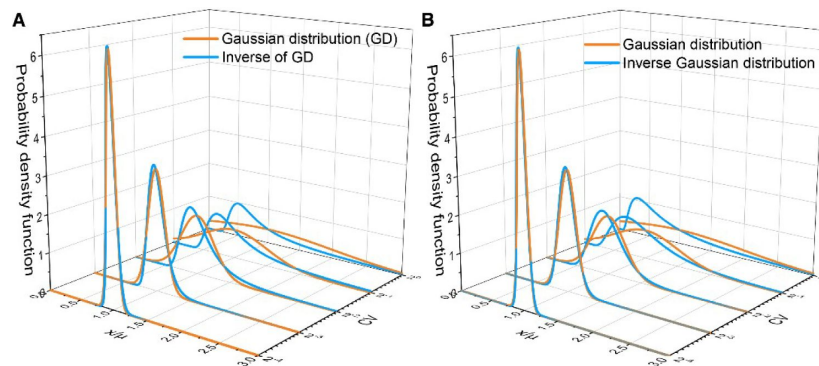
(A–C) Model analysis for carbon utilization in mixtures with amino acids. (A) Coarse-grained model for the case of a Group A carbon source mixed with extracellular amino acids. (B) Model predictions (Eqs. S157, S164–S165) and single-cell reference experimental results (Wallden et al., 2016) of the growth rate distributions for *E. coli* in three culturing conditions. (C) Comparison of the growth rate-fermentation flux relation for *E. coli* in Group A carbon sources between minimum media and enriched media (those with 7AA). (D–E) Influence of translation inhibition on overflow metabolism. (D) A 3D plot of the relations among the fermentation flux, growth rate, and the translation efficiency (Eqs. 79 and S160). (E) Growth rate dependence of acetate excretion rate as K_A varies, with each fixed dose of Cm. The translation efficiency is tuned by the dose of Cm, and the maintenance energy coefficient is set to be 0 (i.e., $w_0 = 0$). (F) The coarse-grained model for Group A carbon source utilization. This model includes more details to compare with experiments. (G) Comparison of the in vivo and in vitro catalytic rates for enzymes within glycolysis and the TCA cycle (see Appendix-table1 for details). (H) The energy efficiencies of respiration and fermentation pathways vary with growth rate as functions of the substrate quality of pyruvate (Eqs. S93 and S96).



Appendix-figure 3.

Relative protein expression of central metabolic enzymes under various types of perturbations

(A-D) Relative protein expression under κ_A perturbation. (A) Experimental data (Hui et al., 2015) of the catalytic enzymes for each step of glycolysis. (B) Experimental data (Hui et al., 2015) of the catalytic enzymes for each step of the TCA cycle. (C) Model predictions (Eq. S118, with $w_0 = 0$) and experimental data (Hui et al., 2015) of representative genes from glycolysis. (D) Model predictions (Eq. S118, with $w_0 = 0$) and experimental data (Hui et al., 2015) of representative genes from the TCA cycle. (E-J) Relative protein expression under ϕ_Z perturbation. (E, F, I) Model predictions and experimental data (Basan et al., 2015) of representative genes from glycolysis. (G, H, J) Model predictions and experimental data (Basan et al., 2015) of representative genes from the TCA cycle. (E-H) Results of ϕ_Z perturbation with $w_0 = 0$ (Eq. S120). (I-J) Results of ϕ_Z perturbation with $w_0 = 2.5$ (Eq. S121). (K-N) Relative protein expression upon energy dissipation. (K-L) Model fits (Eqs. S127 and S123) and experimental data (Basan et al., 2015) of representative genes from glycolysis. (M-N) Model fits (Eqs. S127 and S123) and experimental data (Basan et al., 2015) of representative genes from the TCA cycle.



Appendix-figure 4.

Asymptotic distributions of inverse Gaussian distribution and the inverse of Gaussian distribution

(A) Comparison between the inverse of Gaussian distribution and the corresponding Gaussian distribution for each value of coefficient of variation (CV) (Eqs. S140 and S145). (B) Comparison between the inverse Gaussian distribution and the corresponding Gaussian distribution for each value of CV (Eqs. S142 and S146). Both inverse Gaussian distribution and the inverse of Gaussian distribution converge to Gaussian distributions when CV is small.

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Reviewer #1 (Public Review):

Summary:

Cell metabolism exhibits a well-known behavior in fast-growing cells, which employ seemingly wasteful fermentation to generate energy even in the presence of sufficient environmental oxygen. This phenomenon is known as Overflow Metabolism or the Warburg effect in cancer. It is present in a wide range of organisms, from bacteria and fungi to mammalian cells.

In this work, starting with a metabolic network for *Escherichia coli* based on sets of carbon sources, and using a corresponding coarse-grained model, the author applies some well-based approximations from the literature and algebraic manipulations. These are used to successfully explain the origins of Overflow Metabolism, both qualitatively and quantitatively, by comparing the results with *E. coli* experimental data.

By modeling the proteome energy efficiencies for respiration and fermentation, the study shows that these parameters are dependent on the carbon source quality constants K_i (p.115 and 116). It is demonstrated that as the environment becomes richer, the optimal solution for proteome energy efficiency shifts from respiration to fermentation. This shift occurs at a critical parameter value $K_A(C)$.

This counterintuitive result qualitatively explains Overflow Metabolism.

Quantitative agreement is achieved through the analysis of the heterogeneity of the metabolic status within a cell population. By introducing heterogeneity, the critical growth rate is assumed to follow a Gaussian distribution over the cell population, resulting in accordance with experimental data for *E. coli*. Overflow metabolism is explained by considering optimal protein allocation and cell heterogeneity.

The obtained model is extensively tested through perturbations: 1) Introduction of overexpression of useless proteins; 2) Studying energy dissipation; 3) Analysis of the impact of translation inhibition with different sub-lethal doses of chloramphenicol on *Escherichia coli*; 4) Alteration of nutrient categories of carbon sources using pyruvate. All model perturbation results are corroborated by *E. coli* experimental results.

Strengths:

In this work, the author employs modeling methods typical of Physics to address a problem in Biology, standing at the interface between these two scientific fields. This interdisciplinary approach proves to be highly fruitful and should be further explored in the literature. The use of *Escherichia coli* as an example ensures that all hypotheses and approximations in this study are well-founded in the literature. Examples include the approximation for the Michaelis-Menten equation (line 82), Eq. S1, proteome partition in Appendix 1.1 (lines 68-69), and a stable nutrient environment in Appendix 1.1 (lines 83-84). The section "Testing the model through perturbation" heavily relies on bacterial data. The construction of the model and its agreement with experimental data are convincingly presented.

Weaknesses:

In Section Appendix 6.4, the author explores the generalization of results from bacteria to cancer cells, adapting the metabolic network and coarse-grained model accordingly. It is argued that as a consequence, all subsequent steps become immediately valid. However, I remain unconvinced, considering the numerous approximations used to derive the equations, which the literature demonstrates to be valid primarily for bacteria. A more detailed discussion about this generalization is recommended. Additionally, it is crucial to note that the experimental validation of model perturbations heavily relies on *E. coli* data.

<https://doi.org/10.7554/eLife.94586.1.sa2>

Reviewer #2 (Public Review):

Summary

This paper has three parts. The first part applied a coarse-grained model with proteome partition to calculate cell growth under respiration and fermentation modes. The second part considered single-cell variability and performed population average to acquire an ensemble metabolic profile for acetate fermentation. The third part used model and simulation to compare experimental data in literature and obtained substantial consistency.

Strengths and major contributions

- (i) The coarse-grained model considered specific metabolite groups and their inter-relations and acquired an analytical solution for this scenario. The "resolution" of this model is in between the Flux Balanced Analysis/whole-cell simulation and proteome partition analysis.
- (ii) The author considered single-cell level metabolic heterogeneity and calculated the ensemble average with explicit calculation. The results are consistent with known fermentation and growth phenomena qualitatively and can be quantitatively compared to experimental results.

Weaknesses

- (i) If I am reading this paper correctly, the author's model predicts binary (or "digital") outcomes of single-cell metabolism, that is, after growth rate optimization, each cell will adopt either "respiration mode" or "fermentation mode" (as illustrated in Figure Appendix - Figure 1 C, D). Due to variability enzyme activity k_i^{cat} and critical growth rate λ_C , each cell under the same nutrient condition could have either respiration or fermentation, but the choice is binary.

The binary choice at the single-cell level is inconsistent with our current understanding of metabolism. If a cell only uses fermentation mode (as shown in Appendix - Figure 1C), it could generate enough energy but not be able to have enough metabolic fluxes to feed into the TCA cycle. That is, under pure fermentation mode, the cell cannot expand the pool of TCA cycle metabolites and hence cannot grow.

This caveat also appears in the model in Appendix (S25) that assumes $J_E = r_E J_{\text{BM}}$ where r_E is a constant. From my understanding, r_E can be different between respiration and fermentation modes (at least for real cells) and hence it is inappropriate to conclude that cells using fermentation, which generates enough energy, can also generate a balanced biomass.

- (ii) The minor weakness of this model is that it assumes a priori that each cell chooses its metabolic strategy based on energy efficiency. This is an interesting assumption but there is no known biochemical pathway that directly executes this mechanism. In evolution, growth rate is more frequently considered for metabolic optimization. In Flux Balanced Analysis, one could have multiple objective functions including biomass synthesis, energy generation, entropy production, etc. Therefore, the author would need to justify this assumption and

propose a reasonable biochemical mechanism for cells to sense and regulate their energy efficiency.

My feeling is that the mathematical structure of this model could be correct, but the single-cell interpretation for the ensemble averaging has issues. Each cell could potentially adopt partial respiration and partial fermentation at the same time and have temporal variability in its metabolic mode as well. With the modification of the optimization scheme, the author could have a revised model that avoids the caveat mentioned above.

Discussion and impact for the field

Proteome partition models and Flux Balanced Analysis are both commonly used mathematical models that emphasize different parts of cellular physiology. This paper has ingredients for both, and I expect after revision it will bridge our understanding of the whole cell.

<https://doi.org/10.7554/eLife.94586.1.sa1>

Reviewer #3 (Public Review):

Summary:

In the manuscript "Overflow metabolism originates from growth optimization and cell heterogeneity" the author Xin Wang investigates the hypothesis that the transition into overflow metabolism at large growth rates actually results from an inhomogeneous cell population, in which every individual cell either performs respiration or fermentation.

Weaknesses:

The paper has several major flaws. First, and most importantly, it repeatedly and wrongly claims that the origins of overflow metabolism are not known. The paper is written as if it is the first to study overflow metabolism and provide a sound explanation for the experimental observations. This is obviously not true and the author actually cites many papers in which explanations of overflow metabolism are suggested (see e.g. Basan et al. 2015, which even has the title "Overflow metabolism in *E. coli* results from efficient proteome allocation"). The paper should be rewritten in a more modest and scientific style, not attempting to make claims of novelty that are not supported. In fact, all hypotheses in this paper are old. Also the possibility that cell heterogeneity explains the observed 'smooth' transition into overflow metabolism has been extensively investigated previously (see de Groot et al. 2023, PNAS, "Effective bet-hedging through growth rate dependent stability") and the random drawing of k_{cat} -values is an established technique (Beg et al., 2007, PNAS, "Intracellular crowding defines the mode and sequence of substrate uptake by *Escherichia coli* and constrains its metabolic activity"). Thus, in terms of novelty, this paper is very limited. It reinvents the wheel and it is written as if decades of literature debating overflow metabolism did not exist.

Moreover, the manuscript is not clearly written and is hard to understand. Variables are not properly introduced (the M-pools need to be discussed, fluxes (J_E), "energy coefficients" (η_E), etc. need to be more explicitly explained. What is "flux balance at each intermediate node"? How is the "proteome efficiency" of a pathway defined? The paper continues to speak of energy production. This should be avoided. Energy is conserved (1st law of thermodynamics) and can never be produced. A scientific paper should strive for scientific correctness, including precise choice of words.

The statement that the "energy production rate ... is proportional to the growth rate" is, apart from being incorrect - it should be 'ATP consumption rate' or similar (see above), a non-trivial claim. Why should this be the case? Such statements must be supported by references. The observation that the catabolic power indeed appears to increase linearly with growth rate was made, based on chemostat data for *E. coli* and yeast, in a recent preprint (Ebenhöh et al,

2023, bioRxiv, "Microbial pathway thermodynamics: structural models unveil anabolic and catabolic processes").

All this criticism does not preclude the possibility that cell heterogeneity plays a role in overflow metabolism. However, according to Occam's razor, first the simpler explanations should be explored and refuted before coming up with a more complex solution. Here, it means that the authors first should argue why simpler explanations (e.g. the 'Membrane Real Estate Hypothesis', Szenk et al., 2017, Cell Systems; maximal Gibbs free energy dissipation, Niebel et al., 2019, Nature Metabolism; Saadat et al., 2020, Entropy) are not considered, resp. in what way they are in disagreement with observations, and then provide some evidence of the proposed cell heterogeneity (are there single-cell transcriptomic data supporting the claim?).

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